Structural Relaxation in Glass-Forming Liquids at Extreme Pressure: Photon Correlation Spectroscopy and Constrained Thermodynamic Surface Modeling

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ABSTRACT

Structural relaxation in glass-forming liquids is strongly dependent on both temperature and pressure, yet direct measurement of the structural α-relaxation under multi-gigapascal compression has remained experimentally challenging. This work extends depolarized photon correlation spectroscopy (DPCS) to pressures approaching 5 GPa in a diamond anvil cell, representing the highest-pressure implementation of the technique and enabling direct measurement of the structural α-relaxation in a pure liquid at extreme compression. This capability provides access to dynamical regimes previously unexplored by photon correlation spectroscopy and establishes a platform for probing glassy dynamics at high pressure. To describe these measurements, a constrained relaxation-time surface framework is developed within the Vogel–Tammann–Fulcher (VTF) formalism. A polynomial pressure dependence of the fragility parameter as well as the zero-mobility temperature captures the evolution of dynamics across broad regions of thermodynamic phase space. The surfaces are constrained by atmospheric-pressure data and by an independently measured isochrone obtained using the TGP technique, which probes mechanical relaxation under an imposed pressure gradient and anchors the dynamics near the glass transition. For glycerol, the framework captures deviations from simple scaling associated with pressure-induced suppression of hydrogen bonding. For the van der Waals liquid cumene, a pressure-modified VTF relation enforces the required high-pressure asymptotic approach to vanishing mobility. Together, these experimental and modeling developments provide a coherent framework for measuring and describing structural relaxation in glass-forming liquids under extreme compression.

DEDICATION

Smirfibus quorum caeruleae ineptiae Gargamelem semper nimis occupatum tenuerunt ne mundo verum damnum inferret. Deo qui nos non percutit, etiam cum interpellationem eius oramus in ludis pediludii, plane conscii orbem terrarum plenissimum esse pueris esurientibus. Meaeque impotentiae dicandi aliquid revera eloquens et conveniens.

EPIGRAPH

All models are wrong, but some are useful.

–George E. P. Box

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1 Introduction

1.1 Scientific Context and Motivation

The dramatic slowing of structural dynamics in supercooled liquids as they approach the glass transition remains one of the central problems in condensed matter physics. Over a relatively narrow temperature interval the characteristic structural relaxation time τ_α may increase by more than ten orders of magnitude while the liquid remains structurally disordered. Understanding how this dynamical slowdown emerges from molecular motion and how it evolves under varying thermodynamic conditions remains a fundamental challenge in the physics of glass formation.

A convenient way to visualize this phenomenon is through the Angell plot, in which the logarithm of the structural relaxation time or viscosity is plotted as a function of inverse reduced temperature T_g/T. In this representation liquids with widely differing glass transition temperatures may be directly compared on a common scale. Strong glass-forming liquids exhibit nearly Arrhenius behavior and appear approximately linear on such a plot, whereas fragile systems display pronounced curvature as the glass transition is approached. Representative examples are shown in Fig. 3.1, where systems of differing fragility can be compared directly.

The power of the Angell representation lies in its ability to compress an enormous dynamic range into a simple graphical form. As temperature decreases toward T_g, the structural relaxation time may increase from picoseconds to hundreds of seconds. The slope of the curve near T_g defines the fragility of the liquid, a parameter widely used to characterize the degree of deviation from Arrhenius behavior. In scaled representations of the Angell plot, such as that shown in Fig. 3.2, both

the glass transition and high-temperature limits are normalized, enabling consistent comparison between viscosity measurements and structural relaxation data across many experimental probes.

Importantly, many glass-forming liquids exhibit a similar limiting behavior at high temperatures, where relaxation dynamics approach a common Arrhenius-like regime, as illustrated in Fig. 3.1. The scaled Angell representation leverages this convergence by explicitly normalizing both the high-temperature limit and the glass transition point. In doing so, it provides a more rigorous and universal definition of fragility, wherein the minimum possible scaled fragility is, by construction, $m^* = 1$. This normalization removes some ambiguity associated with differing experimental probes and absolute timescales, allowing deviations from Arrhenius behavior to be compared on an equal footing across materials. Though this is a minor yet valuable contribution to the idea of fragility, this work is certainly not the first to suggest such an adaptation. Scaled fragility and associated Angell plots are discussed in detail in Sec. 3.1.

A more weighty contribution...****Most experimental investigations of structural relaxation have historically been performed along isobaric temperature scans. In this traditional approach the Angell plot represents a single trajectory through thermodynamic space corresponding to measurements performed at fixed pressure. While this framework has proven enormously useful, it implicitly treats the relaxation behavior of liquids as a family of independent one-dimensional curves.

In reality, structural relaxation is fundamentally a function of both temperature and pressure. When viewed more generally the relaxation time may be considered a continuous thermodynamic surface $\tau_\alpha(P, T)$ defined across pressure–temperature space. In this perspective the familiar Angell plot corresponds simply to a slice through that surface along a single isobar. Conceptually, one may imagine extending the traditional Angell representation into a third dimension, forming a

relaxation-time surface whose contours represent isochrones of constant structural relaxation time. A schematic illustration of this interpretation is shown in Fig. **??**.

This surface-based viewpoint provides a natural framework for examining how glassy dynamics evolve under compression. For example, the fragility of a liquid—traditionally defined as the slope of the Angell plot at the glass transition—may be interpreted geometrically as a directional derivative of the relaxation-time surface along a particular thermodynamic path. Different trajectories through the (P, T) surface therefore correspond to different apparent fragilities, highlighting the importance of considering pressure as an intrinsic variable in the study of glass formation.

Despite its conceptual importance, direct experimental exploration of structural relaxation across the full pressure–temperature domain remains challenging. High-pressure measurements are typically conducted in diamond anvil cells (DACs), where the available sample volume is extremely small. Many traditional probes of structural relaxation struggle under these conditions because of limited signal strength, restricted optical access, or the difficulty of maintaining precise temperature control under compression.

1.2 Contributions of This Work

The work presented in this dissertation addresses these challenges through a combination of thermodynamic modeling, analysis of existing experimental datasets, and the development of new experimental capabilities aimed at extending the study of glassy dynamics into the gigapascal pressure regime.

Two molecular glass-forming liquids are examined: glycerol, a strongly hydrogen-bonded system, and cumene, a liquid dominated primarily by van der Waals interactions. Studying these

contrasting systems allows the effects of compression on structural relaxation to be examined across distinct intermolecular interaction regimes.

A central theme of this work is the construction of pressure-dependent relaxation-time surfaces $\tau_\alpha(P,T)$ describing structural relaxation across pressure–temperature space. These surfaces combine experimental measurements obtained along different thermodynamic paths—including isothermal pressure scans, viscosity measurements, and isochronal constraints—into a continuous representation of structural relaxation. Within this framework the familiar Angell plot may be interpreted as a single trajectory through the broader relaxation surface.

The first set of contributions presented in this dissertation concerns the construction and analysis of such relaxation-time surfaces for glycerol. In this work literature viscosity measurements are combined with thermodynamic constraints derived from the glass-transition pressure (TGP) method to produce a continuous representation of $\tau_\alpha(P,T)$. The TGP technique itself was developed previously and provides an experimental method for determining the pressure–temperature locus of the glass transition. In this dissertation the TGP isochrone is incorporated directly into the surface modeling framework, providing an important constraint on the relaxation-time surface in the region surrounding the glass transition. Anchoring the surface near $T_g(P)$ is essential for obtaining meaningful measurements of fragility and for ensuring that the modeled relaxation behavior remains physically consistent near the glass-transition boundary.

In addition to incorporating the TGP constraint into the relaxation-surface modeling procedure, improvements were made to the experimental implementation of the TGP method. These include partial automation of the experimental procedure used to determine the TGP isochrone. This automation improves the precision and reproducibility of the measurements and enables the acquisition of higher-quality isochronal data near the glass transition. In the longer term such

improvements may allow the determination of multiple closely spaced isochrones near $T_g(P)$, providing additional constraints on the slope of the relaxation surface in the vicinity of the glass transition.

A second contribution of this work is the determination of the characteristic relaxation timescale associated with the TGP glass-transition isochrone. This timescale is obtained through two independent approaches: one based on thermodynamic scaling relations and a second derived from constrained relaxation-time surface modeling. Agreement between these methods provides a consistent dynamical interpretation of the TGP criterion within the broader framework of glass-transition measurements.

The final and most experimentally significant contribution of this dissertation is the extension of depolarized photon correlation spectroscopy (DPCS) into a diamond anvil cell environment for the direct measurement of structural relaxation under high pressure. Depolarized photon correlation spectroscopy measures time-dependent fluctuations in the intensity of scattered light arising from molecular reorientation and structural relaxation processes in liquids. By analyzing the autocorrelation function of the scattered intensity, characteristic structural relaxation times can be determined directly in the time domain.

Although photon correlation techniques have long been used to study glass-forming liquids at ambient pressure, their application within diamond anvil cells presents substantial experimental challenges due to the extremely small scattering volume and limited optical access imposed by the high-pressure environment. In this work these challenges were overcome through the development of a dedicated optical configuration enabling reliable DPCS measurements in liquids compressed to pressures approaching the gigapascal regime.

To the author's knowledge, this work represents the first application of depolarized photon

correlation spectroscopy to the measurement of structural relaxation dynamics in a pure liquid contained within a diamond anvil cell. This experimental development enables direct measurement of α-relaxation under extreme compression and provides a new tool for probing glassy dynamics in the high-pressure regime.

1.3 Organization of the Dissertation

The remainder of this dissertation is organized as follows.

Chapter 2 provides background on glass-forming liquids and the glass transition. This chapter introduces the structural and dynamical characteristics of amorphous materials and surveys common experimental probes used to investigate glassy dynamics, including light scattering, viscometry, dielectric spectroscopy, and the glass-transition pressure (TGP) method.

Chapter 3 develops the theoretical framework used throughout the remainder of the dissertation. Topics include the formal definition of fragility, the relationship between molecular fluctuations and scattered electromagnetic fields, and the relaxation models used to extract structural relaxation times from experimental measurements.

Readers already familiar with the theoretical and experimental background of glassy dynamics may choose to skim these chapters without loss of continuity, as the primary original contributions of this dissertation begin in the chapters that follow.

Chapter 4 presents the construction of a pressure-dependent relaxation-time surface for glycerol. Literature viscosity measurements are combined with thermodynamic constraints derived from TGP experiments to produce a continuous representation of $\tau_\alpha(P, T)$. The resulting surface provides a framework for examining the evolution of fragility under compression.

Chapter 5 describes the development of high-pressure depolarized photon correlation spec-

troscopy in a diamond anvil cell and its application to the study of structural relaxation in cumene. This chapter details the experimental design of the instrument, the analysis of photon correlation data obtained under high-pressure conditions, and the incorporation of these measurements into the relaxation-surface framework.

Together these chapters demonstrate how structural relaxation measurements, thermodynamic constraints, and surface-based modeling can be combined to provide a unified description of glassy dynamics across pressure–temperature space.

2 Background

An article, published in 2018 by David Main in the Atlantic, asserts that "Glass has changed the world like no other substance" despite often being generally taken for granted [1]. Main does an excellent job delving into the history behind human adaptation of glass, from prehistoric times when volcanic glasses were used as tools, to its likely initial manufacture and spread throughout Mesopotamia more than four millennia ago, followed by the development of glass blowing, which arose in first-century Jerusalem, and the appearance of clear glass allowing for more functional, rather than solely ornamental, implementations. While many materials have significantly impacted technological and cultural development, glass ranks among the most influential—comparable to metal alloys, petrochemicals, and in recent times, semiconductors. In the sections that follow, we will first define what constitutes a glass and explore the structural and dynamic characteristics that distinguish glasses from other states of matter.

In modern times, an astonishing variety of glasses are manufactured on an industrial scale. Particularly notable are silicate-based glasses, which can be doped with diverse substances—such as lead, germanium oxide, zirconium, titanium, or barium—to precisely control their refractive index. Adjusting the refractive index has both ornamental and practical applications. For instance, lead oxide increases optical dispersion, enhancing the visual aesthetics of decorative glass, while in fiber-optic cables, doping allows higher flexibility and reduced optical leakage by manipulating the critical angle of internal reflection. [2, 3].

Beyond optical properties, glasses can also be engineered for specific thermal or mechanical characteristics. Boron trioxide doping, for example, is frequently used to reduce thermal expan-

sion, while aluminum oxide can enhance tensile strength and heat capacity. [4, 5]. Such tailored glasses have profound implications for practical applications like waste disposal. A particularly promising application involves using glass matrices for immobilizing nuclear waste, dramatically reducing risks of contamination by preventing waste materials from leaching into groundwater. [6, 7].

The classification of glass extends well beyond simple silicate systems. Amorphous metallic alloys, often referred to as metallic glasses, have been extensively researched since their discovery roughly half a century ago due to their unique mechanical strength and magnetic characteristics [8, 4]. Spin glasses represent another important class of non-silicate glasses, characterized by their magnetic domains undergoing transitions analogous to structural glass transitions—making them intriguing subjects of study since the mid-1970s [9, 10, 11].

Water serves as an intriguing example of a substance that forms a glass only under exceptional conditions. Its high heat capacity and strong hydrogen-bond network typically allow crystallization to outpace vitrification. Thus, forming glassy water requires rapid cooling to its glass transition temperature, around 137 K, on a millisecond timescale. However, glassy water has been produced under laboratory conditions and is found to exhibit unusual characteristics, including a tendency to transition directly from a glassy state into a crystalline state upon heating [12]. Glassy water has also been found, predominantly *via* neutron scattering experiments, to exhibit long-range structural order while lacking local order—an atypical feature for a glass, as will be discussed in Sec. 2.1.1 [13, 14].

It is, perhaps, strange to think of water as being a glass, yet evidently it can be and exhibits remarkable properties when in a glassy state. As unusual a material as it is, glassy water is not solely relegated to laboratory experiments. Water in the Kuiper belt has slowly nucleated over

billions of years to form comets and other celestial bodies. This nucleation occurs so slowly, and at such low temperatures, that individual molecules lack sufficient mobility to reorient into a crystalline structure, thus forming, one monolayer at a time, an amorphous glassy ice. A marvelous article on the topic of glassy water, published in *Physics Today*, asserts that "glassy water may be the most common form of water in the universe" [15]. Indeed, given that water is the second most abundant molecule in the universe following hydrogen, glassy water may well be the most common glass in the universe.

This is not the only instance of atypical behavior. The melting point of water exhibits a negative slope above the triple point due to its unusual lattice structure, resulting in expansion upon freezing. Even at exceedingly low temperatures, crystalline water can be melted *via* compression if sufficiently high pressures are applied. It is this phenomenon that enables the pressure-induced amorphization of crystalline ice first observed by Mishima *et al*. in the mid-1980s [16]. Here, a glass is formed *via* pressure-induced melting of ice well below the glass transition temperature. In this instance, amorphization depends on the unusual property of water to melt under increasing pressure.

However, pressure-induced amorphization has also been observed in crystalline silica, a more typical glass-forming system, as well as in the SiO_2 analog GeO_2 [17, 18]. More broadly, pressure-induced amorphization (PIA) has now been reported in scores of materials spanning network oxides, molecular solids, and organic systems. While relatively open-network structures are a common motif among systems exhibiting PIA, they are not a strict requirement. For example, benzene undergoes PIA when sufficiently high pressures disrupt its ring structure, forming amorphous hydrocarbon configurations. More recently, pressure-induced charge amorphization has been reported in materials such as $BiNiO_3$, wherein the lattice remains crystalline while the electronic

degrees of freedom become glassy. These diverse examples illustrate that PIA represents amorphization from an initially crystalline state under applied pressure and may arise through multiple structural or electronic mechanisms depending on how pressure reshapes the underlying energy landscape [19].

There also exist countless organic substances that readily form amorphous glassy solids that permeate the literature. They find applications in medicine, for example the use of glassy scaffolds for drug delivery or even potentially taking advantage of the glassy state of a drug itself where the lower Gibbs energy associated with increased structural entropy may give rise to greater water solubility and thus increase the efficacy of delivery [20, 21]. Polymethyl methacrylate (PMMA), a strong thermoplastic often referred to as an acrylic glass and marketed under the brand names Plexiglass, Acrylite, and Lucite among others, has found a wide variety of applications since its invention by Dupont in the early 1930's including automobile manufacture, aircraft windows, marker boards, and even the ceiling of the Houston Astrodome ballpark due to its high transparency, resistance to external elements such as UV light and water, and other beneficial properties [22]. Polylactic acid (PLA) is another glass-forming organic thermoplastic which has found extensive application in rapid prototyping due to the ability of such materials to be softened in a controlled manner upon heating [23]. Metallic glasses also share this characteristic and thus are quickly capturing interest for their potential to allow for direct 3D printing of metallic parts [24].

The material qualities that define a glass are still somewhat contested today, however functional definitions do exist and will be the topic of the following section. Here it may suffice to say that the term glass describes a vast array of materials spanning countless applications from window panes to, perhaps somewhat surprisingly, even cotton candy—a sucrose glass [25]. In 2015 the Carnegie Science Center of Pittsburgh had a marvelous display spotlighting the indis-

pensable role that glasses play in our day-to-day lives. It was somewhat disappointing to me that a prominent display case labeled "Is it Glass" did not contain any cotton candy—arguably almost as disappointing as the fact that such a fascinating topic as to what constitutes a glass, so poorly understood by the public, was relegated to a small non-imposing display case.



Figure 2.1: Exhibit on display at the Carnegie Science Center (2015)

Sucrose, or common sugar, is in fact an excellent glass-forming material despite typically being encountered in its crystalline form. A familiar example of a sucrose glass—one that for many evokes childhood memory—is peanut brittle. During its preparation a mixture of sugar, corn syrup, and water is heated well above the boiling point of pure water. As boiling proceeds, water is progressively removed and the composition continuously evolves. Because water acts as a strong plasticizer for sucrose, its removal causes the glass transition temperature T_g of the mixture to rise steadily throughout the cooking process.

At early stages, when the water content remains high, pulling filaments from the syrup produces strings that readily flow and collapse back into the pot. If cooled to room temperature at this stage, the mixture would form a viscous liquid rather than a glass, as its T_g lies well below ambient temperature. As evaporation continues, viscosity increases dramatically and the glass transition temperature approaches room temperature. When thin filaments drawn from the pot rapidly stiffen and snap, this indicates that the evolving composition has reached a state for which cooling to ambient conditions will carry the system below its glass transition temperature.

Thus, the critical transformation in peanut brittle is not simply cooling through T_g, but rather the continuous elevation of T_g through removal of the plasticizing water. Pulling the syrup provides a practical means of gauging this evolving composition and determining when the material, upon cooling, will vitrify into a rigid glass rather than remain rubbery or viscous.

If you have never seen peanut brittle made, set this dissertation down for a spell and go make a batch, for in the sections that follow, we will be delving into the details of what constitutes a glass as well as the molecular dynamics of the transition from a liquid to a glassy state. As is the case with all dissertations, beyond this introduction the material will become less colloquial and more technical. Having a plate of peanut brittle to enjoy as you read could certainly do no harm and may even serve to bestow on you a certain experiential intuition regarding the glass transition process. Besides, as Pamela Travers would almost certainly assert, "A glass made of sugar helps the medicine go down [26]."

2.1 Glasses and Glass-Forming Liquids

Before one can have a meaningful discussion of glassy dynamics and the glass transition, one must define what constitutes a glass. Zarzycki offers two potential definitions of glass—one

operational and the other structural [27]. From an operational standpoint, one might define a glass as a liquid rapidly thermally quenched into a non-crystalline solid phase. Alternatively, from a structural perspective, a glass may be defined more generally simply as a non-crystalline solid—ignoring the means by which this state was obtained. As Zarzycki delineates, both of these definitions fall short of providing a sufficient description of the glassy state. The simple structural perspective is too broad, encompassing materials such as gels that do not share the physical attributes of a glass. Conversely, his operational definition is too specific, omitting states that are structurally identical yet were arrived at *via* means other than thermal quenching.

Such alternative pathways include compression, which will be central to the topics of Ch. 4 and 5; pressure-induced amorphization as demonstrated by Mishima or Hemley *et al*. [17, 16]; and cold deposition, as previously discussed regarding water ice on comets formed *via* accretion [15]. His conclusion, and the general consensus today, is that the solution lies in a definition of the glassy state which incorporates the structural perspective with the added caveat that glasses constitute a non-equilibrium solid-state system—a merger of structural and dynamic descriptions. We will begin by examining the structural description of glasses and then proceed to consider their dynamic characteristics in detail.

2.1.1 Structural Description

Zarzycki's definition still requires further clarification. Is a glass a liquid, solid, or a separate phase of matter in its own right? One may be tempted to assert that glass is nothing more than a highly viscous liquid whose viscosity may be rapidly diverging however will still flow—albeit on a remarkably long timescale. Indeed, one would not be hard pressed to find, even among physicists, those who believe that cathedral stained glass is thicker at the bottom of the pane due to the

glass having flowed over the centuries under the influence of gravity—a very common misconception. In reality, medieval glaziers commonly installed panes with the thicker edge at the bottom for structural stability, reflecting the manufacturing process rather than viscous flow. The best models of molecular dynamics in glassy systems must be heavily extrapolated to address such timescales; nevertheless, even extremely conservative estimates place the average timescale required for these silicate glasses to flow appreciably at ambient temperatures several orders of magnitude beyond the age of the Universe [28].

In light of this, to assert that a material in a glassy state is 'fluid' in any colloquial sense may seem absurd. However, one's incredulity does not a valid definition make, and one cannot dismiss the fact that a glass's radial distribution function is often indistinguishable from that of the liquid state as conceptually illustrated in Fig. 2.2.

The radial distribution function, alternatively referred to as the pairwise correlation function, is a quantity defined as the expectation value of the particulate number density in a region as a function of distance from an origin placed at any arbitrarily representative particle or molecule within a bulk volume of material. This is best conceptualized as the ensemble average of the distance between all other particles or molecules as measured from each individual particle or molecule within a bulk material.

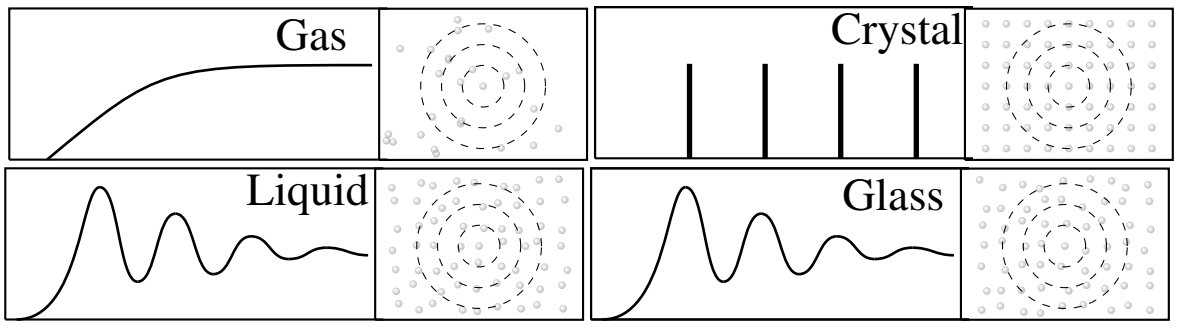


Figure 2.2: Typical radial correlation functions for a gas, liquid, glass, and crystal. This figure is derivative of that originally published by Johari. [29]

Significantly simplified two-dimensional representations of radial distribution functions and the structures giving rise to them for a gas, liquid, crystal, and glass are provided in the before-mentioned Fig. 2.2. Note that the exact particle locations for this figure were determined implementing a random number generator and thus are not representative of the ensemble average resulting from any specific intermolecular potential. This figure is relevant to a discussion of the characteristics typical of pairwise correlation functions for these four material states only in the most general of terms. The two-dimensional analog of a crystalline structure is shown in greater detail here in Fig. 2.3 where the characteristic radii between horizontally and vertically adjacent particles are labeled r_1. The characteristic radii corresponding to the nearest neighbor, next nearest neighbor, and next-next nearest neighbor are labeled r_1, r_2 and r_3 respectively. These will exhibit rapidly decreasing correlation amplitudes for higher order nearest neighbors. The corresponding radial distribution function exhibits very sharp periodic peaks associated with these infinitely repeating lattice spacings. The exact appearance of a radial distribution function for any real crystalline bulk material can clearly be quite complex and will vary from one material to the next as it depends directly upon the crystal symmetry, however it will always consist of such well defined periodic peaks. The arguably oversimplified crystalline pair correlation spectra shown in Fig. 2.2 ignores all peaks associated with these off diagonal contributions from the lattice structure to provide an even further conceptually idealized spectra. Strictly speaking, such a simple correlation spectra could only ever arise from a one dimensional distribution of evenly spaced particles however it represents the behavior of a repetitive lattice structure sufficiently well for the purposes of this discussion, and in fact it is not uncommon for solid-state physics texts to similarly describe the band structure along a single direction of a 3D crystalline structure.

Conversely, gases exhibit no such periodic order. Due to the strong yet exceedingly short

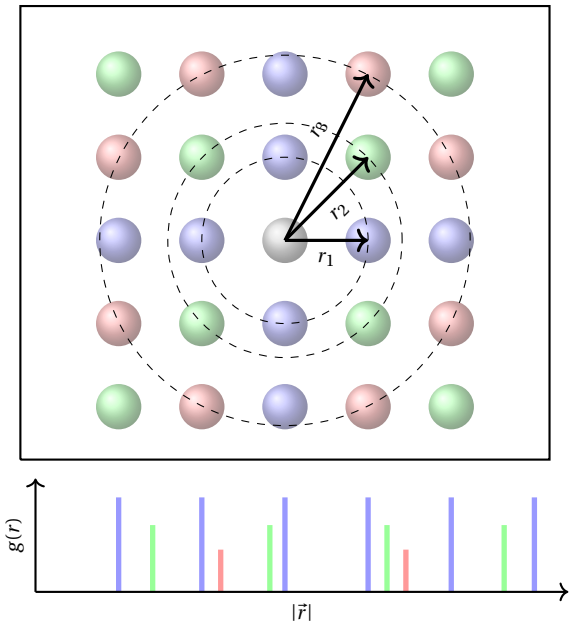


Figure 2.3: A crystalline solid exhibits one or more characteristic radii, for which adjacent individual particles or molecules appear at integer multiples. The corresponding radial distribution function for such a material will consist of sharp periodic peaks.

range Coulombic repulsion, molecules will spend very little time in extremely close proximity, existing mostly in a free state where intermolecular interactions are negligible. As such, their pairwise correlation function will very rapidly approach a uniform value corresponding to the average particle density. An idealized representation of this is provided in the top left window of Fig. 2.2.

From a structural perspective, both liquids and glasses will exhibit radial distributions functions that appear almost as if they consisted of a damped superposition of the gas and crystalline states. This arises due to the fact that the intermolecular interactions between the constituent molecules of both liquids and glasses are not negligible. Short-range order emerges as bonds between adjacent constituent molecules are formed. As this order is at the same time imperfect relative to the crystalline structure, short range, and in the case of a liquid also of short duration, it will manifest itself in the broadening and dampening of the periodic peaks of the corresponding

pairwise correlation function. Representative idealized liquid and glass radial distribution functions are provided in Fig. 2.2 as well. Note the periodic peaks that correspond to an imperfect crystalline structure in short order giving way to an utter lack of long range order indicative of a gas—albeit at significantly higher density.

Similarities between the radial correlation functions for a liquid and a glass are worthy of note. Structurally, they are nearly—if not entirely—identical. However, they differ vastly in their dynamics. For a liquid, the radial correlation function can be thought of either as arising from a time average of the molecular positions or, equivalently, from an ensemble average calculated from the positions of all individual particles at a single instant in time. Conversely, the dynamics of a glass—being an amorphous solid—are such that its radial correlation function is best understood solely in terms of an ensemble average, since constituent molecules do not move appreciably on experimentally meaningful timescales. Thus, we encounter the curious condition of two material states possessing identical structure yet fundamentally different dynamics. Having explored the structural similarities and differences between glasses and liquids, we now turn our attention to their dynamic properties, particularly how these dynamics reflect differences in molecular motion and energy landscapes.

2.1.2 Dynamic Description

Molecules in a gas state have high enough energy on average that they do not really "feel" the potential energy landscape produced by their neighbors. They take on a relatively random and highly degenerate spatial distribution where intermolecular interactions are limited to the strong yet exceedingly short range Coulombic repulsion of their electronic shells. Thus the pairwise correlation function for a well thermalized gas will exhibit very little structure beyond a rapid

increase in correlation, arising from this Coulombic repulsion repelling other molecules at close proximity, quickly leveling off to a region of constant number density, beyond the influence of this repulsive force, that will extend indefinitely as demonstrated in Fig. 2.2.

Gases and crystals are, in this sense, polar opposite scenarios in that the former has sufficient thermal energy that the potential landscape arising from the intermolecular forces is almost irrelevant as molecules spend very little time in close enough proximity to any other molecule for even the dominant Coulombic interaction to play a major role in their dynamics, yet the latter has sufficiently low thermal energy such that dynamics are dominated by intermolecular forces beyond Coulombic repulsion—higher order terms such as dipole or other types of interactions depending on the nature of the constituent atoms making of the crystal. Molecules in a crystalline state certainly *feel* their potential energy landscape, yet are not free to explore it as most of that parameter space is energetically inaccessible. Thus they are locked into discrete, highly symmetric arrangements associated with the lowest energy structural state of the bulk system. These are the two extremes of which the liquid state is an intermediate.

Within the energetically intermediate liquid state, molecules or particles have sufficient energy to explore their potential energy landscape yet still be heavily influenced, and to some extent confined, by said landscape. Short lived bonds between molecules form, increasing the average time in which they are in close proximity and giving rise to cooperative motion. However these nearest neighbor bonds *are* very short lived—the activation energy required to break the bonds being on the order of the thermal energy, $k_B T$, of the material. The bonds conforming to the lowest energy orientations corresponding to the crystalline structure, though not exclusively occurring, will clearly be statistically favored. Thus liquids exhibit the rapid emergence, and subsequent disintegration, of short-range order that may be described as spatially distorted crystalline arrange-

ments. One would expect that the radial distribution function of a liquid will therefore exhibit structures that resemble the crystalline peaks over a short range, though broadened due to thermal motion and the absence of long-range positional order. However, over sufficiently large separations, these short-lived local nucleations bear no correlation from one region to the next and, as such, the radial correlation function for a liquid approaches unity at large radii—reflecting the absence of long-range order while maintaining the characteristic number density of the liquid phase. The radial correlation function for a liquid displayed in Fig. 2.2 shows what would be expected for a state that is intermediate between a gas and a crystalline solid, *i.e.* broadened crystalline peaks at short range that quickly wash out to a uniform average number density at larger radii.

Another way to describe the liquid state of a glass-forming system, beyond purely energetic considerations, is to consider the timescales on which the potential energy landscape is explored by the system—*i.e.* its structural relaxation time. It should be noted that a variety of relaxation processes occur in any glassy system, not all of which are directly associated with the bulk structural rearrangement. These include thermal relaxations as well as processes arising from diffusion and orientational dynamics, the complexity of which depends strongly on the internal degrees of freedom available to the constituent molecules.

Even relatively simple glass formers typically exhibit two broad dynamical regimes: fast β-relaxations, which generally display relatively little dependence on temperature and pressure, and the α relaxation process, which is highly sensitive to thermodynamic conditions and may vary by thirteen or more orders of magnitude between the high-temperature or low-pressure liquid and the glassy state. The α relaxation is the structural relaxation mode of the system, associated with the cooperative motion of many nearby particles and involving mechanisms such as the breaking of nearest-neighbor cages, molecular reorientation, and diffusive rearrangement. It is this structural

α relaxation process that is the focus of this dissertation.

In the following section we will discuss several probes of these relaxation dynamics, including Photon Correlation Spectroscopy in Sec. 2.3.2. Time-domain correlation spectra obtained using this technique on glass-forming systems are generalized in Fig. 2.4 to illustrate this dynamic behavior under varying thermodynamic conditions. Here, the correlation time, τ, is simply the average time required for a bulk material to relax having been subjected to some form of mechanical perturbation—inherently tied to the imaginary component of the bulk modulus. Generally speaking, the fast β-relaxation will be relatively unaffected by changing thermodynamic parameters. However, as the temperature of a liquid decreases or, equivalently, as pressure increases, the α relaxation time increases continuously over many orders of magnitude. In the high-temperature or low-pressure liquid, τ_α is typically on the order of picoseconds, whereas at the glass transition it reaches approximately 10^2 seconds. The rate at which τ_α varies with temperature or pressure depends strongly on the fragility of the liquid, with fragile systems exhibiting especially pronounced changes as T_g is approached, while strong liquids display a more gradual evolution. Understanding how this structural relaxation time grows as thermal energy is reduced or free volume is diminished under compression is central to describing the nature of the glass transition.

In the most simplistic of terms, glassy materials exhibit the rigidity of a crystalline state, having insufficient thermal energy to readily explore their potential energy landscape, yet maintain the semi-ordered structure of their liquid state. Their radial correlation functions, as characterized in Fig. 2.2, will be very similar in structure to that of the liquid state, often exhibiting short range crystalline structure albeit with copious irregularities, transitioning into rapidly diminishing structural correlation over longer distances. A very marked difference between a system in a liquid *vs*. a glassy state is the apparent violation of ergodicity. Although the radial correlation function

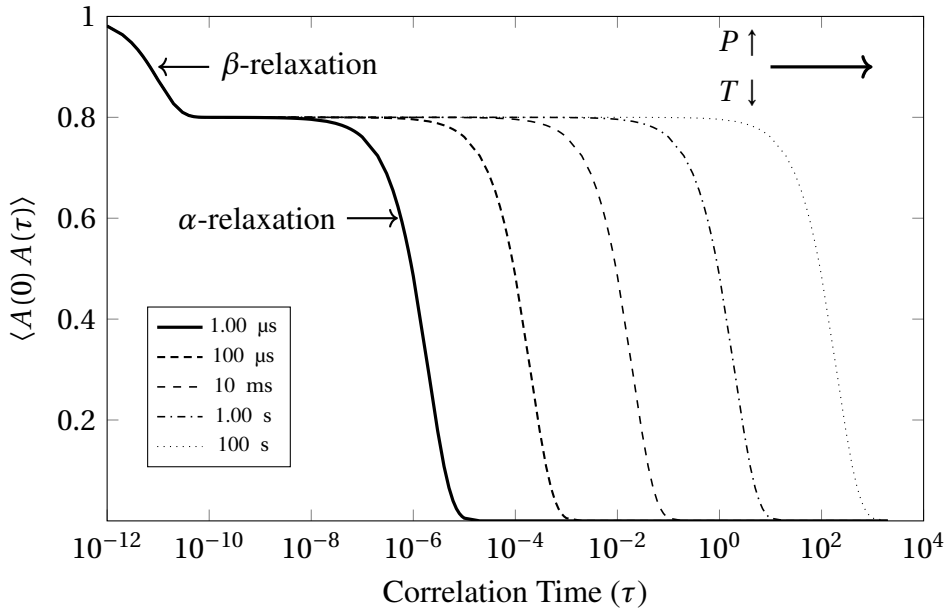


Figure 2.4: An idealized set of autocorrelation curves typical of what might be obtained for a glass-forming system. The characteristic structural relaxation time, τ_α, which will be discussed later in greater detail, falls near the foot of each curve. As temperature is decreased or pressure is increased, the faster β-relaxation remains relatively fixed; however, the α-relaxation shifts to longer times as structural relaxation slows and an increasing fraction of the energy landscape becomes energetically inaccessible.

of a glass may be similar to the liquid state in general structure, it may not be quite as identical as implied by Fig. 2.2. Glassy systems will exhibit mechanical hysteresis stemming from their exceedingly slow structural relaxations. This along with the lower thermal energy of these systems results in intermolecular bonds being much longer lived than in the liquid state which may lead to semi-crystalline domains growing quite large. Cooperative motion of these domains will play a much greater role in the dynamics of glassy systems than in a liquid state. These considerations, among others, leave open the question as to whether glass exists as a separate phase of matter or as nothing more than a highly viscous fluid—a question we will now address.

2.2 The Nature of the Glass Transition

A material undergoing a change in phase will experience a dramatic and discontinuous shift in bulk properties such as density, specific heat, bulk modulus, thermal expansivity, *etc*. Ehrenfest more precisely defined phase transitions as points in the pressure-temperature-volume phase space of the material where discontinuities in the derivatives of the Gibbs free energy with respect to any external thermodynamic parameter emerge. Transitions between any two states described in Fig. 2.2 with the exception of the glass transition will constitute what Ehrenfest described as a first-order phase transition in that it will involve discontinuities in first-order derivatives of the free energy, or equivalently enthalpy, for a transition at constant pressure [30]. The glass transition, however, is distinct from the others in that no such first-order discontinuity exists. Fig. 2.5 shows the plot of enthalpy or specific volume as a function of temperature for an idealized glass-forming system maintained at constant pressure. At the freezing temperature, T_f, the enthalpy of fusion—equivalently the latent heat of fusion—corresponds to a discontinuous change in enthalpy. Such a discontinuity in the first derivative of the Gibbs free energy is the defining characteristic of a first-order phase transition and is apparent in Fig. 2.5. In principle, any liquid capable of forming a crystalline structure, if brought to this temperature, provided with a single nucleation point, and allowed to cool slowly enough to maintain thermalization, will form a single-crystal solid that constitutes the global minimum on the potential energy landscape as represented in Fig. 2.6. This conceptually idealized projection of the potential energy landscape, adapted from the review by Debenedetti and Stillinger [31], represents the potential energy as a function of a single configurational coordinate. It is, obviously, not possible to aptly describe the potential landscape for any real system graphically as it will span a parameter space whose dimension will be on the order of

the number of constituent molecules residing in the bulk multiplied by their individual degrees of freedom. However this overly simplified single dimensional representation is quite sufficient as a conceptual illustration.

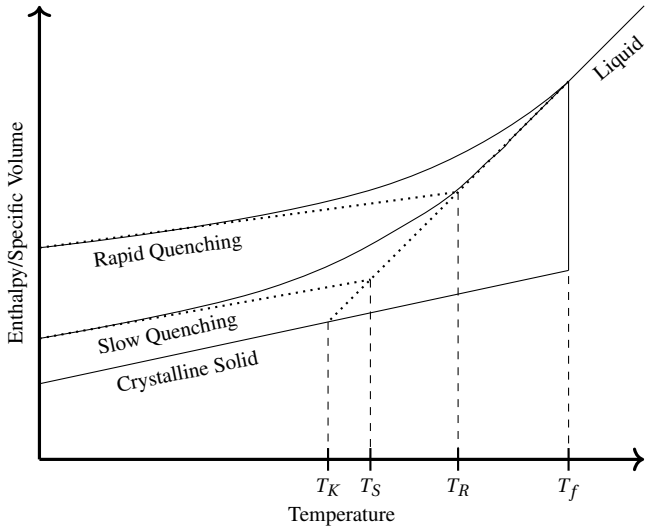


Figure 2.5: An idealized representation of a typical plot obtained *via* differential scanning calorimetry to determine the glass transition [32]. The smooth crossover from the linear trend of the liquid state to that of the glassy state, exaggerated here for clarity, would be expected in the absence of any higher-order transition in the Ehrenfest sense. The dashed lines indicate extrapolations of the equilibrium liquid branch at temperatures sufficiently above T_g and of the amorphous solid branch at temperatures sufficiently below T_g for glasses formed *via* rapid and slow thermal quenching. The Kauzmann temperature for the limiting case is also shown, as it motivates the argument that such smooth extrapolations may be non-physical.

If cooled below the freezing temperature while crystallization is suppressed, the enthalpy will continue to decrease with the removal of internal energy and accompanying thermal contraction until, in the limit of absolute zero, only a single energetically accessible state remains, corresponding to zero entropy. This stands in stark contrast to a glassy state where a certain degree of entropy, associated with structural disorder, is locked in. This can occur when a liquid is thermally quenched very rapidly. Rapidly, in this context, implies that thermal energy is removed quickly enough that the system cannot thoroughly explore its potential energy landscape. Con-

sequently, it becomes trapped in metastable local minima, failing to reach the global minimum associated with the crystalline ground state. Fig. 2.6 labels one such local minimum as an "ideal glass." It should be emphasized, however, that while the crystalline state represents the unique global minimum of the potential energy landscape, this landscape itself depends on the positions and orientations of all constituent particles and is therefore extraordinarily complex. It contains an immense number of local minima, any of which may trap a rapidly quenched system. The "ideal glass" is thus understood as a particular deeply trapped metastable minimum rather than the true thermodynamic ground state. Many of the local minima correspond to metastable basins separated by finite energy barriers. Transition states, properly understood as saddle points between adjacent minima, are unstable along at least one configurational direction and mediate transitions between these metastable states. Supercooled liquids occupy such metastable basins, separated by finite energy barriers. A familiar example is sodium acetate heat packs, which remain in a metastable liquid state until a small perturbation triggers rapid crystallization. Some systems do not so readily crystallize and, upon rapid thermal quenching, settle into an amorphous solid locked within a local energy minimum that rests above that of the crystalline state, but significantly below what would be considered a transition state. For such a system, the entropy does not approach zero as enthalpy decreases because the energy landscape contains an enormous number of distinct metastable basins of comparable depth. Upon rapid quenching, the system becomes confined within one such basin—microscopically reflected in molecules trapped within nearest-neighbor cages—but the multiplicity of alternative basins gives rise to a residual configurational entropy. The barriers separating these basins are sufficiently large that inter-basin transitions are effectively suppressed on experimental timescales.

This is the glass transition and is perhaps well described again by Fig. 2.5 where the dis-

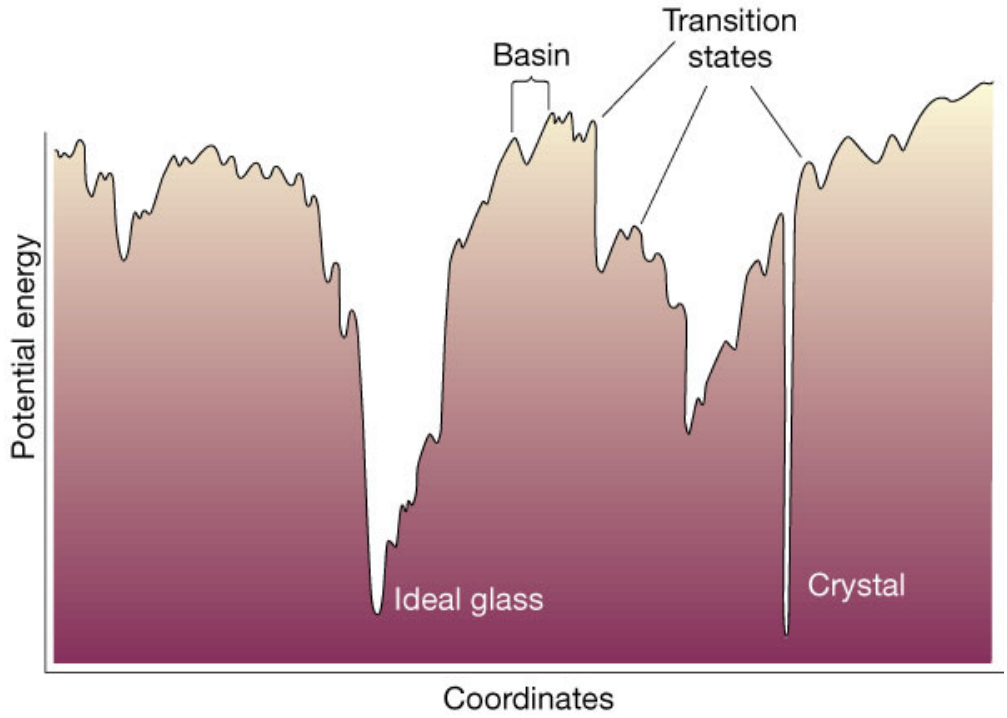


Figure 2.6: The two dimensional representation of an energy landscape for an idealized glass-forming system provided by Debenedetti and Stillinger [31].

continuity in first order derivatives of the free energy are not present but rather a rapid yet smooth transition does occur as temperature decreases. The nature of the glass transition—whether it constitutes a legitimate second-order phase transition or simply reflects rapid vitrification of the liquid state—could, in principle, be experimentally resolved in this context. If the transition is indeed smooth, lacking any discontinuity in higher-order derivatives, then the glass "transition" is merely a kinetic effect associated with a rapidly vitrifying liquid. However, if the smooth crossover implied by Fig. 2.5 does not accurately reflect the true behavior and instead a discrete discontinuity in slope exists, then the glass transition represents more than rapid vitrification; it would constitute a second-order transition in the Ehrenfest sense into a distinct thermodynamic state. Such measurements, however, do not readily provide the precision necessary to resolve a discontinuity of this kind conclusively. Nevertheless, there remains a compelling argument for the necessity of

a second-order transition consistent with the observed thermodynamic behavior of glass-forming systems.

Assume for a moment that the glass "transition" is a purely kinetic vitrification process. As depicted in Fig. 2.5, upon super-cooling below the freezing point, T_m, enthalpy or specific volume will quickly approach a trend that tracks with that of the crystalline solid yet will remain systematically above the corresponding values of the crystal enthalpy as temperature continues to decrease. The exact values of enthalpy, specific volume, heat capacity, or many other bulk properties of the glassy material will depend on its thermal history including how rapidly it was quenched. Extremely rapid thermal quenching may result in a much less ordered glassy system as molecules have virtually no time to seek out even relatively pronounced local minimums on the potential surface. Slightly slower quenching may allow local order to arise thus giving bulk properties closer to that of the crystalline state. For a first order phase transition, such as from a liquid to a crystalline solid, there is a specific well defined temperature at which this transition occurs. It is desirable to define such a transition temperature for the glass transition as well. One such definition is demonstrated in Fig. 2.5 as the intersection of the extrapolations of the linear trends in Enthalpy or specific volume as a function of temperature from well within the glassy state and from the supercooled liquid state. These extrapolations are shown as dotted lines in Fig. 2.5 and their intersections marked as vertical dashed lines corresponding to the characteristic transition temperatures, T_S and T_R, for a glass-forming system undergoing slow and rapid thermal quenching, respectively. These points of intersection clearly cannot suffice as a definition for the glass transition temperature, T_g, as they will inherently constitute a range of values strongly dependent upon the rate of cooling as, from an energy landscape perspective, the rate of cooling will determine to what extent the constituent molecules have time to explore the energy landscape before becoming energetically locked into a

local minimum corresponding to a glassy state associated with a specific local minimum on the potential energy landscape. The more rapidly a system is quenched the more likely it is to be locked into a state that is energetically much further from the crystalline state as well as having a significantly higher residual entropy as temperature approaches absolute zero corresponding to the higher level of structural disorder. A standard metric experimental protocol must be agreed upon if one is to define a singular glass transition temperature, T_g. The most common method of defining T_g, at least at atmospheric pressure, has been *via* calorimetric measurements which directly probe the heat capacity of the liquid as temperature is scanned at a controlled, generally agreed upon rate through the glass transition—typically ten Kelvin per minute. At these rates a value of T_g is found consistently to correspond to viscosities on the order of 10^{13} poise or an $\alpha-$relaxation time on the order of 100 seconds [33]. The temperatures associated with these viscosities or characteristic structural relaxation times have been generally accepted as the standard definition of the glass transition temperatures, T_g, throughout the literature and are those to which this work will conform. Further details of calorimetric methods will not be described here as these methods do not prove practical at very high pressures and were not implemented to obtain any of the data appearing in this work. However, this thermodynamic description of the glass transition remains pertinent as it illustrates, quite effectively, one argument for the glass transition being a second order phase transition, as opposed to a purely kinematic process of rapid vitrification as will now be discussed.

The curve labeled "Slow Quenching" in Fig. 2.5 is not, in any strict sense, a limiting case for a given glass-forming material; in principle, one may always cool more slowly. In practice, spontaneous nucleation events often induce crystallization in supercooled liquids, as observed in systems such as propylene carbonate and salol [34]. Similar spontaneous crystallization has also been observed in our laboratory in super-pressed salol under high-pressure conditions, where a

crystalline front advanced gradually across the sample on timescales of days. However, for strong glass-formers, bulk crystallization can be suppressed even under extremely slow cooling rates.

Under such conditions, one may instead envision the formation of increasingly large locally ordered domains distributed throughout the material. These domains can progressively restrict the mobility of neighboring regions, ultimately arresting long-range structural rearrangement and producing a glassy state. The microscopic mechanisms that frustrate large-scale crystallization remain an active area of investigation [35].

If cooling could proceed arbitrarily slowly while still avoiding macroscopic crystallization, the bulk thermodynamic properties would be expected to approach those of the crystalline state. Extrapolation of the linear trend shown in Fig. 2.5 would then imply that the enthalpy or specific volume of the liquid could fall below that of the crystal. Such behavior would correspond to a state possessing lower entropy than the crystalline phase—an outcome known as the Kauzmann paradox [36].

Because this scenario is thermodynamically untenable, the approach to the crystalline state must either remain asymptotic or be interrupted by a phase transition. Gibbs and DiMarzio proposed that a true second-order Ehrenfest transition could occur at the Kauzmann temperature, T_K, thereby resolving the paradox [37, 38]. Experimentally, however, spontaneous crystallization typically intervenes before T_K can be reached, preventing direct verification of this proposal.

The argument outlined above highlights a long-standing temptation within the glass literature—namely, to interpret the apparent approach toward the Kauzmann temperature as evidence for an underlying thermodynamic phase transition. In many theoretical treatments, this transition is hypothesized to occur either at an extrapolated temperature such as T_K or, in some frameworks, at a temperature above T_g that would mark the onset of distinct glassy order. Mean-field and random

first-order transition models, for example, predict such behavior under certain assumptions. Moreover, in disordered systems such as spin glasses, genuine thermodynamic phase transitions are well established. The situation for structural glasses, however, remains less definitive. At present, the experimental evidence does not conclusively establish whether the structural glass transition—at T_g, above it, or at any extrapolated temperature such as T_K—corresponds to a true thermodynamic phase transition.

The persistence of the higher-order interpretation stems in part from an extrapolative paradox whose logical structure bears resemblance to Zeno's famous paradox of motion. In Zeno's construction, a traveler attempting to reach a wall must first traverse half the distance, then half of the remaining distance, and so on *ad infinitum*. Because this reasoning generates an infinite sequence of steps, it appears to forbid arrival. The flaw, of course, lies not in the subdivision of space, but in neglecting the dynamics of the process: at finite velocity, each successive interval requires half the time of the previous one, and the total traversal time forms a convergent geometric series. Once the temporal structure is properly accounted for, the paradox dissolves.

The Kauzmann paradox may be viewed as an inverted analogue of this reasoning. If one assumes that a supercooled liquid can be cooled arbitrarily slowly while remaining in thermodynamic equilibrium down to some limiting temperature, then extrapolation suggests a thermodynamic inconsistency—the entropy of the liquid falling below that of the crystal. The tension arises not from the mathematics of extrapolation, but from extending equilibrium assumptions into a regime where kinetic arrest may intervene. Introducing a thermodynamic phase transition to resolve this contradiction represents one possible interpretation, and such a scenario cannot be ruled out a priori. Nevertheless, for structural glasses, no unambiguous order parameter, symmetry breaking, or universal scaling behavior has been identified that would clearly signal such a transition.

Even if a genuine thermodynamic transition were to occur at some temperature below or even modestly above T_g, its signatures might be sharply defined only asymptotically close to that temperature. At experimentally accessible conditions, any associated anomalies could be progressively weakened and plausibly obscured by kinetic effects or experimental limitations. This possibility helps explain the enduring appeal of the hidden-transition hypothesis.

The central difficulty, however, is not that thermodynamic anomalies are absent near the laboratory glass transition—clear changes in quantities such as the specific heat are routinely observed—but that these signatures do not uniquely compel the existence of an underlying thermodynamic phase transition. The anomalies observed at T_g are broadened, rate dependent, and consistent with kinetic arrest, making it difficult to distinguish between a true thermodynamic singularity and a dynamically induced crossover. In the absence of a well-defined order parameter or predictive framework specifying its character, the higher-order interpretation remains a plausible but unverified element of the broader theoretical landscape.

In reality, as a system is cooled toward T_K under conditions of equilibrium, there exist two physically viable mechanisms by which the extrapolated Kauzmann paradox is avoided long before any such crisis is reached. The first is crystallization, which becomes increasingly thermodynamically favorable as temperature is lowered below the melting point, owing to the growing free-energy difference between the crystalline and supercooled liquid states. In principle, this enhanced thermodynamic driving force may reduce nucleation barriers and permit crystallization through pathways that do not rely exclusively on energetically activated bond breaking. In particular, entropically driven rearrangements along nearly equipotential directions of the potential-energy landscape may allow for limited structural reorganization even as thermal activation becomes suppressed.

However, the kinetic accessibility of such crystallization pathways generally diminishes as temperature is further reduced. As cooperative domain sizes grow and structural relaxation becomes increasingly collective, rearrangements require the correlated motion of ever larger regions of the material, while the available thermal energy necessary to facilitate bond breaking and large-scale reorganization is progressively denied. Under these conditions, both energetically activated and entropically driven pathways to crystallization become increasingly constrained, rendering bulk crystallization dynamically improbable despite its thermodynamic favorability.

The second, and more likely, mechanism by which the paradox is avoided is the loss of equilibrium itself. Structural relaxation time grows rapidly upon cooling, the system inevitably falls out of equilibrium at a temperature $T_g > T_K$, thereby locking in configurational entropy without inducing any thermodynamic singularity. This departure from equilibrium may occur arbitrarily sharply and at temperatures arbitrarily close to T_K, while remaining fully continuous in all experimentally accessible thermodynamic quantities.

The oft-invoked assertion that one may always quench more slowly, and thus asymptotically approach equilibrium down to T_K, is therefore no more compelling than Zeno's conclusion that the wall can never be reached. In the limit of infinitesimal cooling rates, the time required to reach T_K diverges. One does not reach T_K within experimental timescales, nor within geological timescales, nor within the lifetime of the universe.

The resolution lies not in postulating a hidden thermodynamic transition, but in recognizing the loss of ergodicity discussed previously. As relaxation times grow and eventually exceed accessible timescales, the system falls out of equilibrium and can no longer explore phase space in a manner consistent with equilibrium thermodynamics. The apparent paradox arises from extending equilibrium reasoning into a regime where ergodicity has effectively broken down.

In this sense, the Kauzmann construction does not by itself compel the existence of a higher-order thermodynamic phase transition. Rather, it highlights the limitations of equilibrium extrapolation in systems whose relaxation times diverge.

2.3 Probes of Glassy Dynamics

In this section, we will introduce and briefly discuss a selection of widely used experimental techniques—such as depolarized photon correlation spectroscopy (DPCS), depolarized Brillouin light scattering (DBLS), and dielectric spectroscopy (DS)—that provide complementary insight into the molecular dynamics underpinning the glass transition.

Although the calorimetric method of defining the glass transition temperature, T_g, as described above is theoretically compelling and remains the standard at atmospheric pressure, it does require a large amount of data, taken with tight experimental control of the cooling/heating rates, in order to arrive at reproducible values for the glass transition temperature, T_g. As mentioned above, it is experimentally advantageous to define the glass transition temperature in terms of viscosity or, equivalently, the characteristic structural relaxation time. The characteristic timescales on which molecular dynamics evolve in a glass-forming system vary from on the order of picoseconds for very high-temperatures or in low pressure environments, to thousands of seconds at low temperatures or in high pressure environments. Due to this extreme dynamic range, a single experimental probe for tracking these molecular dynamics will likely never exist; and in fact, experimental gaps within the literature typically exist for dynamic data on any given glass-forming system. Be that as it may, with the aid of a proper mathematical model and the implementation of several different experimental probes, each within their applicable dynamic window, excellent descriptions of these dynamics can be ascertained over exceedingly large dynamic ranges encompassing the glass

transition for many materials.

At this point, the astute reader may be asking a simple question: what are "these dynamics"? Specifically, the study of the glass transition relies on experimental probes capable of tracking changes in molecular or atomic locations and orientations. Some may accomplish this directly, perhaps *via* some scattering mechanism that tracks thermal diffusion, or more indirectly *via* tracking changes to thermodynamic bulk properties such as viscosity which arises as a direct consequence of said diffusion. This work will exclusively focus on probes of viscosity, η, and the characteristic structural α-relaxation time, τ_α. These two bulk properties are intrinsically related *via* the Maxwell relation which observes that, for small deformations or applied strain, $\tau_\alpha = \frac{\eta}{G_\infty}$, where G_∞ is the elastic sheer modulus. Fig. 2.7 illustrates how the stress resulting from an adiabatically applied strain relaxes over time due to viscous flow of the bulk material.

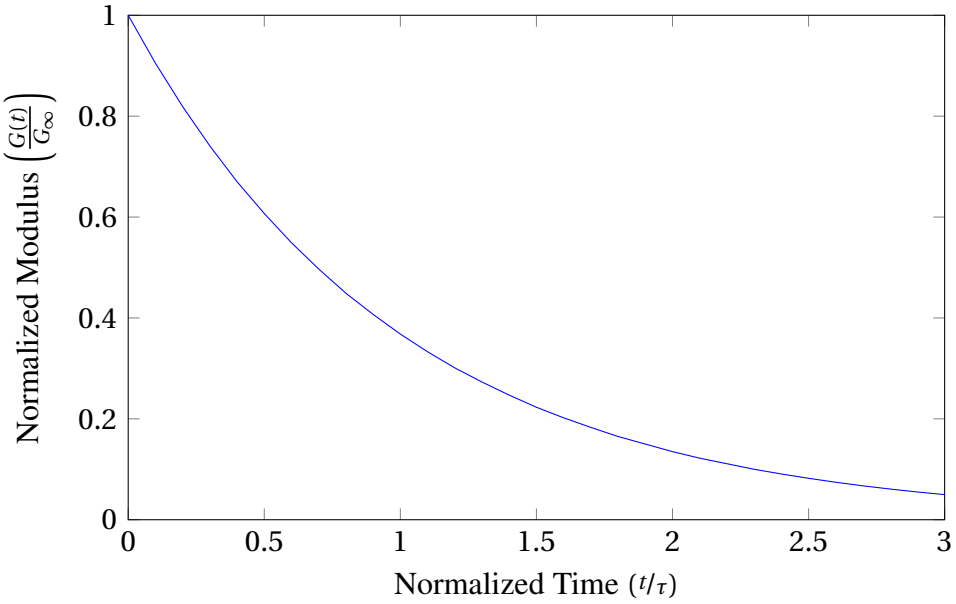


Figure 2.7: An idealized stress relaxation response following the rapid imposition of a small step strain, ϵ_0, in the linear viscoelastic regime. The stress, $\sigma(t)$, is normalized by $G_\infty\epsilon_0$, where G_∞ is the instantaneous shear modulus. The characteristic relaxation time over which viscous flow mediates mechanical stress relaxation is $\tau = \eta/G_\infty$.

However direct measurements of structural relaxation and viscosity are certainly not the only window into glassy dynamics. Fig. 2.5 illustrates a classical dilatometric determination of the glass transition temperature, in which the underlying molecular dynamics are not directly probed. Instead, the specific volume is monitored as a function of temperature. In calorimetric measurements, the corresponding signature appears in the temperature dependence of the heat capacity, C_p. Both approaches track bulk thermodynamic quantities that emerge from the collective molecular dynamics of the system. Fig. 2.2, though not comprised of actual data, is representative of that which has been obtained *via* direct structural probes such as neutron scattering. However neutron scattering has provided more than just structural snapshots. Though it will not be discussed in great detail here, as it is not directly relevant to the data presented in this work, inelastic and quasielastic neutron scattering have also served as probes of molecular dynamics in glassy systems. These techniques are sensitive to motions on picosecond to nanosecond timescales, enabling the study of vibrational excitations and fast relaxational processes [39, 40, 41].

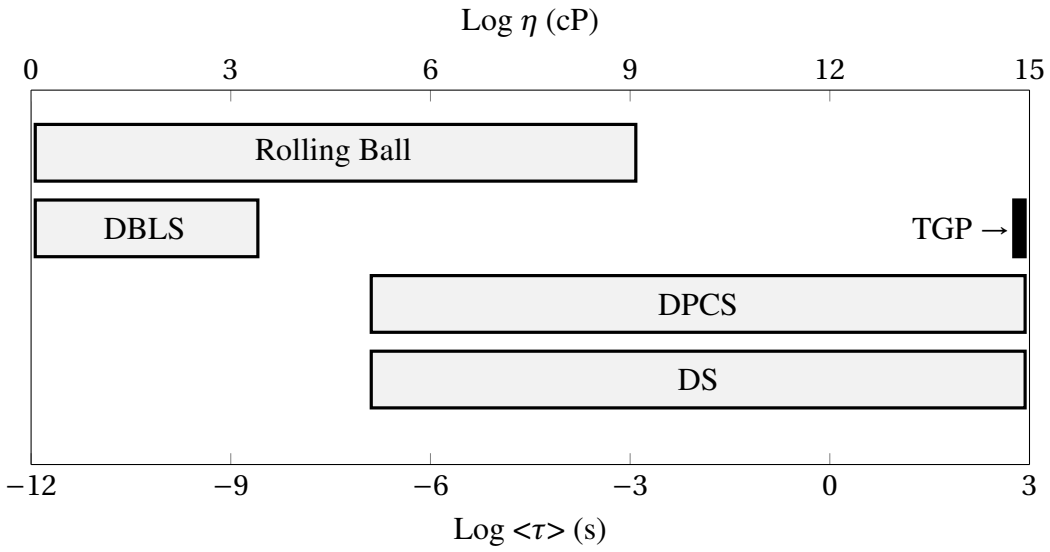


Figure 2.8: Accessible dynamic ranges for various probes of viscosity η and structural relaxation time τ_α. Rolling Ball: rolling-ball viscometry (probe of η); DBLS: depolarized Brillouin light scattering; DPCS: depolarized photon correlation spectroscopy; DS: dielectric loss spectroscopy; TGP: technique described in Sec. 2.3.4.

The experimental probes referenced or implemented in this work are summarized in Fig. 2.8. These include DPCS, DBLS, rolling ball viscometry, TGP analysis, and DS. The physical mechanism behind each of these probes will be described briefly below as well as their dynamic windows and other limitations to their applicability pertinent to this work. These descriptions will be kept intentionally brief as an entire dissertation could easily be written on any one of these probes alone, and indeed, Ch. 4 is in effect a reanalysis of one such dissertation outlining a means of extending a previously existing probe, rolling ball viscometry, into a diamond anvil cell allowing for viscosity measurements at higher pressures.[42] Following this, Ch. 5 will provide a similar extension, this time of photon correlation spectroscopy, into the diamond anvil cell. Significantly more detail regarding these specific probes will be provided in their respective chapters.

2.3.1 Depolarized Brillouin Light Scattering (DBLS)

Electromagnetic interactions with matter lie at the heart of most experimental probes of structure and dynamics. Notable exceptions include ultrasonic techniques and neutron scattering, which rely on mechanical or nuclear interactions rather than electromagnetic coupling. Nevertheless, even neutron scattering possesses an electromagnetic aspect: the neutron's appreciable magnetic moment renders it uniquely sensitive to ordered magnetic structures, such as spin fluctuations in superconducting materials [43].

Light scattering methods probe how condensed matter responds to incident electromagnetic radiation, giving rise to a scattered field that carries information about the structure and dynamics of the bulk material. A generalized schematic for a light scattering experiment is shown in Fig. 2.9. Polarized monochromatic laser light is collimated or focused through a sample volume, and a fraction of this light is scattered, collected, and directed onto a suitable spectral analyzer.

In the experiments presented in this dissertation, an intracavity etalon was employed to ensure single-mode laser operation with a spectral width of approximately 10 MHz, as required for high-resolution Brillouin measurements. It is worth noting that other light scattering techniques, such as Raman scattering, commonly utilize multi-mode lasers with linewidths on the order of $\sim$ GHz—still only about one part in 10^6 of the optical carrier frequency. The scattered light in this work is analyzed using either a triple-pass tandem Fabry-Pérot interferometer (DBLS) or a photon correlation spectrometer (DPCS).

A light source with exceedingly narrow spectral width is necessary for DBLS, as this probe resolves very small frequency shifts in the scattered light arising from dynamic processes within the sample. These processes will be described in more detail in Sec. 3.2 and 3.3; for now, it suffices

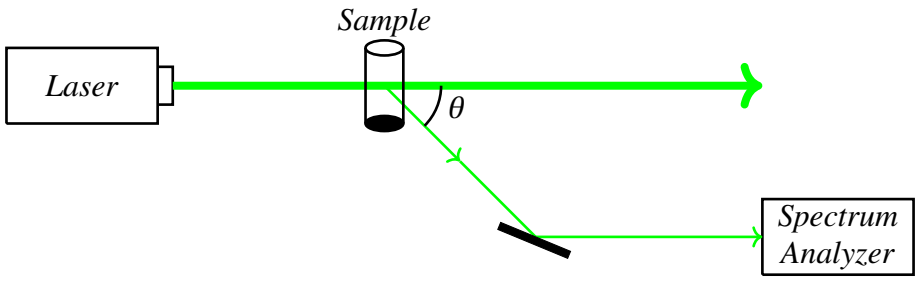


Figure 2.9: A simple forward scattering DBLS experimental design in which light from a monochromatic source, such as a single mode locked AR^+ laser, is scattered from a sample and collected at a known angle. The collected light is then directed into a spectrum analyzer, such as a triple pass tandem Fabry-Pérot interferometer.

to note that light is scattered whenever there is a local temporal or spatial variation in the dielectric susceptibility of the medium. Such variations may arise from density fluctuations, induced dipoles, or orientational dynamics—anything that alters how the charges within the material respond to an applied electromagnetic field. If the susceptibility variation is purely spatial and static—for example, in a material with frozen-in density inhomogeneities—the scattered field retains the frequency of the incident beam, giving rise to elastic (Rayleigh) scattering. In fluids, however, density fluctuations are inherently time dependent. Any temporal modulation of the dielectric susceptibility must therefore produce a corresponding distribution of frequencies in the scattered field. Even crystalline solids exhibit such time-dependent fluctuations due to lattice vibrations. Longitudinal and transverse acoustic modes, as well as higher-frequency optical phonons, appear as frequency-shifted features in the scattered spectrum.

Depolarized Brillouin Light Scattering (DBLS), as implemented in this work, refers to the spectral analysis of depolarized scattered light using a high-resolution Fabry-Pérot interferometer. Brillouin scattering arises from the interaction of light with thermally excited acoustic modes in the material—long-wavelength density fluctuations corresponding to small wavevector phonons in the GHz frequency regime. This distinguishes Brillouin scattering from Raman spectroscopy, which

probes optical phonons or intramolecular vibrational modes at much higher (THz) frequencies and whose larger energy shifts are typically resolved using grating-based spectrometers.

In glass-forming liquids, the depolarized Brillouin spectrum contains not only acoustic sidebands but also a central quasielastic component. It is this central quasielastic feature, arising from time-dependent fluctuations in the anisotropic part of the dielectric susceptibility, that is of primary interest in the present work. This spectral component reflects structural relaxation processes within the bulk material and provides a direct probe of the α-relaxation over a dynamic window approximately bounded by

$$10^{-12}\,\mathrm{s} \lesssim \tau_\alpha \lesssim 10^{-9}\,\mathrm{s},$$

as shown in Fig. 2.8.

While analysis of the linewidths of longitudinal acoustic (LA) Brillouin modes can provide access to viscoelastic damping and relaxation phenomena over a broader dynamic range [44], such approaches probe relaxation indirectly through mechanical response functions. In contrast, the DBLS measurements discussed here focus specifically on the quasielastic depolarized spectrum as a more direct probe of structural α-relaxation.

It is important to distinguish this spectrally resolved DBLS method from photon correlation spectroscopy (DPCS), which also probes α-relaxation but does so through analysis of temporal fluctuations in scattered intensity rather than through frequency-domain spectral analysis. The two techniques therefore access distinct dynamic windows and rely on fundamentally different measurement formalisms.

2.3.2 Depolarized Photon Correlation Spectroscopy (DPCS)

Depolarized photon correlation spectroscopy (DPCS) is a light scattering technique, similar to DBLS described above, yet performed in the time domain. Both methods provide access to the α-structural relaxation dynamics within a glass-forming liquid, however, they cover vastly different dynamic ranges. Whereas DBLS lends itself to probing very fast relaxations, on the order of tens to thousands of picoseconds, DPCS is generally applicable in the slow to intermediate dynamic regime, probing characteristic relaxation times ranging from tens of nanoseconds to thousands of seconds. A typical experimental setup for DPCS is shown in Fig. 2.9 —differing from DBLS only in the form of analyzer implemented. In fact, data taken for both DBLS and DPCS, analyzed in Ch. 5, were all acquired on a single experimental setup in which a flip mirror allowed the collected scattered light from a sample to be directed either into a tandem triple-pass Fabry–Perot interferometer for frequency analysis or, alternatively, into a digital autocorrelator for temporal analysis.

There are two general experimental domains relevant to PCS—heterodyne and homodyne autocorrelation. It is important to define these terms here as there is some discrepancy within the literature regarding their use. For the purposes of this work, homodyne PCS involves temporal correlation analysis of light scattered from a sample, where great care is taken to ensure that the detected signal arises solely from spontaneous density fluctuations within the material. Particular attention is given to minimizing parasitic elastic scattering and specular reflections from dust, cell windows, culets, or table surfaces, which could act as a coherent local oscillator and introduce heterodyne contributions to the measured correlation function. The intensity of this scattered light will fluctuate as a result of any temporal dependency associated with these dynamic processes.

As such, this experimental arrangement is often referred to as "self-beating," a designation that only becomes meaningful when contrasted with the alternative PCS configuration—heterodyne detection.

In a heterodyne experimental arrangement, such as is represented in Fig. 2.10, a fraction of the source laser is split off before it is passed through the scattering sample and then re-mixed before being directed onto the detector for autocorrelation. In this arrangement, the fluctuations in detected intensity do not arise from the total number of scattering events, but rather from a shift in phase between light arising from scattering events and that of the local oscillator—a portion of the laser being mixed in providing a phase reference for the scattered light to beat against. Below is shown one possible heterodyne experimental arrangement. It is also common simply to collect light from the sample such that the specular reflection from the container can itself provide a local oscillator. All that is required to operate within a heterodyne regime is that the local oscillator intensity be significantly greater than that of the light scattered from the sample. Conversely, if one wishes to operate in a homodyne regime, one must ensure that any specular reflections or contributions from static scattering are negligible compared to the light scattered by the dynamic processes of interest within the sample. This often requires significant spatial filtering as well as exceedingly pure samples and clean experimental environments.

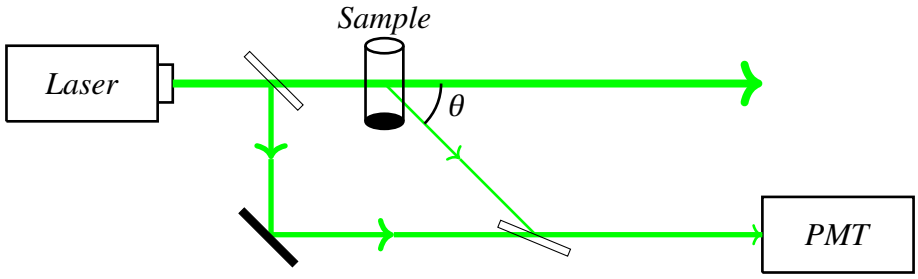


Figure 2.10: Heterodyne experimental arrangement for PCS. A local oscillator is mixed with the scattered signal before being redirected onto the photomultiplier or other detector.

In both experimental arrangements the intensity fluctuations at the detector are due to the same fluctuations in the local density or dielectric constant within the scattering volume in the sample, however, the two methods measure different correlation functions of these fluctuations. Thus, proper analysis of resulting data requires that one be firmly within one or the other experimental domain. It is generally very undesirable to be within an intermediate homodyne/heterodyne scenario. If a homodyne experiment is desired one must often take great care to assure that only light arising from the scattering events of interest are collected. Dust and contaminates, both within the sample and on the sample container, or even undesired specular reflections may give rise to undesired heterodyne components which may make proper data analysis challenging or even impossible. These concerns are not always easily mitigated, and great care is required to identify the regime in which one is operating, as will be discussed in Sec. 5.2.1, where a method for extending PCS to high-pressure is described in detail.

2.3.3 Rolling Ball Viscometry

Rolling or falling Ball viscometry provides a means of direct mechanical measurement of viscosity. Many different types of high pressure vicometers and their range of application are described in more detail by Cook, Herbst, and King[45]. These methods generally implement a spherical ball as it moves through a viscous fluid under the influence of gravity or other forces. Placing this ball within a diamond anvil cell allows access to very high pressures however the viscosities that can be measured are obviously limited to those low enough for a macroscopic ball to progress through the material during the course of a reasonable experimental timescale. In practice this means that rolling and falling ball viscometry in a diamond anvil cell is limited to approximately 10^7 cP—eight orders of magnitude away from the glass transition. In the work previously

cited, Cook *et al*. managed to extend this viscosity range an additional two and a half orders of magnitude, up to approximately 10^9 cP, *via* placing the diamond anvil cell viscometer onto a centrifuge which greatly increases the g-forces driving the ball through the fluid[45]. However this still limits the probe to a viscosity range many orders of magnitude away from the glass transition.

Though this probe cannot provide measurements in the vicinity of the glass transition it does, as shown in Fig. 2.8, provide access to a very large dynamic range of viscosities—up to nine orders of magnitude. There are some experimental concerns even within its range of application. Specifically, rolling ball viscometry relies on a correction, or "wall factor," that accounts for dissipative losses as the ball rolls down the diamond surface within the cell. This correction factor is assumed to be constant as experimental variables such as pressure and temperature are changed. It is also generally assumed that the probe ball is rolling down the diamond cutlet without slipping. It is very difficult, if not impossible, to directly verify the degree to which these assumptions hold over the entire range of experimental parameters. However this work will show that data for the glass former glycerol obtained by Cook *et al*. *via* rolling ball viscometry appears to be reasonably consistent over its applicable range with data taken *via* other probes and non-related methods.

2.3.4 TGP

The TGP process is a very straight forward means of measuring the isochronal curve of the glass transition temperature T_g as a function of Pressure. A liquid is loaded into a diamond anvil cell and brought to a temperature and pressure such that it is well within the solid glass phase. It has been found advantageous to pre-indent the gasket, which hardens it, thus making higher pressures more achievable. Two or more rubies are arranged throughout the sample as shown in the far left DAC in Fig. 2.11. A pressure gradient is then applied to the DAC *via* asymmetric application of

force as indicated by the dashed Isobaric Lines.

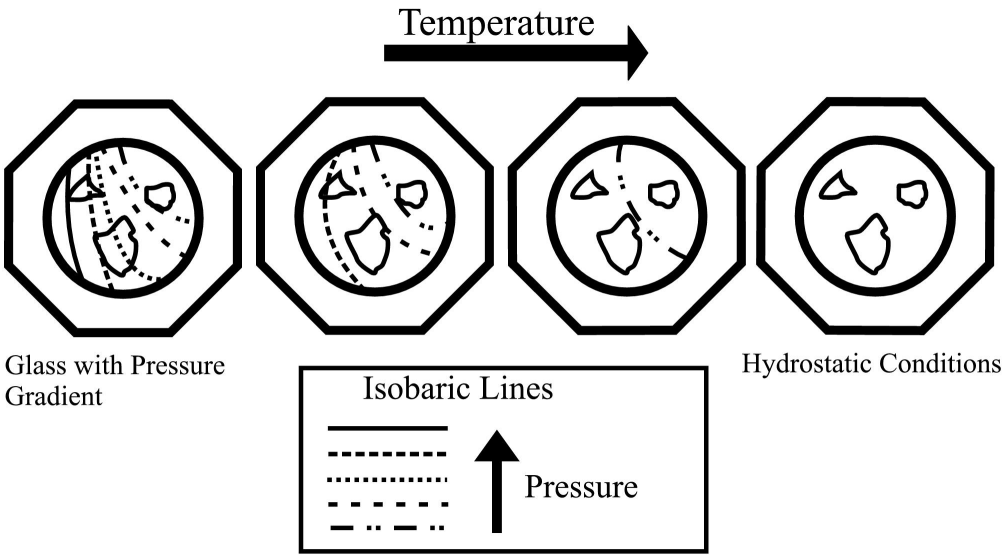


Figure 2.11: Pressure gradients established within a diamond anvil cell decrease as temperature increases and the system approaches the glass transition from the glassy state, ultimately attaining hydrostatic conditions. This figure is reproduced from Ref. [46].

Temperature is then increased incrementally, typically in 5 to 10 degree steps initially and then smaller steps as the glass transition is approached as evident *via* the pressure gradients giving way to hydrostatic conditions above T_g. At each temperature, pressure is measured from a consistent location on each ruby to again check for pressure gradients. Temperature is then increased to a new set point and a small set time is allowed for the sample to reestablish thermal equilibrium. A means of determining the approximate temperature at which the gradients dissolve and the material transitions into a liquid have been developed previously and described in Ref. [46, 47]. The values obtained for the glass transition temperatures as a function of pressure are fit to an Andersson-Andersson equation [48] provided along with data for glycerol and cumene in App. B and C respectively. The automation of this process will increase its efficacy in the future and may even allow it to provide more than a single isochronal curve near and around the glass transition.

2.3.5 Dielectric Loss Spectrometry

Frequency Response Analysis encompasses various methods capable of measuring molecular dynamics over an exceptionally broad dynamic range. Among these, Dielectric Spectroscopy (DS) is particularly important and will be extensively discussed in this dissertation. DS takes advantage of molecular dipole moments, which either inherently exist or are induced by an applied oscillating electric field.

As molecular orientations within the glass-forming liquid respond to an applied external field, dielectric spectroscopy (DS) measures both the real and imaginary components of the complex dielectric constant. When the frequency of the applied field is systematically swept—from fractions of a hertz to several gigahertz—the imaginary component plotted as a function of frequency forms the dielectric loss spectrum. Peaks in this loss spectrum correspond directly to molecular relaxation processes occurring within the sample.

Alternatively, DS measurements may be performed at fixed frequency while varying temperature or pressure, with both components of the dielectric response recorded as functions of T or P. In either protocol, the rate at which frequency, temperature, or pressure is varied plays an important role in determining the apparent relaxation behavior. By monitoring how the positions of loss maxima evolve under controlled thermodynamic conditions, researchers gain access to molecular dynamics such as the structural relaxation time (τ_α), as well as faster secondary processes, commonly referred to as β-relaxations.

However, dielectric spectroscopy provides considerably more information than merely the characteristic structural relaxation times inferred from dielectric loss peaks. Both the real and imaginary components of the complex dielectric constant contain dynamical information. When

analyzed together, these components may be represented in the complex plane to form characteristic Cole–Cole or Cole–Davidson plots [49]. Such representations facilitate visualization of relaxation behavior and, when combined with appropriate empirical response functions, enable extraction of additional parameters. Among these is the stretching exponent β, which reflects the breadth of the underlying distribution of relaxation times and the degree of dynamical heterogeneity within the glass-forming liquid. Representative examples illustrating both Debye-like and stretched-exponential behavior are presented in Ch. 4 (see, for example, Fig. 4.2).

Given its capability to provide a wide array of parameters and its extensive dynamic range, dielectric spectroscopy is one of the most powerful and widely implemented experimental methods in glass transition research. Indeed, due to its comprehensive coverage of both structural and secondary relaxations, DS serves as an indispensable tool for studying the complex molecular dynamics of glass-forming liquids, a role this dissertation will explore further.

3 Supporting Theory

This chapter will provide the working definition of fragility as originally defined by Austin Angell and that is pervasive throughout the literature [50]. This classic definition of fragility will be implemented in this work to compare results with those previously obtained in other studies. However, a scaled definition of fragility—here referred to as the *scaled fragility*—will also be introduced. This definition normalizes the conventional Angell fragility such that a purely Arrhenius liquid assumes a value of unity, providing a more physically transparent reference point for comparing dynamic behavior across systems. The scaled fragility further facilitates direct comparison of results obtained via different experimental probes—some measuring viscosity, η, and others measuring the characteristic structural relaxation time, τ_α—by embedding the dynamic range explicitly within the definition. Beyond this, the focus will turn to a simple derivation of the scattered radiation field, providing important theory linking this field to the dynamics of the liquid under study. Next, we will discuss the spectral structure of light scattered from a simple fluid and how this translates into the time domain in the case of intensity autocorrelation. Finally, mathematical models that describe these scattered fields and allow extraction of thermodynamic parameters of interest are presented.

3.1 Fragility

Fragility is a useful parameter for describing the dynamic behavior of a glass-forming liquid as the glass transition is approached. As defined by Angell, fragility characterizes how strongly a liquid's molecular dynamics or viscosity deviate from Arrhenius behavior upon cooling toward

the glass transition. The viscosity and associated structural relaxation time of a subset of glass-forming liquids known as Arrhenius systems, which will be described in Sec. 3.1.1, exhibit a nearly exponential dependence on inverse temperature. Other glass-forming liquids display pronounced super-Arrhenius behavior and are referred to as fragile systems. In all systems, the viscosity and associated α-relaxation dynamics change by many orders of magnitude as the glass transition is approached. While Angell's original definition of fragility emphasizes temperature, comparable dynamical changes occur under compression, setting the stage for the pressure-dependent investigations presented in this work. For these reasons, fragility is defined in terms of the slope of the logarithm of viscosity or structural relaxation time as a function of inverse temperature, with each system normalized by its own glass transition temperature T_g, as given in Eq. 3.1.

$$m = \left(\frac{\partial \log_{10} \eta}{\partial \left(T_g / T \right)} \right)_{T=T_g} \tag{3.1}$$

To further understand the motivation behind this definition of fragility, we introduce the Angell plot, shown in Fig. 3.1, as applied to three typical glass-forming liquids exhibiting varying degrees of fragility. The horizontal axis of a classic Angell plot is the inverse temperature normalized to the glass-transition temperature, T_g, under isobaric conditions. This ensures that all liquids, regardless of their individual glass-transition temperatures, converge at $T_g/T = 1$ on the right side of the Angell plot, corresponding to a viscosity of $\eta_g = 10^{13}$ poise or an α-relaxation time of $\tau_g = 100$ seconds—the values for which the glass transition is here defined.

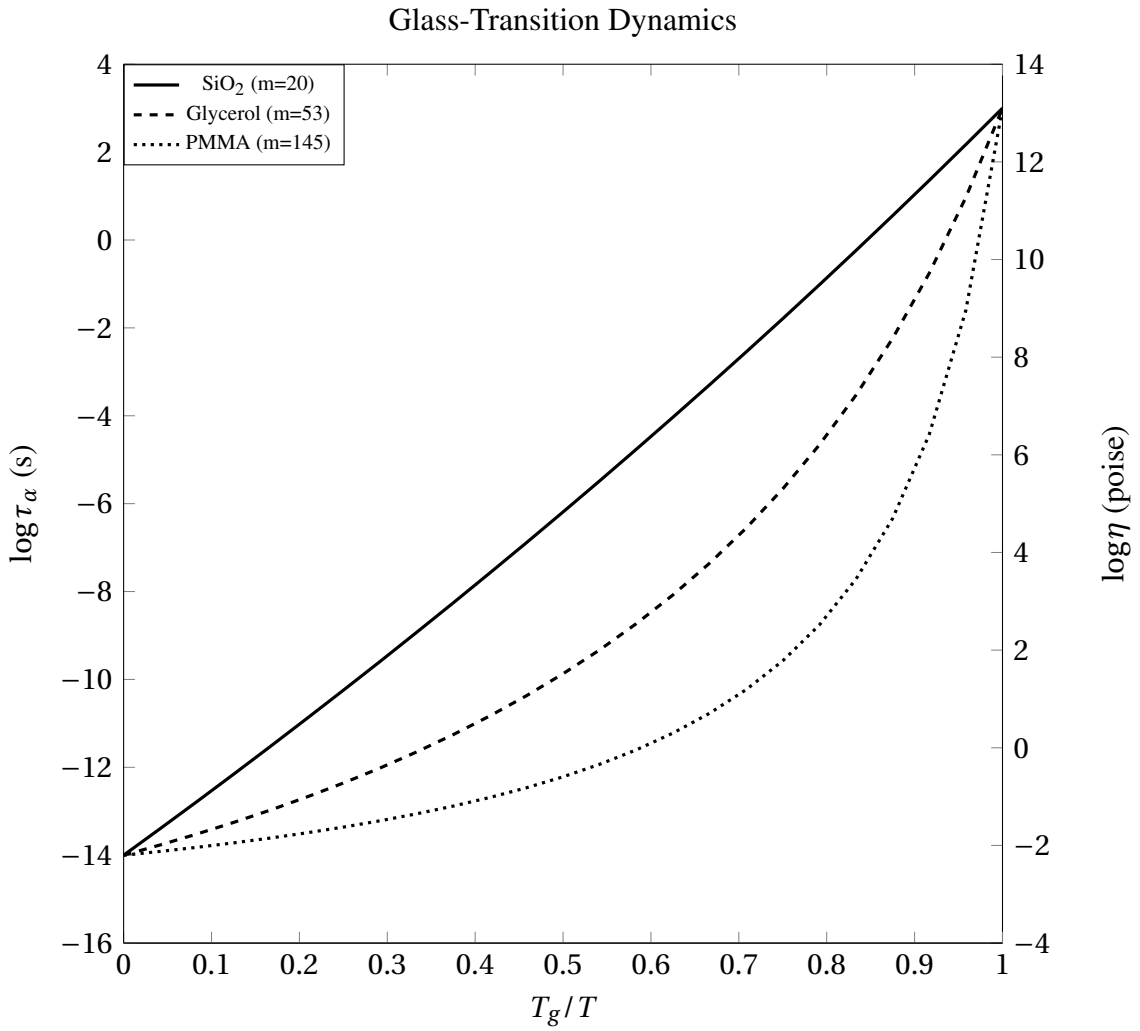


Figure 3.1: Examples of Angell plots illustrating the fragility of three representative glass-forming systems [50]. In this representation, each system is scaled by its respective glass transition temperature T_g, allowing direct comparison on a common framework. Strong (Arrhenius) liquids display approximately linear behavior, whereas fragile liquids exhibit pronounced super-Arrhenius curvature as T_g is approached.

If one defines the high-temperature asymptotic limit as the regime in which the average thermal energy of any given molecule greatly exceeds the characteristic intermolecular interaction energies, then intermolecular constraints cease to dominate the structural dynamics of the liquid. In this regime, the system approaches behavior characteristic of weakly interacting particles, and relaxation is governed primarily by intrinsic molecular parameters rather than cooperative rearrange-

ments. Although the liquid phase itself will ultimately lose thermodynamic stability at sufficiently high temperature, within the accessible liquid domain the relaxation dynamics progressively lose sensitivity to collective structural constraints.

As intermolecular barriers become energetically insignificant relative to $k_B T$, the system samples the potential energy landscape with minimal constraint. Structural relaxation is therefore controlled primarily by internal molecular properties such as moments of inertia and local conformational flexibility. Most small organic molecules span only a few angstroms, and even larger macromolecules exhibit local flexibility on comparable length scales. Consequently, the characteristic microscopic times associated with molecular motion tend to fall within similar ranges across many liquids. It is therefore not surprising that, under a given isobaric condition, liquids exhibit high-temperature asymptotic relaxation times τ_α confined to a relatively narrow window—typically from hundreds of femtoseconds to a few picoseconds.

Fig. 3.1 shows the Angell plot for three distinct glass-forming systems. Silicon dioxide serves as a classic example of an Arrhenius, or what Angell termed a "strong" system. Glycerol provides an example of a liquid intermediate in fragility, while the polymer PMMA exhibits highly super-Arrhenius behavior characteristic of a fragile system. Although these materials differ significantly in their intermolecular bonding—from network-forming covalent structures to hydrogen-bonded and polymeric systems—fragility itself does not necessarily reflect bond strength. Rather, it characterizes how strongly the relaxation dynamics depend on temperature as the glass transition is approached.

The Angell plot provides a compact means to compare the dynamics of different glass-forming systems on a single graph while simultaneously visualizing their fragilities. Note that, as defined in Eq. 3.1, fragility is geometrically represented as the slope of the Angell plot at $T_g / T = 1$,

i.e. , as the system approaches the glass transition from the liquid state. Thus, the relative fragilities of different liquids—or of a single liquid under varying pressure—can be determined at a glance, as demonstrated by Cook *et al.* [51].

Because the horizontal axis of a proper Angell plot progresses from zero to one as T_g/T increases, and most liquids share a comparable high-temperature asymptotic viscosity on the order of 10^{-2} Poise, the minimum fragility attainable under the conventional definition corresponds to the slope of a straight line connecting this asymptotic viscosity to that at which the glass transition is defined to occur. Taking $\eta_g \equiv 10^{13}$ Poise, a nearly Arrhenius (strong) liquid therefore exhibits a fragility on the order of $m \approx 15$.

Although the conventional fragility is defined as the slope of the Angell plot at T_g, the choice of T_g itself—fixed by selecting a particular dynamical value—implicitly embeds the dynamic range into the definition. As a result, even a perfectly Arrhenius liquid does not assume a value of unity. It is therefore more natural to rescale the ordinate of the Angell representation and redefine fragility accordingly, such that an Arrhenius liquid serves as the reference case with scaled fragility equal to one.

$$m^* = \frac{1}{\log_{10}\left(\frac{\eta_g}{\eta_\infty}\right)} \left(\frac{\partial \log_{10}\left(\frac{\eta}{\eta_\infty}\right)}{\partial\left(T_g/T\right)} \right)_{T=T_g} . \tag{3.2}$$

In practice, this redefinition merely rescales the classical Angell fragility by a constant normalization factor. It introduces no additional complexity into the analysis, yet it enforces well-defined high- and low-temperature limits in the Angell representation and removes ambiguity in interpretation: a purely Arrhenius system exhibits a scaled fragility of precisely $m^* = 1$. By embedding the dynamic range explicitly within the definition, the scaled fragility also facilitates con-

sistent comparison between viscosity data and structural relaxation measurements.

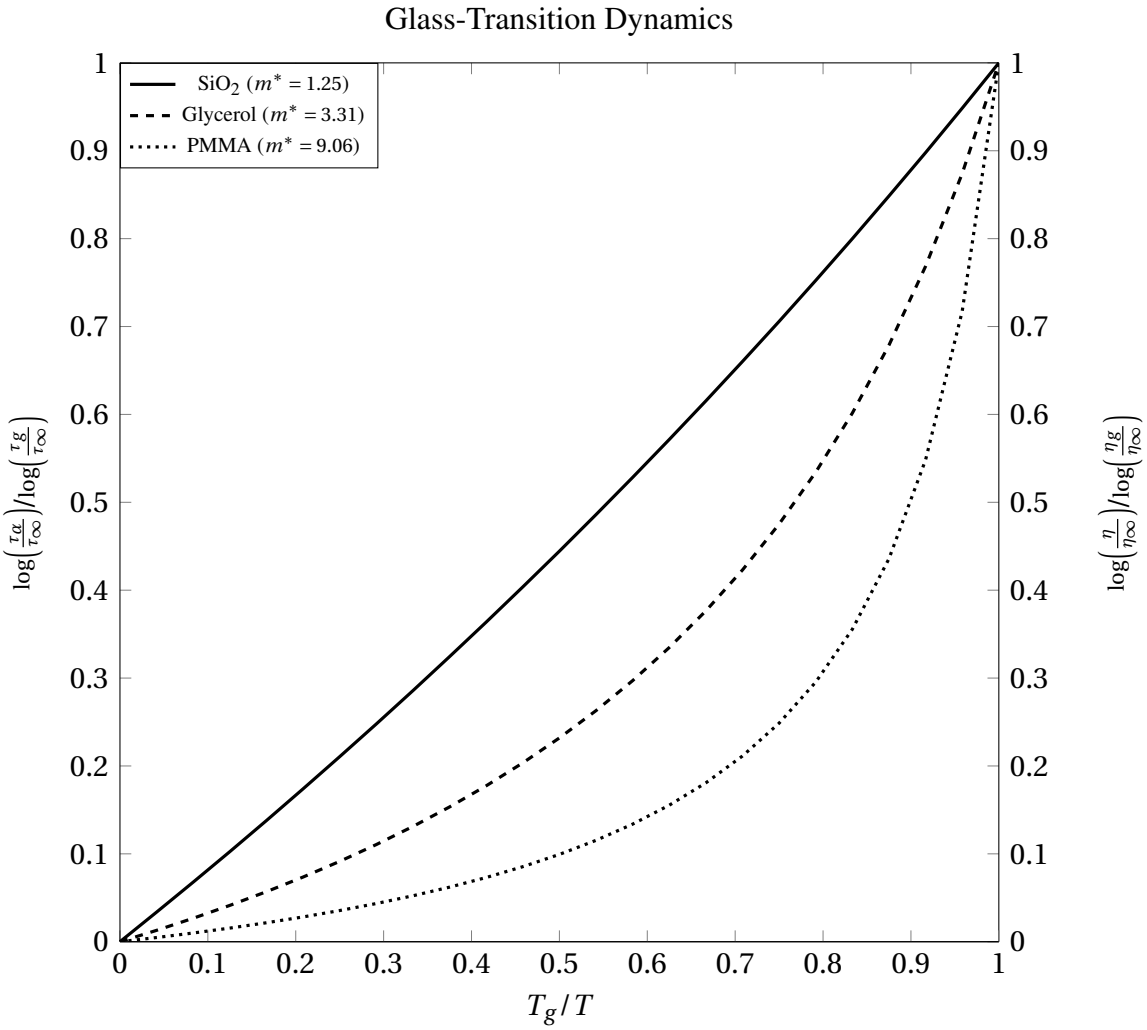


Figure 3.2: Scaled Angell plots for the same three liquids shown in Fig. 3.1, constrained at both the glass transition and in the high-temperature limit. In this representation, the dynamic range is normalized such that a purely Arrhenius liquid exhibits a scaled fragility of $m^* = 1$.

The scaled Angell plot shown in Fig. 3.2 retains the same qualitative behavior as Fig. 3.1. In the classical Angell representation, the high-temperature asymptotic viscosities or relaxation times are not constrained to coincide, so the curves remain vertically offset at low T_g/T. By contrast, the scaled representation explicitly enforces a common dynamic range between the high-temperature asymptote and the glass-transition point. In this representation, both viscosity and structural re-

laxation data are placed on the same scaled axis, enabling direct and consistent comparison across probes.

3.1.1 Arrhenius Systems

Arrhenius, or "strong," glass-forming liquids are those whose primary structural (α) relaxation exhibits an approximately temperature-independent activation energy as the glass transition is approached. For a nearly Arrhenius system, the primary structural relaxation is characterized by an approximately temperature-independent activation energy (or, equivalently, activation volume under pressure). The probability that a region of the material possesses sufficient thermal energy to undergo structural rearrangement at a temperature T therefore follows Boltzmann statistics. Consequently, because the characteristic structural relaxation time and viscosity are governed by these thermally activated events, they are expected to exhibit an exponential dependence on inverse temperature as follows:

$$\tau_\alpha = \tau_\infty \exp\left[\frac{\Delta E}{k_b T}\right]. \tag{3.3}$$

Here τ_∞ denotes the high-temperature asymptotic relaxation time, as previously defined, corresponding to the regime in which the thermal energy $k_B T$ greatly exceeds the characteristic intermolecular interaction energies and the potential energy landscape no longer contributes appreciably to molecular motion. A classic example is silicon dioxide, or SiO_2, shown in the Angell plot of Fig. 3.1. Note the nearly linear relationship between viscosity (or structural relaxation time) and inverse temperature, as expected on a logarithmic plot for a strong glass former described by Eq. 3.3. It is evident from either Fig. 3.1 or Fig. 3.2 that a minimum fragility exists, which in

the classical definition depends on the chosen glass-transition criterion but generally lies between $m = 15$ and $m = 16$. This minimum reflects the fixed dynamic range embedded in the conventional Angell representation. By contrast, in the scaled Angell formulation and its associated definition of fragility, the minimum possible fragility for any glass-forming system is precisely $m^* = 1$, corresponding to perfectly Arrhenius dynamics.

3.1.2 Non-Arrhenius Systems

It should be clear that most systems do not exhibit purely Arrhenius behavior. Even the classic strong glass formers, such as SiO_2, exhibit some level of curvature on an Angell plot, and a fragility of $m \approx 20$ ($m^* \approx 1.25$) is significantly above that expected from a single discrete activation energy described by Eq. 3.3. Realistically, any molecule or particle that does not possess perfect spherical symmetry is likely to exhibit more than one mode of relaxation, giving rise to the deviations from Arrhenius dynamics seen in Fig. 3.1 and Fig. 3.2.

Glass-forming liquids that deviate strongly from Arrhenius behavior are referred to as "fragile" glass formers. In these systems, the effective activation energy governing structural relaxation is temperature dependent rather than constant. Fragile liquids exhibit a relatively small apparent activation energy in the high-temperature limit, but as the glass transition is approached from the liquid state the effective barrier increases, leading to pronounced super-Arrhenius behavior. A PMMA melt is a classic example of a fragile glass-forming liquid, whereas glycerol is typically described as having intermediate fragility.

Fragility in non-Arrhenius systems may also be pressure-dependent. Glycerol is a prime example, as its fragility increases with increasing pressure—the result of hydrogen bonding being suppressed. Ch. 4 will explore this pressure dependency of glycerol fragility in detail.

However, the isobaric dynamics of such systems exhibit pronounced super-Arrhenius temperature dependence, deviating strongly from a simple Arrhenius (single-barrier) description. These dynamics are commonly modeled using the Vogel–Fulcher–Tammann (VFT) equation, first proposed by Tammann and Hesse in 1929 to describe the temperature dependence of viscosity in polar liquids [52]. The modern form of this equation, most prevalent in the literature, is given by

$$\tau_\alpha = \tau_\infty \exp\left[\frac{DT_0}{T - T_0}\right]. \tag{3.4}$$

Here T_0 is often referred to as the zero-mobility temperature, representing the extrapolated temperature at which the structural relaxation time τ_α diverges. In this limit, the configurational degrees of freedom of the system would cease to evolve on experimental timescales, corresponding to a vanishing configurational mobility within the framework of the VFT model. The parameter D is referred to as the fragility parameter, as it is closely related to fragility and is often used directly as an alternative definition of fragility.

3.2 Dielectric Fluctuations and Light Scattering

Any time light travels through a transparent medium, the electromagnetic fields couple with the charges within the medium, inducing electronic moments. This interaction can be quite complicated, resulting in a scattered field that requires a multipole expansion to describe properly. However, in the limit where the wavelength of the incident field is large compared to the characteristic microscopic length scales within the medium, the electric dipole term generally dominates and the induced polarization may be treated in the quasi-static (long-wavelength) limit, where spatial variations of the field across a scattering region are negligible.

The general form of the scattered field, though technically a quantum effect, can be derived via classical electrodynamic theory and is well described in general terms by both Jackson and Landau–Lifshitz, while Berne and Pecora offer a more practical rendition of Landau's derivation specifically in the context of dynamic light scattering[53, 54, 55]. In all cases, an incident monochromatic plane wave is assumed to propagate through a scattering medium as depicted in Fig. 3.2.

An incident monochromatic plane wave, indicated by the vertical blue lines, propagates in the $+\hat{z}$ direction with wavevector

$$\vec{k}_{in} = \frac{2n\pi}{\lambda_{in}},$$

where λ_{in} is the incident wavelength in vacuum and n is the refractive index of the medium. Light scattered by the medium is collected at a scattering angle θ_S relative to the incident direction, as indicated by the dashed arrows, with outgoing wavevector

$$\vec{k}_{out} = \frac{2n\pi}{\lambda_{out}}.$$

Typical laser wavelengths used in light-scattering experiments are on the order of $\lambda_{in} \approx$ 500 nm, corresponding to optical frequencies near 6×10^{14} Hz. For a representative refractive index of $n \approx 1.4$, the magnitude of the incident wavevector is therefore

$$k_{in} \approx \frac{2(1.4)\pi}{500 \times 10^{-9}\,\mathrm{m}} \approx 1.8 \times 10^{7}\ \mathrm{m}^{-1}.$$

In inelastic Brillouin light scattering, the scattered light typically differs in frequency from the incident light by at most a few gigahertz, while in quasi-elastic photon correlation spectroscopy

(PCS), the frequency shifts range from sub-hertz to megahertz. In either case, these shifts are negligible compared to the optical carrier frequency. It is therefore an excellent approximation to assume $\lambda_{out} \approx \lambda_{in}$, so that $|\vec{k}_{out}| \approx |\vec{k}_{in}| \equiv k$. Under this assumption, the scattering vector

$$\vec{q} = \vec{k}_{out} - \vec{k}_{in}$$

has magnitude, via the law of cosines,

$$q = 2k \sin\left(\frac{\theta_S}{2}\right).$$

This long-wavelength approximation, in which the incoming and outgoing wavelengths are effectively identical, constitutes one of the primary assumptions underlying the analysis that follows.

For typical scattering angles employed in Brillouin light-scattering experiments, q is on the same order as k_{in}, and the frequencies of the Stokes and anti-Stokes Brillouin shifts are determined by the acoustic velocity v of the medium according to

$$\nu = \frac{qv}{2\pi}.$$

In PCS experiments on simple fluids, the relevant density fluctuations occur over spatial scales that are small compared to the wavelength, provided the system is not near a critical point where macroscopic critical fluctuations would dominate [56]. In this limit, the electromagnetic field is effectively uniform across each microscopic scattering region, and all molecules within that region experience essentially the same local electric field. The total scattering volume in an experiment then consists of many such statistically independent regions, and the scattered light

collected represents the coherent superposition of the fields radiated by each region. This physical picture of light scattering from thermodynamic density fluctuations was first articulated by Einstein in his early treatment of critical opalescence [57].

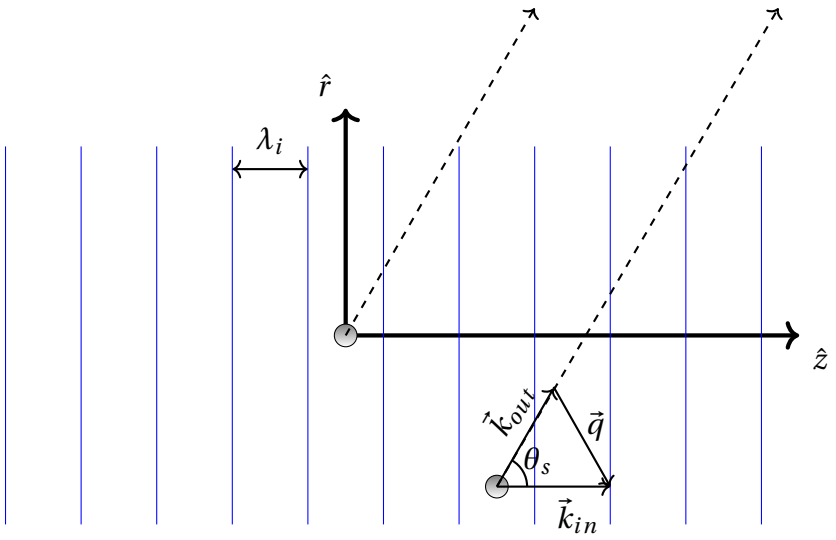


Figure 3.3: Vertical lines represent a plane wave propagating in the $\hat{z}$ direction into a region of dispersed scattering bodies. The time-dependent dipole moments induced in each molecule or particle act as independent sources of a scattered field. Here we assume a diffuse scattering medium such that secondary scattering events contribute negligibly to the net scattered field[55].

For scattering to serve as an effective structural probe, the medium must be optically thin such that higher-order multiple scattering contributions are negligible. In this dilute (or optically thin) limit, the detected scattered field is linear in the dielectric fluctuation $\delta\epsilon$, and light collected at an angle θ_s may be associated with a well-defined scattering vector $\vec{q} = \vec{k}_{out} - \vec{k}_{in}$. Because the incident wavevector $\vec{k}_{in}$ is known and the collection geometry fixes $\vec{k}_{out}$, the scattering vector $\vec{q}$ is likewise well defined and will play a central role in the analysis that follows.

To build intuition for the light-scattering process, one may first consider a system in which the local number density of scatterers fluctuates in time due to thermal motion. In a dilute suspension of particles, such fluctuations arise from Brownian motion. In condensed phases, including

simple liquids, analogous fluctuations occur as molecules move and create short-lived variations in local density within the bulk.

Thus, to describe how light is scattered from a bulk medium, one must recognize that the dielectric permittivity of any real material fluctuates—both temporally and spatially—due to the random thermal motions of the constituent particles or molecules, as well as any structural inhomogeneities. This can be modeled as a static "average" permittivity around which local fluctuations occur as follows:

$$\epsilon(\vec{r}, t) = \epsilon I + \delta\epsilon(\vec{r}, t). \tag{3.5}$$

Here ϵ is the scalar dielectric permittivity of free space, and $\delta\epsilon$ describes deviations from this constant, which in general depend on both space and time and may include off-diagonal elements, providing a completely general description of local permittivity in any transparent medium.

From this point we derive the wave equation for the scattered field to understand what gives rise to it. Landau–Lifshitz and Jackson provide excellent classical derivations, while Berne and Pecora give a simplified treatment specific to light scattering[53, 54, 55]. As illustrated in Fig. 3.2, if a plane wave traveling through a medium is to produce a scattered field, the resulting field in the medium is the superposition of the incident and scattered fields:

$$E = E_i + E_s, \tag{3.6}$$

$$D = D_i + D_s, \tag{3.7}$$

$$H = H_i + H_s. \tag{3.8}$$

Whatever form the scattering fields assume, they must satisfy Maxwell's equations for fields within a medium:

$$\nabla \times E_s = -\tfrac{1}{c}\tfrac{\partial H_s}{\partial t}, \tag{3.9}$$

$$\nabla \times H_s = \tfrac{1}{c}\tfrac{\partial D_s}{\partial t}, \tag{3.10}$$

$$\nabla \cdot H_s = 0, \tag{3.11}$$

$$\nabla \cdot D_s = 0. \tag{3.12}$$

From Eq. 3.9 and Eq. 3.10 one obtains:

$$\nabla \times \nabla \times E_s = -\frac{1}{c^2}\frac{\partial^2 D_s}{\partial t^2}. \tag{3.13}$$

The net instantaneous displacement vector from Eq. 3.7 is related to the electric field vector from Eq. 3.6 by the dielectric tensor from Eq. 3.5:

$$D_i + D_s = (\epsilon I + \delta\epsilon(\vec{r}, t)) \cdot (E_i + E_s). \tag{3.14}$$

Solving for the scattering field E_s and neglecting the $\delta\epsilon \cdot E_s$ term that would give rise to secondary scattering of much smaller magnitude:

$$E_s = \frac{1}{\epsilon} D_s + \frac{1}{\epsilon}(\delta\epsilon \cdot E_i). \tag{3.15}$$

This scattering field must satisfy Eq. 3.13:

$$\nabla \times \nabla \times \frac{D_s}{\epsilon} + \nabla \times \nabla \times \frac{(\delta\epsilon \cdot E_i)}{\epsilon} = -\frac{\epsilon}{c^2}\frac{\partial^2 D_s}{\partial t^2}. \tag{3.16}$$

Utilizing Lagrange's cross-product identity and Eq. 3.12, this simplifies to:

$$\nabla^2 D_s - \frac{\epsilon}{c^2}\frac{\partial^2 D_s}{\partial t^2} = -\nabla \times \nabla \times (\delta\epsilon \cdot E_i)\,. \tag{3.17}$$

Eq. 3.17 is a wave equation for the scattering displacement vector D_s with a source term proportional to $\delta\epsilon \cdot E_i$. This derivation assumes only that fields satisfy Maxwell's equations and that the dielectric permittivity is described by the tensor in Eq. 3.5.

3.3 Spectrum of a Simple Fluid

Until this point, the scattering field has been discussed in very general terms. Moving forward the discussion will be focused on the application at hand–light scattering as a probe of molecular dynamics. As such, the incident field, E_i, will be assumed to be a monochromatic constant intensity light source passed through a transparent fluid from which scattered light will be collected along a specific scattering vector as shown below.

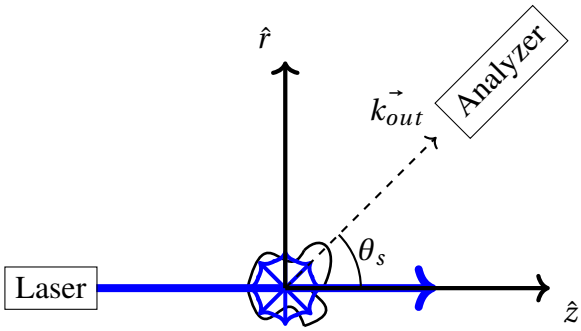


Figure 3.4: Generalized light scattering geometry

It is also important to note that, within the context of light scattering, the wavelength of the incident field, E_i, will always be very large when compared to intermolecular spacings within the scattering region. This was assumed in the derivation of Eq. 3.17 previously as scattering events that arise from dynamic variations in dielectric susceptibility emerge stochastically from the independent motions of large numbers of constituent molecules or particles. Within any fluid above absolute zero there will be local regions where bulk thermodynamic properties such as entropy, temperature, or pressure will fluctuate about the average value. It is these fluctuations, as alluded to previously, that give rise to the scattering field. Thus, it is reasonable to invoke hydrodynamic and thermodynamic theory to arrive at a model for the spectrum of light scattered from a simple fluid. Such a model was first derived by Komarov and Fisher in their 1962 publication[58]. In this work Komarov adapted neutron scattering models derived by Leon van Hove in 1954 to describe light scattering spectra[59]. It should be noted that this derivation is also provided in excellent detail by Berne and Pecora [55]. The resulting spectrum of this scattered light was shown by Komarov to be proportional to the spatial and temporal Fourier transform of the generalized pair correlation function. While Section 2.1.1 introduced the static radial distribution function $g(r)$

describing equilibrium structure, it is necessary here to consider its dynamical extension.

Following the work of van Hove, the static pair correlation function is generalized to the time-dependent van Hove correlation function $G(\vec{r}, t)$, which describes the probability of finding a particle at position $\vec{r}$ at time t given that a particle was located at the origin at time $t = 0$. The dynamic structure factor $S(\vec{q}, \omega)$ is then obtained as the spatial and temporal Fourier transform of $G(\vec{r}, t)$.

The scattered intensity may therefore be written as

$$I'(\vec{r}, \omega) = I_0 N \omega^4 \sin^2 \gamma \, S\left(\vec{q}, \omega\right). \tag{3.18}$$

This result is, of course, quite similar to the result for Rayleigh scattering from a diffuse gas with the addition of the structure factor, $S\left(\vec{q}, \omega\right)$, which contains information regarding molecular dynamics of the system. Mountain later extended Komarov and Fisher's results and, via application of the linearized hydrodynamic equations, the Navier-Stokes equation, and the energy transport equation, derived a model for the frequency dependent structure factor arising from a classical fluid comprised of spherically symmetric molecules as follows:

$$S\left(q, \omega\right) = \frac{c_p - c_v}{c_p} \frac{2\lambda q^2 / \rho_0 c_p}{\left(\lambda q^2 / \rho_0 c_p\right)^2 + \omega^2} + \frac{c_v}{c_p} \left[\frac{\Gamma q^2}{\left(\Gamma q^2\right)^2 + \left(\omega + C_0 q\right)^2} + \frac{\Gamma q^2}{\left(\Gamma q^2\right)^2 + \left(\omega - C_0 q\right)^2} \right]. \tag{3.19}$$

Here ρ_0, c_p, c_v, C_0, and λ are the equilibrium density, specific heat at constant pressure, specific heat at constant volume, low frequency limit of the sound velocity, and the thermal conductivity respectively.[60] The first, left-most term of Eq. 3.19 represents the broadened central peak of light scattering from entropy fluctuations which occur at constant pressure. In a simple fluid, as there

are no internal degrees of freedom, this takes the form of diffusion or breaking of nearest neighbor cages—*i.e.* structural relaxation. The second, or right-most term of Eq. 3.19 represents light scattered by propagating acoustic waves. As this is an adiabatic process, this occurs at constant entropy. This term forms two peaks symmetrically located around the central peak with frequency shifts from E_i proportionally to the speed of sound. These are often referred to as the longitudinal acoustic modes. This model is sufficient to describe the dynamics of a simple fluid in the sense described by Mountain: a single-component, monatomic system in which internal degrees of freedom may be neglected and the dynamics are governed entirely by density fluctuations. However, typical glass-forming systems possess additional internal and configurational degrees of freedom, giving rise to additional modes of structural relaxation beyond those captured by this description.

Many glass forming systems are significantly more complex, exhibiting internal degrees of freedom, intermolecular hydrogen bonding, and other additional considerations. Polymer systems can fold and flex. Even simple molecular systems limited to van der Waals interactions, such as cumene, have rotational degrees of freedom that give rise to additional relaxation modes. Such considerations will add to the complexity of this structure factor. Mountain followed this previous work with a derivation of a structure factor accounting for internal degrees of freedom weakly coupled to the originally considered translational degrees of freedom via a single representative orientational relaxation time, τ as follows:

$$
\begin{aligned}
S(q,\omega) = \frac{c_p - c_v}{c_p}\frac{2\lambda q^2/\rho_0 c_p}{\left(\lambda q^2/\rho_0 c_p\right)^2 + \omega^2} + \\
\left[\frac{\left(c_\infty^2 - c_0^2\right)k^2 - \left(v^2/c_0^2 - 1\right)\left(c_0^4/v^4\tau^2 + c_0^2 q^2\left(\frac{c_p - c_v}{c_p}\right)\right)}{c_0^4/v^4\tau^2 + v^2 q^2}\right]\left[\frac{\frac{2c_0^2}{v^2\tau}}{\frac{c_0^4}{v^4\tau^2} + \omega^2}\right] + \\
\left[\frac{\left[1 - \frac{c_0^2}{v^2}\left(\frac{c_p - c_v}{c_p}\right)\right]\left[v^2 q^2 + \frac{c_0^2}{v^2\tau^2}\right] - \left(c_\infty^2 - c_0^2\right)q^2}{\frac{c_0^4}{v^4\tau^2} + \frac{v^2}{q^2}}\right] \times \\
\left[\frac{\Gamma_B}{\Gamma_B^2 + \left(\omega - vq\right)^2} + \frac{\Gamma_B}{\Gamma_B^2 + \left(\omega + vq\right)^2}\right].
\end{aligned}
\tag{3.20}
$$

Here the longitudinal acoustic modes represented by the third term are structurally indistinguishable from the second term of Eq. 3.19 though now the coefficient is τ dependent.[61] Likewise the first term, associated with translational diffusion, remains unchanged from before. However, most notably, there now emerges a secondary central feature symmetric about the incident frequency, associated with this internal relaxation mode. Thus the orientational dynamics directly affect the structure of this central feature. There is a vast amount of insight provided by this model. However the primary consideration relevant to this work lies in this additional central term of Eq. 3.20 above. Consider the following simplified expression of this structure.

$$
f(\omega) = \left(\frac{2\Gamma}{\Gamma^2 + \omega^2}\right) \tag{3.21}
$$

This function, as before mentioned, will be symmetric about the axis $\omega = 0$ and will have a characteristic half width half maximum value of Γ as shown in Fig. 3.3 below.

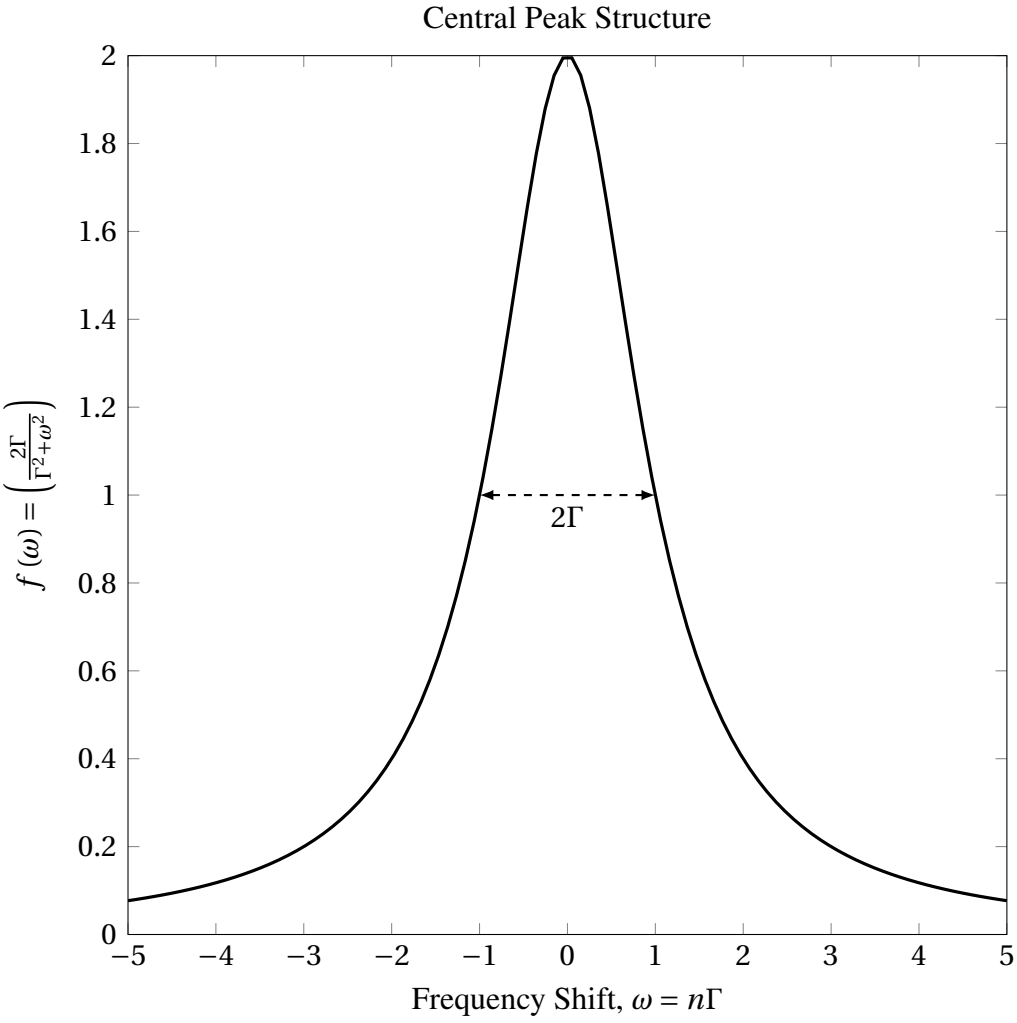


Figure 3.5: Here we have the generalized structure of the central features from Eq. 3.19 and3.20. The x-axis here is in terms of multiples of the generalized parameter Γ from Eq. 3.21 which also defines the half width-half maximum value for this feature.

Placing aside the specific values of any particular bulk thermodynamic property, both the first and second term of Eq. 3.20 have the same basic structure in this model. Generally speaking, the structural relaxation associated with translational and orientational diffusion can occur on vastly different timescales however both will contribute to the width of this central feature. It bears mention that, for a simple monatomic fluid, only the translational diffusion term contributes to this central feature and $\Gamma = \frac{\lambda}{\rho_0 c_p} q^2 = D_T q^2$ where D_T is the thermal diffusivity and there exists a de-

pendency on the scattering vector. However, if the orientational diffusion contribution dominates the width of the central feature then $\Gamma = \frac{{c_0}^2}{v^2\tau} = \frac{D_\tau}{\tau}$ and there is no dependence on the scattering vector.

The work of Demoulin, Montrose, and Ostrowsky provides valuable insights and foundational support for understanding the scattering phenomena described in this chapter. Specifically, Demoulin et al. employed digital-correlation spectroscopy, analogous to photon correlation spectroscopy (PCS) utilized in this dissertation, to investigate structural relaxation in viscous liquids over time scales from 1 µs to 1 s. Their work explicitly analyzed the Rayleigh scattering of light from undercooled glycerol, relating the measured correlation function to the isothermal structural relaxation dynamics of the liquid[62].

Demoulin et al. characterized three distinct regimes governing the Rayleigh line shape arising from density fluctuations: thermal diffusion dominance, structural relaxation dominance, and an intermediate regime involving contributions from both processes. Of particular interest here is what Demoulin et al. described as the "Mountain component," which appears prominently under conditions where the structural relaxation timescale is significantly slower than the phonon modes but faster than thermal diffusion processes.

This Mountain component corresponds directly to the central feature described earlier in the structural factor from Eq. 3.20, particularly when orientational diffusion significantly influences its width. Demoulin et al. modeled the relaxation function using a Cole-Davidson distribution—precisely the approach taken in this dissertation—to quantify the relaxation dynamics, further validating the methodology adopted here. Their analysis determined that the Cole-Davidson distribution, characterized by a width parameter, provided a robust fit for their data.

Thus, the work presented in this dissertation parallels Demoulin et al.'s in both the choice

of methodology (PCS) and the theoretical framework (Cole-Davidson distribution), thereby establishing continuity and credibility in the application of these techniques to characterize structural relaxation in glass-forming liquids. Furthermore, Demoulin et al.'s findings underscore the importance and practicality of employing digital autocorrelation techniques to probe relaxation phenomena, reinforcing the methodologies implemented throughout this dissertation.

3.4 Mathematical Models

In analyzing experimental data from dynamic light scattering experiments, it becomes necessary to implement mathematical models to characterize structural relaxation processes accurately. Four commonly used models—Debye relaxation, Cole-Cole, Cole-Davidson, and Williams-Watts—will be introduced, described, and compared in this section, with emphasis on their typical usage in either the time or frequency domain, and the interrelation between domains provided by Lindsey and Patterson.

3.4.1 Debye Relaxation

The Debye relaxation model represents the simplest case, assuming a single exponential decay characterizes the relaxation process. In the time domain, this corresponds to:

$$\phi(t) = e^{-t/\tau_0}. \tag{3.22}$$

where τ_0 is a single characteristic relaxation time. In the frequency domain, Debye relaxation is described by:

$$\phi(\omega) = \frac{1}{1 + i\omega\tau_0}. \tag{3.23}$$

This model assumes a uniform relaxation process without distribution of relaxation times. Although appealing for simplicity, the Debye model fails to describe the complex relaxation behaviors observed in most glass-forming liquids.

3.4.2 Cole-Cole Relaxation

Recognizing deviations from Debye behavior, the Cole-Cole model introduces a symmetrical broadening of the relaxation time distribution, characterized by an additional parameter α:

$$\phi(\omega) = \frac{1}{1 + (i\omega\tau_{cc})^{1-\alpha}}, \quad 0 \leq \alpha < 1. \tag{3.24}$$

where τ_{cc} is a central relaxation time and α measures the symmetric broadening of the relaxation time distribution around τ_{cc}. This symmetrical distribution is particularly useful in frequency-domain dielectric measurements exhibiting slight deviations from ideal Debye behavior.

3.4.3 Cole-Davidson Relaxation

The Cole-Davidson model extends Debye relaxation by introducing an asymmetric broadening of the relaxation time distribution, characterized by the parameter β:

$$\phi(\omega) = \frac{1}{(1 + i\omega\tau_{cd})^{\beta}}, \quad 0 < \beta \leq 1. \tag{3.25}$$

This model typically finds application in frequency-domain analyses, where it accurately captures relaxation processes characterized by an effective upper bound to the relaxation time

distribution. Demoulin et al. effectively employed this model to describe structural relaxation phenomena in viscous liquids like glycerol using digital-correlation spectroscopy in the frequency domain.

3.4.4 Williams-Watts (Stretched Exponential) Relaxation

The Williams-Watts model, commonly known as the stretched exponential, is primarily used to fit autocorrelation data in the time domain:

$$\phi(t) = e^{-(t/\tau_{ww})^{\beta_{ww}}}, \quad 0 < \beta_{ww} \leq 1. \tag{3.26}$$

In this expression, τ_{ww} defines a characteristic relaxation time, and β_{ww} measures the stretching of the exponential function. Unlike Cole-Davidson, the Williams-Watts distribution of relaxation times decays approximately exponentially at longer times without a sharp cutoff. It has become the preferred model in photon correlation spectroscopy for analyzing cooperative and complex relaxation dynamics.

3.4.5 Connecting Time and Frequency Domain Models

Although Cole-Davidson and Williams-Watts models are typically employed in different domains, frequency and time respectively, Lindsey and Patterson established a detailed theoretical relationship between these two distribution functions [63]. Their work provides empirical relations allowing parameters from one domain to be related directly to parameters in the other, thereby facilitating the comparison and interpretation of data analyzed by either method. This interrelation is crucial as it offers practical guidelines for consistent and comparative analysis across diverse

experimental studies and methodologies.

3.4.6 Comparison, Contrast, and Justification

The Debye model, while simple, rarely suffices for most complex fluid relaxations. Cole-Cole introduces symmetric broadening suitable for moderately complex systems, while Cole-Davidson addresses systems with pronounced asymmetry and defined cutoff behaviors in the frequency domain. The Williams-Watts model provides the most flexibility in describing relaxation processes observed in the time domain, especially those involving cooperative molecular dynamics or hierarchical structural relaxation mechanisms.

As previously mentioned in Sec. 2.3, multiple probing techniques are necessary to obtain relaxation data spanning multiple orders of magnitude. These probes yield data in both frequency-domain and temporal autocorrelation formats. To ensure accurate interpretation and comparison across these varying methodologies, the parameters from these fits must be translatable, emphasizing the value of the relationships provided by Lindsey and Patterson. However, Demoulin preferred directly fitting his time-domain autocorrelation data to the Cole-Davidson distribution via a transform. We will adopt Demoulin's approach extensively in analyzing our own time-domain data in the chapters that follow.

4 The α-Relaxation Dynamics of Glycerol

Glycerol is perhaps one of the most widely studied glass forming systems. As a hydrogen bonding liquid, it exhibits more complicated behavior than a simple van der Waals liquid such as cumene discussed in Ch. 5. A topic of significant interest within the literature over the past twenty five years has been the behavior of fragility as a function of pressure in such hydrogen bonding systems. Whereas most molecular liquid systems exhibit either a decreasing fragility as pressure increases or very little pressure dependence at all, glycerol and other hydrogen bonding liquids have been observed to show an increase in fragility which has been attributed to the suppression of hydrogen bonding under pressure [64, 65, 66]. This chapter will explore the current state of the literature on this topic, apply a new model to the dynamics of glycerol over a wide range of pressure and temperature with the aid of literature data obtained through multiple methods, reassess the efficacy of viscosity data obtained by Cook *et al*. that has come under some scrutiny, further compare resulting thermodynamic scaling regimens with previous literature results, as well as provide a more precise measure of the pressure dependence of fragility in glycerol.

4.1 Earlier Works Regarding Glycerol Dynamics

As previously asserted, glycerol is one of the most widely studied glass forming systems, with the temperature dependence of its viscosity being investigated as far back as 1926 in the work of Tammann and Hesse [52]. However the modern history of glycerol, as relevant to this work, arguably begins with Davidson and Cole's investigation of the temperature dependence of the dielectric relaxation in glycerol circa 1951 [49]. Davidson and Cole found that not all associated

liquids can be modeled with a simple single-relaxation or Debye equation as shown in Eq. 4.1 [67].

$$\epsilon^* = \epsilon_\infty + \frac{\epsilon_0 - \epsilon_\infty}{1 + i\omega\tau_0}, \tag{4.1}$$

where ϵ_∞ and ϵ_0 are the susceptibilities in the infinite-frequency and steady-state limits respectively. Though 1-propanol was found to be well described by a simple Debye relationship, glycerol exhibited a higher degree of dispersion as vitrification accelerates and required unrealistically small values for its molecular volume in order for the observed relaxation time to conform to that predicted by the Debye hydrodynamic model for hard spheres. Davidson and Cole correctly surmised that this was due to the existence of additional modes of relaxation available to glycerol that simply cannot exist for the simpler and more rigid 1-propanol molecule. To deal with this anomalous dielectric dispersion, Davidson and Cole suggested modeling such associated systems with the empirically derived "skewed" Debye like model

$$\epsilon^* = \epsilon_\infty + \frac{\epsilon_0 - \epsilon_\infty}{(1 + i\omega\tau_0)^\beta}, \tag{4.2}$$

where β is the "skewing parameter." Note that Eq. 4.2 reduces to Eq. 4.1 when $\beta = 1$. Starting with Cole and Cole, it became common in the literature of this time to plot the real *vs*. imaginary components of this empirical expression as would have been common for the analysis of an analogous RC circuit[68]. With regards to complex susceptibility, in the context of dielectric spectroscopy, the real part of Eq. 4.2 corresponds to the non-dissipative component and the imaginary part to the dissipative component. This can be modeled, for a simple Debye relaxation where $\beta = 1$, in terms of a three component RC circuit as follows:

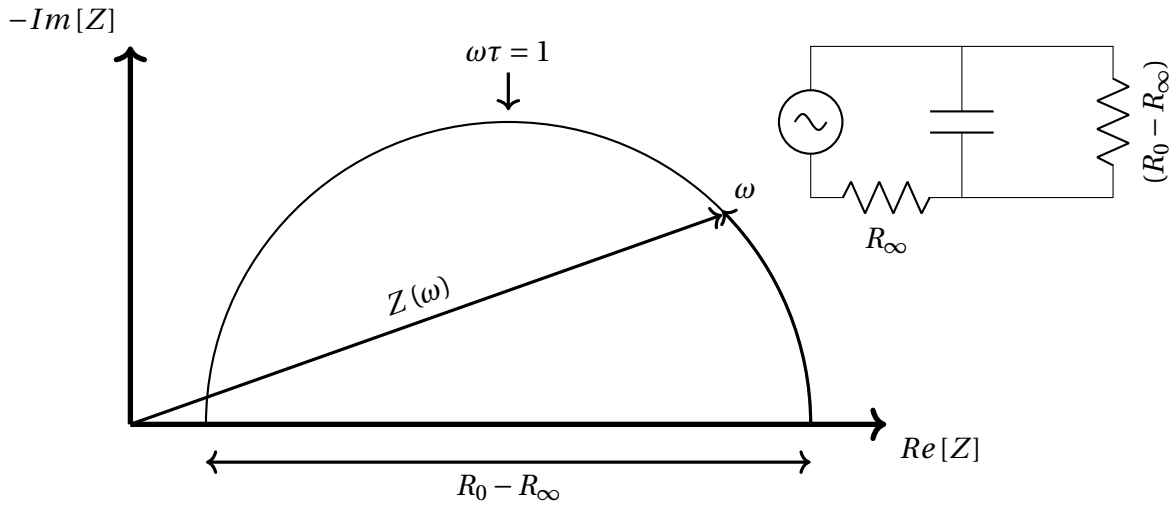


Figure 4.1: A system with a frequency dependent susceptibility and a single characteristic relaxation time τ driven at a given frequency ω can be modeled as an RC circuit as show here in the upper right corner. The complex impedance will be of the form $Z(\omega) = R_\infty + \frac{R_0 - R_\infty}{1+i\omega\tau}$ which is functionally identical to the dielectric susceptibility of the simple Debye relaxation of Eq. 4.1. Though the component which plays the role of irreversible dissipation is reversed in the case of the RC circuit above, both complex dielectric relaxation and this analogous electrical impedance will exhibit maximum dissipation at resonance where $\omega\tau = 1$.

Note that the magnitude of impedance will progress from R_0 to R_∞ as the driving frequency is increased from $\omega = 0$ to $\omega \to \infty$. The impedance of the two limiting cases, R_0 and R_∞, are analogous to the limiting values of the dielectric susceptibility ϵ_0 and ϵ_∞ from Eq. 4.1, respectively.

As previously mentioned, the Cole-Davidson model described by Eq. 4.2 above may be described as a "skewed" Debye model when $0 < \beta < 1$ as it results in a distortion of the corresponding Cole-Cole plot. An example of such a skewed plot of glycerol where $\beta = 0.6$ and n-propanol where $\beta \approx 1$, as obtained by Davidsion, is shown below. More importantly, Davidson provides a closed form expression for this distribution of simple Debye relaxations, first calculated by Fuoss and Kirkwood to describe dipole moments in polymers, which give rise to the Cole-Davidson model of Eq. 4.2 and this is described in more detail in Ch. 5 [69]. For now, we assert that this skewing of the dielectric dispersion, increasing as the value of β decreases, corresponds to a broader distribution

of Debye-like relaxation processes.

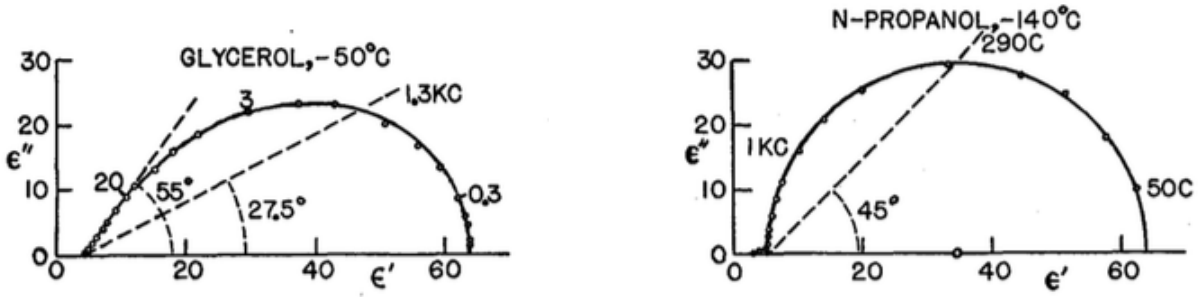


Figure 4.2: Cole-Cole plots of glycerol and n-propanol data (points). Solid lines represent fits to the Cole-Davidson model of Eq. 4.2 with beta-values of 0.6 and ≈ 1, respectively [49].

Davidson and Cole also noted that the value of β is essentially constant at ambient pressure over a temperature range from 40 °C to 75 °C. The implications of this being that the general structure of the distribution of relaxation modes is constant over a range of temperatures for which the dynamics of said relaxations are changing by at least six orders of magnitude. Similar results have been found for many systems over the years and for such systems, this is known as time-temperature superposition. More recently, the same has been found to hold over significant pressure ranges as well, in which case it is referred to as time-temperature-pressure superposition. We found such results for Cumene, are described in Ch. 5—β being essentially constant over a wide range of temperature and pressure.

Though Davidson and Cole only considered the isobaric behavior of glycerol relaxation as a function of temperature, their contribution here provided some of the first data that could be used to estimate fragility—at least at atmospheric pressure. They demonstrated that the strong temperature dependence of glycerol relaxation is poorly described, even over this moderate temperature range, by a classical Arrhenius rate equation but was well fit, within experimental uncertainty, by the more empirical Vogel-Tammann-Fulcher (VTF) equation previously described ~~in~~ Eq. 3.4

and first proposed by Tammann and Hesse to describe glycerol viscosity as a function of temperature [52]. Furthermore, they provided a closed form expression for the implied distribution of simple Debye relaxations underlying Cole-Davidson Eq. 4.2 above. Finally, they noted that the average molecular relaxation time of this distribution for glycerol would correspond to "hard-sphere" molecular dimensions, as provided by the classic Debye hydrodynamic theory, that are far too small to be physically realistic for glycerol, *e.g.* 1×10^{-25} cm^3, implying the existence of additional relaxation modes involving reorientation of molecular sub-units. It really is this observation, above all other contributions, that led to the development and application of free-volume models for viscosity and structural relaxation upon which much of this chapter will rely. Due to the importance of free-volume in this chapter and throughout the literature, particularly with regards to modeling viscosity of systems beyond ambient pressure, a discussion and derivation of the free-volume model will be provided in Sec. 4.2.

Throughout the 1960s, work began to emerge within the literature probing viscosity and structural relaxation of glycerol beyond ambient pressure. One of the first to appear was published in 1964 by McDuffie and Kelly where data for both viscosity, *via* a rolling ball viscometer, and relaxation data, *via* dielectric loss spectroscopy, were obtained. These methods were described previously in Sec. 2.3.3 and Sec. 2.3.5, respectively. McDuffie obtained viscosity data over three separate isotherms and dielectric relaxation data over a fourth which collectively covered a very small temperature range of $\Delta T = 20\,^{\circ}$C, however, ~~he~~ extended these isotherms over what was, at the time, a very large range of pressure from ambient up to 4 kbar. Beyond simply publishing some of the first high pressure data for glycerol viscosity, McDuffie's major contribution in this work was to show that neither the pressure nor temperature dependence of viscosity for glycerol was well described by the Eyring rate theory commonly used at the time and which will be described in the

following section. Also of considerable interest was that the free-volume model, ~~also~~ described in Sec. 4.2 below, was shown by McDuffie to yield a compressibility for glycerol that was not congruent with other high pressure values derived from ultrasonic probes at the time and also that the implementation of pressure-independent thermal expansion coefficients within the free-volume model leads to an over-estimation of the pressure dependence of viscosity. However these discrepancies were quickly addressed the following year, in 1966, by Matheson [70]. His arguably obvious approach was simply to apply a quadratic model to account for the pressure dependence of v_m in the free-volume expression (Eq. 4.29) from Sec. 4.2.1. An identical approach was applied for the temperature dependence for several other liquids however the limited temperature range in McDuffie's data was insufficient to meaningfully model a quadratic temperature dependence for v_m in glycerol. Still, in effectively addressing the shortcomings of early free-volume models exposed by McDufffie *via* extending the model beyond ambient pressure, Matheson's contribution was very significant. What Matheson was in want of, and what was conspicuously lacking in the literature up to this point, would be carefully collected data for glycerol viscosity or structural relaxation spanning a large temperature and pressure range.

This data vacuum was quickly filled by Johari and Whalley in the early 70's when they published a very heavily sighted paper containing dielectric susceptibility measurements of glycerol spanning a dynamic range of six decades. These data also contained a much larger pressure and temperature range than any previous work—ambient to 53 kbar and $-55\,^{\circ}$C to $84\,^{\circ}$C respectively [71]. At lower temperatures Johari and Whalley were able to probe the relaxation dynamics of glycerol very near the glass transition. At higher temperatures, above $T = 250$ K, their data falls around three decades short of approaching the glass transition. However their data has proven to be of superb quality and opened up the door to investigations of the pressure dependence of such

parameters as β, defined in Eq. 4.2, and much latter fragility. Similar contributions were made for several other glass-forming systems and many citations exist in the literature including this work, particularly to assess the pressure dependence of glycerol's fragility [72, 73, 74, 75].

Around a decade latter, in 1986, Forsman *et al*. published a paper also probing dielectric relaxation of glycerol [74]. They had the advantage of nearly a decade of advancement in instrumentation and thus were able to probe significantly faster relaxations, up to 13 MHz. However they did not provide a great deal of data and the pressure and temperature range was already well covered by Johari and Wally's previous work, particularly with regards to investigations into fragility as the faster relaxation times do not significantly add to the efficacy of fragility calculations. Though Forsman was focused more on the broadening of the loss peak as a function of pressure and furthers the discussion of many-body relaxation theory to account for deviations of the loss spectra from Debye theory, this work is worthy of mention here as it references several other papers providing data for glycerol relaxation collected over several decades.

Another work worthy of mention would be the work published by Demoulin in the early 70's [62]. Though data was only taken at ambient pressure and low temperatures, Demoulin probed glycerol relaxations directly in the time domain *via* photon correlation spectroscopy as described in Sec. 2.3.2. Though his data was confined to a relatively limited dynamic range, as autocorrelators of this era were among the first and significantly slower than those currently commercially available, with the aid of ultrasonic measurements a reasonable dynamic range was obtained. The aim of Demoulin was not to extend the dynamic range of data available for glycerol relaxation, but rather to prove the efficacy of digital photon correlation spectroscopy as a probe of relaxation dynamics over what was an exceedingly large dynamic range for the time. In Ch. 5, a method for extending this probe into the range of very high pressures will be presented. It is interesting here

to note that, though the data obtained *via* photon correlation spectroscopy is in the time domain, the Cole-Davidson distribution, being used pervasively throughout the literature to describe the dynamics of structural relaxation probed *via* dielectric spectroscopy and ultrasonics methods, is most naturally expressed in terms of frequency. This work was published before that of Lindsey and Patterson, which worked out a simple means of translating between the Williams-Watts distribution, expressed most naturally in the time domain and commonly used to model data obtained *via* photon correlation spectroscopy, and the Cole-Davidson distribution of Eq. 4.2 [63]. Thus it is not surprising that Demoulin chose to impart significant effort to fit his time domain data to the more common, yet inconvenient, Cole-Davidson distribution so that it may be directly compared to data within the literature. His method for accomplishing this will be extended in Ch. 5, taking advantage of modern computational tools, for exactly the same reasons.

Demoulin's method of PCS will later be well established to constitute a direct measurement of structural α-relaxation within a glass forming system. However another direct probe, this time of viscosity, was introduced before the end of the decade by Geils and Keezer [76]. The rolling ball viscometer as a means of directly measuring viscosity *via* tracking the terminal velocity of a metal ball rolling down an incline, as previously described in Sec. 2.3.3, certainly predated the work of Geils and Keezer, however early implementations placed great restrictions on the geometry of the viscometer requiring that the ratio of the rolling ball diameter to that of the viscometer remain in one or the other limiting extremes—either rolling without slipping down a surface of the viscometer or lacking any significant interaction with the confining surfaces [77, 78]. Geils and Keezer worked out the means of accounting for the correction factors associated with wall effects for this method, allowing for greater flexibility in experimental design to allow for temperature and potentially pressure control, drastically increasing its efficacy as a probe for viscosity of glass

forming systems.

4.2 Free-Volume Theory

The concept of free-volume as a mechanism to explain viscosity was first postulated as far back as 1913 when Batschinski proposed a linear relationship between thermal expansion and viscosity [79]. Though this linear model will obviously suffice over any sufficiently small thermodynamic range, it fails spectacularly in the description of data from high temperatures down through a glass transition. However the general idea of describing viscosity in terms of solid molecular units passing through regions of free-volume is certainly conceptually appealing and was gaining more traction in the literature in the early fifties. Though Batschinski's model was known by this time to be insufficient to describe viscosity behavior beyond a first approximation, and certainly would only be applicable far from any glass transition where viscosity is rapidly diverging, there was no reason to believe the concept of free-volume was invalid. The main competing theory of viscosity at the time was the rate theory proposed by Henry Eyring, which one may assert is physically equivalent to free-volume theory though couched in terms of the energy fluctuations associated with creating that free-volume—in the simplest terms, pushing nearest neighbors out of the way [80]. Eyring proposed that viscosity of any non-associated liquid, to the extent that structural relaxation is assumed to be an energetic process which involves breaking free of potential wells created by neighboring molecules, may be treated in much the same way as any chemical process activated by thermal energy fluctuations and should be well described by a Boltzman-like distribution involving a single activation energy, E_0, and of the following form:

$$\eta = \frac{Nd}{V}(2\pi m k_b T)^{\frac{1}{2}} e^{\frac{E_0}{k_b T}} \tag{4.3}$$

It is not surprising that an associated liquid such as glycerol might fail to be well described by this Arrhenius model. However even most non-associated fragile glass forming systems also conform well to the phenomenological VTF equation. Thus if one is to overcome the apparent discrepancy between Eyring's rate theory of Eq. 4.3 above and the VTF equation, the functional dependence of the activation energy on temperature must be of the form $E(T) \propto \frac{T}{T-T_\infty}$, where T_∞ is the zero mobility temperature at which viscosity becomes infinite. This requirement rather diminishes the physical grounding of the Eyring rate theory as it offers no explanation as to why the temperature dependence of activation energy should take this form. Some have tried to link the temperature dependence of activation energy with local entropy fluctuations and met with some level of success [81]. In fact such a rate theory is the foundation for the Adam-Gibbs equation which has been heavily implemented to describe enthalpy relaxation in many glass forming systems [82, 83]. However the conceptual simplicity and elegance of a free-volume description may suggest that a non-linear free-volume equation, beyond that suggested by Batschinski, may be the most physically meaningful approach to describing viscosity of these non-Arrhenius systems. It was during the same year of 1951 that Doolittle published his work describing viscosity of n-paraffin hydrocarbons *via* an exponential free-volume model, similar in form the the Eyring rate theory, yet dependent upon the relative free-space, $\frac{v_f}{v_0}$ within a bulk liquid rather than an activation energy,

$$\eta = Ae^{B\left(\frac{v_f}{v_0}\right)}, \tag{4.4}$$

where v_f is the volume of free-space per unit of liquid and v_0 is that free-volume density specifically evaluated at some reference temperature—often absolute zero [84].

This is useful as it circumvents the need to explain any functional dependence of a potentially complicated distribution of activation energies and attacks the problem from an arguably conceptually simpler standpoint of free-volume—is there sufficient free-volume, on average, for molecular mobility to progress at a given rate. Only thermal expansion needs to be considered as all other parameters are temperature independent. The Doolittle expression for free-volume of Eq. 4.4 above also provided the groundwork for Williams, Landel, and Ferry to explain the temperature dependence of kinetic constants as a function of temperature near the glass transition [85]. It was a major contribution that launched the free-volume model into common implementation. However Eyring's rate theory explanation was certainly not supplanted by a free-volume approach. Again, structural fluctuations that give rise to local free-volume, *i.e.* the opening of a path through which a caged molecule might diffuse, must be energetically accessible as must be the diffusion process itself. It should be unsurprising that a model relying on a temperature independent single activation energy to describe what, even in simple liquids, amounts to several independent modes of relaxation all of which may have strong temperature dependence, especially as the glass transition is approached, would prove to be every bit as insufficient as Batschinski's linear free-volume model. Eyring's model failed to sufficiently describe diffusion even in some simple crystalline systems [86].

Consider again the potential energy landscape of Fig. 2.6 way back in Ch. 2. Though, to some extent, structural relaxation may be entirely entropy driven as regions held at constant energy may still explore equipotential surfaces, moving from one local minimum within the energy landscape to another will be an energy activated process. Particularly as one approaches a state

here described as an ideal glass, the activation energy associated with hopping between adjacent potential wells will be strongly temperature dependent and the complexity of the parameter space implies there will be a near uniform distribution of potential paths through the parameter space corresponding to physical modes of structural relaxation. Thus Henry Eyring's description of viscosity as an energy activated process underlies the concept of free-volume. Bueche makes note, in his contemporary work on the segmental mobility of polymers, that one is left free to conceptualize relaxation processes in terms of the activation energy, ϵ, needed to break free of nearest neighbor cages or equivalently in terms of a activation volume, ΔV, needed for a molecular unit, or sub-molecular in the case of polymer chains, to pass through [87]. Bueche also importantly notes that, as previously alluded to by Davidson and Cole regarding the failure of the Debye hydrodynamic model to sufficiently explain the relaxation processes of glycerol, an activation volume, ΔV, need not be on the order of the molecular volume as relaxation modes may involve cooperative motions of sub-molecular units.

Bueche's desire to connect free-volume theory, which since the contribution of Doolittle had been shown to aptly model viscous behavior of many liquids over a much larger range of temperatures than when first proposed as a linear model by Batschinski, with the Eyring rate theory is quite understandable. Eyring's rate theory of viscosity, though experimentally found wanting, was strongly rooted in a first-principles derivation and shares an elegant symmetry with the well established thermodynamic theory of chemical reaction rates. If viscosity is, in fact, an energy activated process it should be governed by similar expressions to those governing chemical reactions. Though it may be convenient to model viscosity in terms of free-volume, Bueche argues that the creation of this free-volume comes at an energetic cost. The free-volume model proposed by Doolittle, though experimentally successful, falls short of the Eyring rate theory in its lack of any

theoretical explanation—the exponential form of Eq. 4.4 being entirely phenomenological.

A first-principles theoretical derivation of Doolittle's free-volume model was finally provided in 1959 *via* the work of Cohen and Turnbull [88]. To accomplish this, Cohen envisioned an entropic rather than energetic activation behind structural relaxation. Consider again the parameter space idealized by Fig. 2.6 as the n-dimensional surface that would exist for any real liquid. The energetic processes described by Bueche and Eyring may not be necessary to navigate a sufficiently complex parameter space. Couched in terms of the energy landscape picture, Cohen might suggest that one consider that what perhaps appears to be an insurmountable energy barrier between two adjacent potential wells in Fig. 2.6 in the absence of some introduced activation energy, ϵ, may in fact be accessible *via* multiple paths along equipotential surfaces in higher dimensional parameter space. Evolution along these equipotential surfaces correspond to pure entropy fluctuations at constant energy and can be physically interpreted as a rearrangement of existing free-volume within the bulk material.

Cohen and Turnbull do well to point out that the diffusion constant for spherical particles dispersed in a viscous medium is inversely related to viscosity *via* the Stokes-Einstein relation,

$$D = \frac{k_b T}{3\pi a_0 \eta}, \tag{4.5}$$

where a_0 is the hard sphere cross-section. Though this is derived in the context of low number density macroscopic spherical particles suspended in a viscous fluid, rather than the self diffusion of a pure liquid, it is still contextually relevant to the latter. In fact the use of nano-particles suspended in a liquid to serve as a probe of the molecular dynamics of the host liquid certainly appears in the literature and was considered as a possible method for extending photon correlation

spectroscopy into the diamond anvil cell, the results of which are described in Ch. 5. Suffice it to say for now that there was concern regarding the light arising from density fluctuations within a pure liquid from the very small volume of a diamond anvil cell being sufficient in intensity for photon correlation spectroscopy. As such it was considered that sufficiently small probe particles, perhaps within an order of magnitude of the molecular size of the liquid under study, might provide a sufficient scattering intensity and their diffusion would approximate, or at least track with, that of the molecular self diffusion for the host liquid.

This idea was in part inspired by the work of Caronna *et al*. who implemented x-ray scattering from suspended nano-particles to probe the dynamics of a glass forming liquid [89]. They found that at higher temperatures the particles undergo Brownian motion as predicated by Eq. 4.5 above however, as the glass transition is approached, the probe particles exhibit what the authors referred to as "hyperdiffusive behavior" which they suggest is due to the emergence of cooperative motions of the host liquid. It may be further suggested that, while approaching the glass transition, the effective size of the molecular units involved in structural relaxation increase accordingly as cooperative domains of the host liquid increase in size. This hyperdiffusive behavior will emerge as the size of these cooperative domains approach that of the probe particles.

The point to be had here is that, to the extent that a pure liquid can be modeled as consisting of hard spheres of radius a_0, self diffusion within a pure liquid should be well modeled by the Stokes-Eienstein relationship. Cohen and Turnbull assumed that this would hold true through the glass transition, however, as found much later by Caronna *et al*., this will depend on the emergence of cooperative domains that likely have the effect of imposing a strong temperature dependence on the effective size of the relaxing units, which may range from sub-molecular to macroscopic as the glass transition is approached. The emergence of cooperative domains and their roll regarding

structural relaxation within glass forming systems was not so well understood at this time. None the less, in the case of self diffusion from the perspective of free-volume, the effective size of the relaxing units, weather they be sub-molecular or large cooperative domains, will be intrinsically associated with the viscosity η. It is reasonable to assert that, no matter the functional form, to first approximation self diffusion within a pure liquid should be inversely proportional to the viscosity of the liquid.

The question that remains is weather this viscosity and diffusion result from energy activated processes or pure entropy fluctuations involving a rearrangement of free-volume within the bulk material. Cohen assumes the latter. What will follow here is Cohen's derivation of Doolittle's free-volume formula from first principles that exactly correlates with that of the energy activated rate theory of Eyring—the only difference being that energy constraints and statistical distributions are replaced with equivalent free-volume constraints and distributions. However it should perhaps be pointed out that, though Cohen here describes the process of diffusion, or equivalently viscosity, as arising from pure entropy fluctuations, the derivation that follows does not preclude free-volume being created or destroyed as the result of energetic processes. Generally speaking, a substance whose constituent molecules have sufficient average energy, $k_b T$, to exist in a liquid state will manifest local thermal fluctuations on the order of the heat of fusion giving rise to the creation of vacancies within the liquid—*i.e.* free-volume. Thus, arguably, one should not view free-volume theory as an alternative to an energy activated rate theory, but rather simply a more natural means of expressing such a theory *via* circumventing the need to parameterize the complicated distribution of temperature dependent activated processes that may give rise to the creation of free-volume.

The total free-volume of a bulk liquid, having average thermal energy $k_b T$, will undoubt-

edly fluctuate, however the expectation value of that total free-volume will be as solidly defined as any thermodynamic parameter. Cohen defines the average free-volume per molecule, analogous to the average thermal energy per molecule defined by Eyring, as $\nu_f = V_f/N$, where V_f is the total free-volume and N is the total number of particles in the bulk. Though it was reasonable for Eyring to assume the total energy of an isolated system is constant, there certainly will be fluctuations in the total free-volume, V_f, even in an isolated system, however such fluctuations will be of negligible magnitude and thus would likely prove inconsequential to any derivation. In any case such fluctuations remain irrelevant as V_f is either assumed to be the time-average value of the total free-volume or, assuming ergodicity, ν_f is simply defined as the ensemble average of local free-volume over the entirety of the bulk at any given instant. All of that being sufficiently addressed, we can move on to the derivation of Doolittle's free-volume model derived in a similar fashion as by Cohen.

4.2.1 Derivation of a Free Volume model

Let us consider a bulk fluid of finite volume. In general, there may be an infinite number of ways in which nearest neighbors to any particular molecule may be arranged within the fluid. However there is going to be a net free-volume, V_f, within the bulk at any given time which can be thought to be divided among N constituent molecular (or sub-molecular) units. Consider a set of small yet discrete increments of free-volume, $\Delta\nu$, that may be attributed to the ith molecule within the bulk as illustrated in Fig. 4.3.

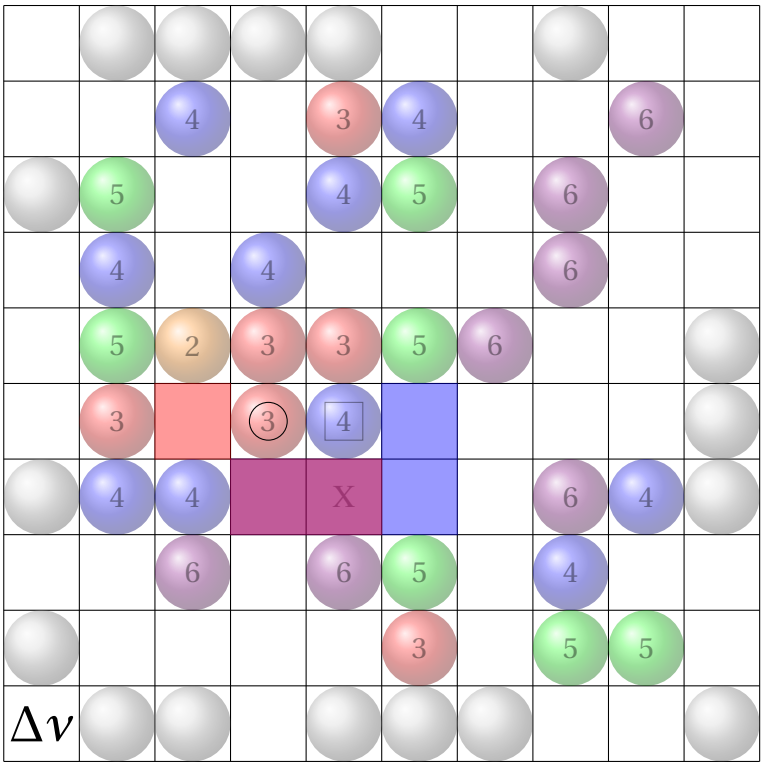


Figure 4.3: Here we have depicted a two dimensional conceptual illustration of free-volume in which the space is divided into small discrete volumes, Δv. Each particle is labeled with its number of associated free-volume increments, i, and color is used to indicate particles which share a common free-volume, v_i. The free-volumes associated with the circled and squared particles is indicated above *via* red and blue shading respectively where $v_\circ = 3\Delta v$ and $v_\square = 4\Delta v$. Thus, as asserted in Eq. 4.6, the total free-volume within the bulk will be given as $V_f = \gamma \sum n_i v_i$ where we are summing over all possible values of v_i, in this case $0 \leq v_i \leq 8\Delta v$. Note the need for the γ-factor to account for the overlapping free-volume indicated by the magenta region above. In this instance, $\gamma \approx 1/4$ implying that any given Δv will be shared with four adjacent particles on average, as happens to be the case for the magenta square marked with an X.

The ith molecule within the bulk may have an associated free-volume $v_i = i\Delta v$, where i is just an integer coefficient, such that $0 \leq v_i \leq V_f$ subject to the following constraints:

$$\begin{aligned} N &= \sum_i n_i \\ V_f &= \gamma \sum_i n_i v_i \end{aligned} \tag{4.6}$$

In the sum over all possible values of free-volume, v_i is the number of molecules having an as-

sociated free-volume of v_i, and $0.5 \leq \gamma \leq 1$ is a geometric factor to account for overlapping free-volume attributed to adjacent molecules. It should be noted that the lower limit of $\gamma = 0.5$ arises from the fact that a typical liquid will only share, on average, no more than half of its surrounding free-volume with at most one adjacent molecule. However, the illustration provided in Fig. 4.3 would, being excessively tightly packed, have a value of γ significantly less than one half. There is no physical reason why γ can not be less than one half and, in fact, for tightly packed granular systems almost certainly will be.

Getting back to our derivation, from here we will assume, similarly to Boltzmann, that any arrangement of free-volume among constituent molecules within the bulk is equally probable so long as the constraints of Eq. 4.6 above are met. Thus the odds of finding the free-volume distributed in any particular arrangement among N molecules such that n_i molecules have an associated free-volume v_i will be given by the total number of permutations of n_i as follows:

$$W = \frac{N!}{\prod_i (n_i!)} \tag{4.7}$$

From here Lagrange multipliers are implemented to find the most probable permutation. Just as is typical to derive the classic Boltzmann distribution we will, for mathematical convenience, maximize $\ln W$ invoking the Stirling approximation.

$$N! \approx \left(\frac{N}{e}\right)^N \tag{4.8}$$

Thus:

$$\begin{aligned} \ln[W] &= \ln[N!] - \sum_i \ln[n_i!] \\ &\approx N(\ln[N] - 1) - \sum_i \ln[n_i!] \end{aligned} \tag{4.9}$$

Note that we can now invoke the Stirling approximation on the second term of Eq. 4.9 resulting from the denominator in Eq. 4.7 and this will be done shortly. One might argue that there is no assurance the number of molecules, n_i, within the bulk having any arbitrary associated free-volume, v_i, is always sufficiently large as to justify the Stirling approximation. However, as is so often the case with thermodynamic quantities, the distribution of volumes, v_i, are distributed sharply around their average value. Therefor, though there certainly will be small values of n_i, neglecting them entirely will have negligible effect on the total free-volume of the system. Now, to maximize Eq. 4.9, we implement the Lagrange multipliers λ and μ as follows:

$$\delta \ln[W] - \lambda \delta N - \mu \delta V_f = 0 \tag{4.10}$$

First we consider $\delta \ln[W]$:

$$\begin{aligned} \delta \ln[W] &= \delta[N(\ln[N] - 1)] - \sum_i \delta[n_i(\ln[n_i] - 1)] \\ &= 0 - \sum_i \ln[n_i]\, \delta n_i - \sum_i \frac{n_i}{n_i} \delta n_i + \sum_i \delta n_i \\ &= -\sum_i \ln[n_i]\, \delta n_i \end{aligned} \tag{4.11}$$

Next we consider the constraints $\lambda\delta N$ and $\mu\delta V_f$:

$$\begin{aligned}\lambda\delta N &= \lambda\sum_i \delta n_i \\ \mu\delta V_f &= \mu\gamma\sum_i v_i\delta n_i\end{aligned} \tag{4.12}$$

Eq. 4.10 then becomes:

$$-\sum_i \left(\ln[n_i] + \lambda + \mu\gamma v_i\right)\delta n_i = 0 \tag{4.13}$$

As λ and μ are arbitrary Lagrange multipliers, asserting the sum to be zero implies that each term must be zero. From this we find the distribution of the number of particles, n_i, having associated free-volume, v_i which maximize the permutations of Eq. 4.7 corresponding the most probable distribution of free-volume within the bulk liquid as follows:

$$\begin{aligned}\ln[n_i] &= -\lambda - \mu\gamma v_i \\ n_i &= e^{-(\lambda+\mu\gamma v_i)}\end{aligned} \tag{4.14}$$

Thus the maximum permutations of Eq. 4.9 is found to be as follows:

$$\begin{aligned}\ln[W_{max}] &\approx N(\ln[N]-1) - \sum_i n_i(\ln[n_i]-1) \\ &= N(\ln[N]-1) - \sum_i e^{-(\lambda+\mu\gamma v_i)}\left(-\lambda-\mu\gamma v_i - 1\right) \\ &= N\ln[N] - N + \lambda\sum_i n_i + \mu\gamma\sum_i n_i v_i + \sum_i n_i \\ &= N\ln[N] + \lambda\sum_i n_i + \mu\gamma\sum_i n_i v_i\end{aligned} \tag{4.15}$$

Returning to the results of Eq. 4.14 and the constraints of Eq. 4.6 we can now express the total free-volume within a bulk liquid as follows:

$$\begin{aligned} V_f &= \gamma \sum_i n_i v_i \\ &= \gamma \sum_i v_i e^{-(\lambda+\mu\gamma v_i)} \end{aligned} \tag{4.16}$$

In the limit where the difference in free-volume among the discrete bins, thus far enumerated as v_i, becomes infinitesimal we may express the total free-volume of the bulk, V_f, as a continuous integral over an infinite number of infinitesimal bins.

$$V_f = \gamma \int_0^\infty v e^{-(\lambda+\mu\gamma v)} \delta v = \frac{e^{-\lambda}}{\mu^2 \gamma} \tag{4.17}$$

Solving this for the Lagrange multiplier, λ, yields the following:

$$\lambda = -\ln\left[\mu^2 \gamma V_f\right] \tag{4.18}$$

The constraints of Eq. 4.6 on the total number of molecules in the bulk can similarly be assessed.

$$\begin{aligned} N = \sum_i n_i &= \sum_i e^{\lambda+\mu\gamma v_i} \\ &= \int_0^\infty e^{-(\lambda+\mu\gamma v)} \delta v = \frac{e^{-\lambda}}{\mu\gamma} \end{aligned} \tag{4.19}$$

From here the average free energy per unit molecule, or the ensemble average $< v_f >$, is found to be as follows:

$$v_f = \frac{V_f}{N} = \frac{\left(\frac{e^{-\lambda}}{\mu^2\gamma}\right)}{\left(\frac{e^{-\lambda}}{\mu\gamma}\right)} = \frac{1}{\mu} \tag{4.20}$$

Returning to our previous solution for λ in Eq. 4.18 above we find the following:

$$\begin{aligned}\lambda &= -\ln\left[\mu^2 \gamma V_f\right] = -\ln\left[\frac{\gamma V_f}{v_f^2}\right] = -\ln\left[\frac{\gamma v_f N}{v_f^2}\right] \\ &= -\ln\left[\frac{\gamma}{v_f}\right] - \ln[N]\end{aligned} \tag{4.21}$$

In the limit where the difference in free-volume among the discrete bins becomes infinitesimal, we may express Eq. 4.14 as follows:

$$\begin{aligned}\delta n &= e^{-(\lambda+\mu\gamma v)} \\ &= \exp\left[-\left(-\ln\left[\frac{\gamma}{v_f}\right] - \ln[N] + \frac{\gamma v}{v_f}\right)\right] \\ &= N\left(\frac{\gamma}{v_f}\right) e^{\frac{-\gamma v}{v_f}}\end{aligned} \tag{4.22}$$

Thus we can express the number density of particles, $\rho(v)$, having an associated free-volume, v to be described as follows:

$$\rho(v) = \frac{\delta n}{N} = \left(\frac{\gamma}{v_f}\right) e^{-\frac{\gamma v}{v_f}} \tag{4.23}$$

where the normalization condition $\int_0^\infty \rho(v)\,\delta v = 1$ is naturally met. As previously mentioned, Cohen and Turnbull approaches this derivation assuming that self diffusion is not an energetically activated process, but rather is a purely structurally entropic process in which free-volume within the bulk can be thought of as being randomly redistributed *via* Brownian type motion along equipotential paths through the energy landscape. This is actually a reasonable assumption to the extent that a liquid is well approximated by a spherically symmetric force field where the intermolecular potential landscape is found to be relatively flat [90]. This assumption will hold true for almost any liquid in the high temperature limit where the potential energy landscape is not really felt by

the constituent molecules. At lower temperatures, certainly as one approaches the glass transition, energy activated process may play a larger role in the active creation of free-volume however the considerations of Cohen and Turnbull remain valid. A diffusion event is going to be inherently dependent on the existence of free-volume sufficient for molecular egress regardless of how that free-volume comes to be present. This makes free-volume arguably a more fundamental consideration in regards to molecular self diffusion than is any energetic rate theory. Thus ignoring for the moment how a constituent particle might find itself in the presence of free-volume v, it may be considered to be moving across a cage of nearest neighbors having a diameter of $\alpha(v)$ that is a function of its associated free-volume, v, and traveling with a velocity u. Cohen and Turnbull then reasonably assert that diffusion proceeds in much the same way as it does within crystals and any molecule's contribution to the diffusion constant will be given as follows [91]:

$$D(v) = \zeta u \alpha(v), \tag{4.24}$$

where ζ is just a geometric factor. As the distribution of free-volume is provided by the results of Eq. 4.23, the sum of all contributions will yield the following constant of self diffusion:

$$D = \int_{v_m}^{\infty} D(v)\, \rho(v)\, \delta v, \tag{4.25}$$

where v_m is the minimum value of free-volume needed to allow for any molecular egress and for which any contribution to diffusion can occur. The value of v_m is generally going to be at least an order of magnitude greater than the average available free volume per molecule, $< v_f >$. As such, Eq. 4.24 is going to be a slowly evolving function on the timescales of any particular diffusion

event. Thus Eq. 4.25 can be simplified as follows:

$$\begin{aligned} D &= \zeta u \alpha(v_m) \int_{v_m}^{\infty} \left(\frac{\gamma}{v_f}\right) e^{-\frac{\gamma v}{v_f}} \delta v \\ &= \zeta u \alpha(v_m) e^{-\frac{\gamma v_m}{v_f}} \end{aligned} \tag{4.26}$$

As Cohen points out, this is equivalent in form to the phenomenological model proposed by Doolittle [84]. In the case of "hard sphere" liquids, $\alpha(v_m)$ will be on the order of the spherical diameter and $u \propto \sqrt{k_b T}$ implies that even the simplest liquids must undergo a glass transition if cooled sufficiently in the absence of crystallization. Of course this model is certainly not limited to this $T^{1/2}$ temperature dependence. Eyring's general concept of a rate theory of diffusion, even extended to include any arbitrary temperature dependence or distribution of activation energies suggested by the broad applicability of the VTF equation, will here give rise to a temperature and pressure dependence of the average free-volume, v_f, related to the coefficient of thermal expansion. However, though energy activated processes may determine the average free-volume within a bulk, it is the existence of this free-volume, not the energy associated with its creation, that is fundamental to diffusion, viscosity, and relaxation processes. Thus Eq. 4.26 can be generalized as follows:

$$D(T,P) = \zeta u \alpha(v_m) e^{\frac{-\gamma v_m}{v(T,P) - v_m}} \tag{4.27}$$

The diffusion of spherical particles dispersed in a viscous medium described by Einstein and Stokes *via* Eq. 4.5 provided above can, to first approximation, be assumed to apply to self-diffusion of non-associated pure liquids well described by the flat potential energy landscape of hard spheres. It is physically reasonable to assume that, even when the potential energy landscape is significantly more complex, the diffusion rate will still remain proportional to fluidity or, equivalently, inversely

proportional to the viscosity. Thus one may arrive at the following simplified model for viscosity as a function of free-volume:

$$\eta(T,P) = C(T)\, e^{\frac{\gamma v_m}{v(T,P)-v_m}} \tag{4.28}$$

Here $C(T)$ is just some temperature dependent constant that will correspond to the viscosity at infinite volume which will be an exceedingly small value. It is worthy of note that viscosity really should trend to zero as free-volume approaches infinity. The fact that this model does not predict this implies that it cannot accurately describe viscosity over an infinite dynamic range regardless of the functional dependence of the parameters. However it does show remarkable agreement for most liquids over a very large dynamic range. As such it is common to expand this model around its value at ambient pressure as follows:

$$\eta(T,P) = \eta_0(T)\, e^{\frac{\gamma v_m}{v_m - v_0(T)}} e^{\frac{\gamma v_m}{v(T,P)-v_m}} \tag{4.29}$$

Here $\eta_0(T)$ and $v_0(T)$ are the temperature dependent viscosity and specific volume respectively at ambient pressure. As this model is generally applied to systems at or above ambient pressure, the physical incongruity regarding the non-zero infinite free-volume viscosity is not of significant concern. As implied by the super-exponential form of Eq. 4.29, these viscosities will increase by many orders of magnitude quite rapidly as either pressure is increased or temperature decreased. As such, it is very common to express Eq. 4.29 in logarithmic form as follows:

$$\ln\left[\frac{\eta(T,P)}{\eta_0(T)}\right] = \gamma v_m \left(\frac{1}{v(T,P)-v_m} - \frac{1}{v_0(T)-v_m}\right) \tag{4.30}$$

Some variation of Eq. 4.30 has been used extensively, with various functional dependencies of

parameters on temperature and pressure, throughout the literature to fit a large variety of associated and non-associated systems, including polymers, with high efficacy for viscosities ranging over several orders of magnitude [45, 92, 93, 94].

4.3 Reduction of extrapolation error in fragility measurements from pressure-limited data *via* implementation of TGP

Viscosity measurements at elevated pressure for glycerol up to about 4 kPa, first obtained by McDuffie and Kelly, was reported in the Journal of Chemical Physics in 1964 [95]. This pressure range was significantly extended by just under an order of magnitude *via* the work of Cook *et al*. in 1992 implementing a diamond anvil cell viscometer attached to a centrifuge [45, 51]. The data of Cook *et al*. were then used to investigate the pressure dependence of fragility over this domain and appeared to show a trend of steadily increasing fragility from approximately $m \approx 55$ ($m^* = 3.2$) at ambient pressure up to $m \approx 170$ ($m^* = 13.6$) at 3 GPa as shown in Fig. 4.4. Note that, though the definition of fragility provided by Eq. 3.2 is arguably more physically meaningful, it may be convenient for the sake of literature comparisons to express some fragilities in terms of the more classical definition of Eq. 3.1. Thus to alleviate any confusion we will adopt a consistent notation of m and m^* to indicate the classical fragility of Eq. 3.1 and the scaled fragility of Eq. 3.2 respectively. Returning now to Fig. 4.4, note the excessive error associated with these fragility calculations arising from the necessity to extrapolate the VTF model over several orders of magnitude up to the glass transition. The fact that this trend towards extreme fragility is not in satisfactory agreement with subsequent publications, particularly at elevated pressures, and the excessive error implying that the fragility measurements at high pressure are essentially meaningless has led to the Cook *et al*. data being generally under-emphasized. Paluch *et al*. drew attention to these concerns

as well as the fact that several of the viscosity measurements were made near the melt temperature, T_m, where VTF model extrapolations are known to be unreliable [73]. The fundamental limitation for Cook *et al*. was that rolling ball viscometery, even when implemented within a centrifuge, can only measure viscosities up to on the order of $10^9 cP$—*i.e*. as much as six orders of magnitude away from the glass transition at which fragility is assessed. The data of Cook *et al*., even with these concerns, does show strong support for an increasing trend in fragility with pressure up to around 2 GPa, which is not unexpected for a hydrogen bonding liquid where increasing fragility can be explained as arising from the diminished free-volume inhibiting hydrogen bonding. Beyond approximately 2 GPa, the minimal uncertainty in these fragilities obtained from propagating the errors in the fit parameters of the VTF equation seems to render the actual values of fragility statistically meaningless. Though the assessment of Cook *et al*., that the fragility continues to increase with pressure beyond this point, my have been somewhat compelling given that the measurements continue along the same well defined linear regression and their standard Angel plots, shown here as modified Angel plots and provided on the right side of Fig. 4.4, seem to consistently show increasing curvature as pressure increases, the higher pressure data is dubious to say the least, not just due to the excessive error but also in that, having reached a pressure of 3.0 GPa, glycerol has become about as fragile as a liquid can become. This linear trend in fragility certainly cannot continue indefinitely yet, at a pressure of 3 GPa, there is no indication of departing from said trend. Furthermore, it is worth noting that the 3 GPa fragility shown in Fig. 4.4 is based on only four data points and, though it is less evident in our modified Angel plot of Fig. 4.4 than it would be in a classical Angel plot, does not converge to a similar infinite temperature viscosity as do data obtained at lower pressures. It is therefor not surprising that this data might yield an excessively high fragility while having error bars that include negative values of fragility—statistically speaking this

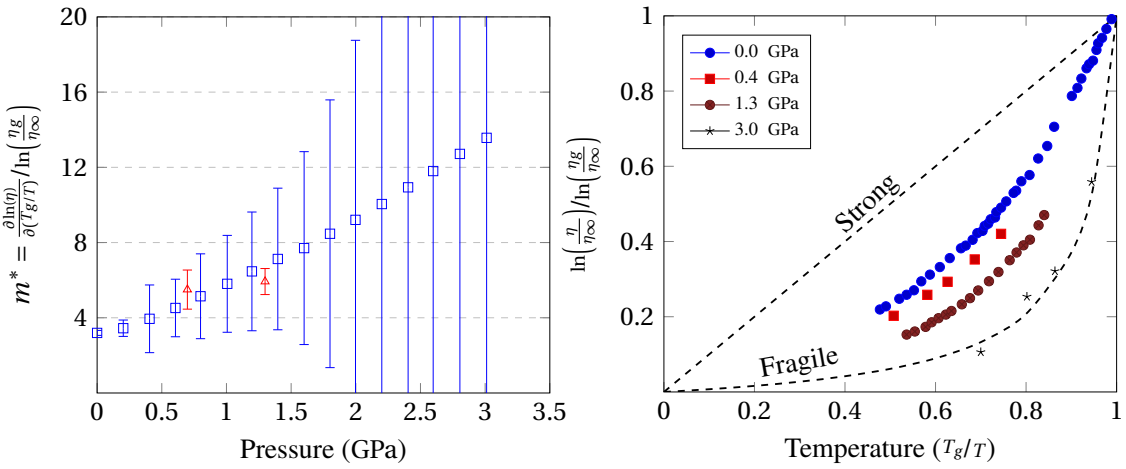


Figure 4.4: Left: Cook's calculations for the isobaric fragility of glycerol as a function of pressure with the glass transition defined at 10^{15} centipoise. Note that these values have been scaled to reflect our redefinition of fragility as defined in Eq. 3.2. This scaling of fragility is trivially accomplished as Cook conveniently provides his fit parameters. Right: Angel plots for Cook's 0.4, 1.3, and 3.0 GPa corrected isobars [51]. The 0.0 GPa isobar data is from Laughlin's 1969 Ph.D. thesis although all values plotted here were directly extracted from Cook's published figures [96].

is a meaningless measurement. In truth, this point should likely have been omitted from the analysis of Cook *et al*. as it relies on only three reasonably reliable data points—the 50.0 °C data point requiring a free-volume model extrapolation of almost 1 GPa. However omitting the 50.0 °C data point would leave only the 22.5, 75.0, and 125.0 °C, two of which still require slight extrapolation to 3 GPa, and require one to perform a non-linear model fit with only three data points providing constraint to a three parameter VTF model—though technically possible, an arguably meaningless endeavor. This was the root cause of their exploding fragility estimations—simply a lack of data to properly constrain the VTF model fits and extreme extrapolation over orders of magnitude as none of this data is within the vicinity of the glass transition.

More recent work has made a well supported case that the fragility of glycerol increases with pressure due to the inhibiting of hydrogen bonding up to approximately 2 GPa at which point it levels off becoming relatively constant. It would be reasonable to assert that the data of Cook *et*

al., as analyzed here, equally well supports this general trend of fragility. Arguably, the analysis supports a general increasing trend in fragility as pressure increases up to approximately 2 GPa even if little confidence can be put into the exact fragility values up to the point that the error begins to diverge rapidly. Beyond 2 GPa little can be confidently asserted from Cook's analysis however it does not contradict a constant fragility when the propagated errors are accounted for.

In order to assess the validity of Cook's high pressure data and the associated fragility, it is necessary to diminish these large error bars. This can be accomplished by addressing one of his two primary underlying issues—extreme data extrapolation. The TGP process developed by our group provides a direct measurement of the glass transition as a function of pressure allowing for a re-analysis of the Cook data without the need for extrapolation of the VTF model over so many orders of magnitude. This TGP data is provided in App. C (Tbl. B.1) along with the associated Anderson-Anderson regression fit. Using these values for the glass transition pressure on any given isotherm and assuming this transition to occur on an isochronal curve associated with a viscosity of $\eta = 10^{15} cps$, the data Cook *et al*. were refit to a classical VTF model.

Note that most of the data of Cook *et al*. were obtained over isotherms at varying pressures. In order to calculate fragility, they fit five sets of isothermal viscosity data to the classic free-volume model of Eq. 4.30. The Tait equation of state, fit to density data provided by Bridgman, was implemented to convert pressure to volume [97]. More details regarding the implementation of the Tait equation of state for glycerol are provided in App. E as Cook *et al*. and this work use this free-volume model to obtain (mostly) interpolated isobaric data as true isobaric data is difficult to obtain experimentally given that pressures will vary significantly within the DAC under thermal expansion. Thus it is this interpolated isobaric data that is used in the fits to the VTF model for fragility analysis.

Fig. 4.5 (left) provides a scaled Angel plot both excluding the TGP glass transition point (solid) as well as the refit passing through the TGP point (dashed) for a 1.3 GPa isobar. Note that the lower five data points associated with both the dashed and solid regressions are in fact the same data from Cook *et al*. —scaled according to their associated VTF fits and glass transition temperatures. The TGP point is also provided and is, on a scaled Angel plot, located at (0,0) by definition. It is immediately clear that both adequately describe Cook's data however omitting the TGP data leads to a significant over-estimation of fragility. This is associated with an over-estimation of the glass transition temperature. Fig. 4.5 (right) shows eight isobaric data sets, consisting of four to five data points each, ranging left to right from 0 GPa to 3 GPa which are obtained from the before mentioned free-volume fits to Cook's published isothermal data. The VTF fits to this data exclude the TGP point and thus help to demonstrate the error that arises from the extrapolation of the VTF equation. Notice that at the lowest pressures (left most curves) there is a systematic under-estimation of the glass transition temperature due to model extrapolation. However, as we move up in pressure (to the right of the figure) we begin to increasingly over-estimate the glass transition temperature and thus over-estimate the fragility. This trend is what is giving rise to Cook's exploding fragility measurements.

From this reassessment of the data of Cook *et al*., we see a very different behavior of fragility as a function of pressure. Though, as demonstrated in Fig. 4.7, the calculations of fragility attained from the Cook *et al*. data enhanced *via* the TGP isochronal data were still not found to be in perfect agreement with more recent measurements, the general trend is found to be somewhat congruent [66, 98]. It should be noted that the actual values of fragility, apart from their trend with pressure, is somewhat dependent on the viscosity at the glass transition. We have chosen to assume the TGP transition correlates with a viscosity of $\approx 10^{15} cps$. A quick assessment of Fig. 4.6

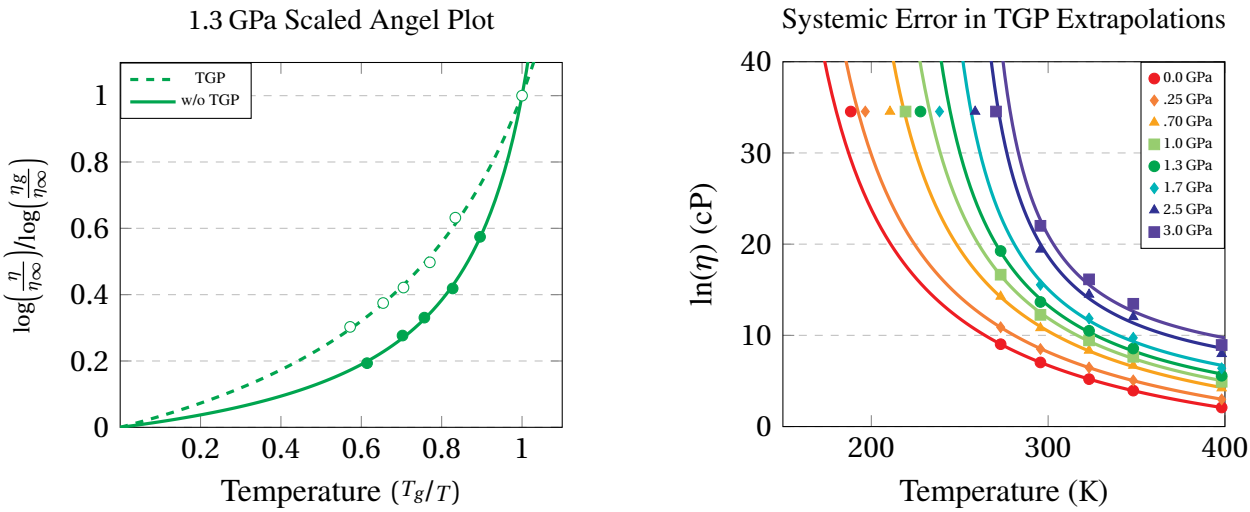


Figure 4.5: Left: A scaled Angel plot of the data of Cook *et al*. clearly demonstrates the over-estimation of fragility occurring at 1.3 GPa due to extreme extrapolation up to the glass transition that is rectified *via* the inclusion of TGP isochronal data. The same data is provided for both the solid and dotted VTF fits—scaled accordingly to their fit parameters and the glass transition temperatures they predict. Right: Isobaric data from Cook *et al*. showing the general trend of under-estimation followed by increasing over-estimation of the glass transition temperature with increasing pressure. The TGP method provides the isochronal data at the glass transition [51].

might lead one to conclude that fragility is initially decreasing from ambient pressure up to approximately 1.5 GPa however, as evidenced by the error of the VTF fit parameters propagated through the fragility calculation as well as keeping in mind that our confidence in these values is still limited *via* insufficient data particularly in the region of the glass transition, this trend should not be assumed statistically significant—the fragility is relatively constant up to around 1.3 GPa. It is worth noting here that, though Cook's viscosity data was predominately taken along isotherms as this was experimentally favorable, he did manage to obtain two pseudo-isobaric runs at 0.7 and 1.3 GPa on which he then used interpolation between free-volume fits to his isothermal data to correct for slight deviations of pressure. Though this process was extremely time intensive, it did provide him with two isobaric runs over a large temperature range addressing the issue of insufficient data points. It should be noted, however, that these data are still limited to well below the glass transi-

tion. Due to the larger number of data points associated with these pseudo-isobaric runs, the error associated with the fragility estimations are significantly smaller and these two points are included (△) in the left side of Fig. 4.4. These two higher-confidence points would alone imply a potentially constant fragility below 1.5 GPa—albeit at a significantly higher scaled fragility ($m^* \approx 5.7$). Beyond 1.5 GPa the fragility steadily increases by $\Delta m^* \approx 7.8$ and then remain approximately constant within uncertainty beyond 2.5 to 3 GPa before rapidly diverging due to extrapolation errors again. This general trend of increasing fragility is, in the loosest sense, in "agreement" with subsequent published works as will be demonstrated. This leads to the conclusion that the viscosity data of Cook *et al*. is perhaps significantly more reliable than may have been assumed and much of its error, as perhaps expected, is predominately the result of these extrapolation issues. Our method of employing TGP data to diminish this extrapolation error significantly decreases the error associated with these fragility estimations however it is still limited—particularly at elevated pressures. More data is needed in the region of interest to further define the slope at the glass transition. It is possible that additional TGP runs with longer or shorter annealing periods could provide this however in the absence of a solid equation of state at high pressure and thermodynamic scaling to provide the effective viscosity at the glass transition to higher precision than the assumed 10^{15}cP, this is not yet a viable means of defining the slope near the glass transition. Other methods such as photon correlation spectroscopy and dielectric loss spectroscopy certainly exist and there is no shortage of data for glycerol dynamics within the literature around the vicinity of the glass transition, though generally limited to cryogenic measurements at low pressures. If the issues with these fragility calculations do in fact stem predominately from the lack of data in critical regions rather than the efficacy of the data itself, then supplementation of their data with other literature data in critical regions should allow accurate determinations of fragility as a function of pressure. Glyc-

erol data taken by several different groups extend over different pressure and temperature regions and include both viscosity and α-relaxation measurements, making glycerol a prime candidate for a surface fragility analysis, similar to what will also be done for cumene in Ch. 5.

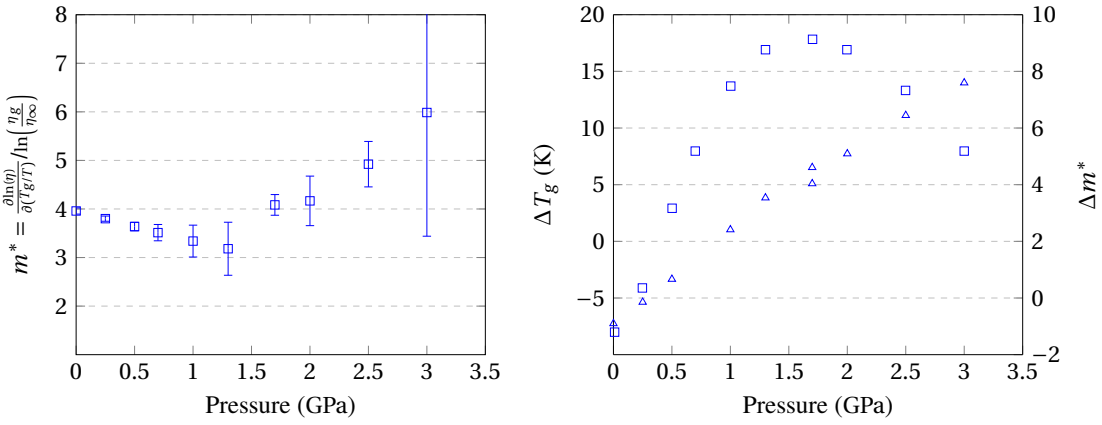


Figure 4.6: Left: Scaled fragility as a function of pressure derived from classic VTF fits to the data of Cook *et al*. with the glass transition point included as determined *via* our TGP isochron. Right: The error in terms of under or over-estimation of the glass transition temperature, ΔT_g, due to extreme extrapolation of the VTF model as demonstrated in Fig. 4.5, is provided (□) along with error in scaled fragility (△) calculated from these extrapolated models.

The right side of Fig. 4.6 shows the under or over estimation of the glass transition temperature as a function of pressure derived from the classic VTF fits to the data of Cook *et al*. without the TGP isochronal data. The initial under-estimation of the glass transition temperature for the two lowest pressures is followed by a trend of increasing over-estimation up to about 1.7 GPa where we continue to over-estimate the transition temperature, albeit to an increasingly lesser degree. However the error in fragility associated with this over-estimation in glass transition temperature continues to increase in a fairly linear fashion leading to the excessively high fragilities at increased pressure observed by Cook *et al*. in Fig. 4.4.

To add further validation to Cooks data at lower pressures, the same analysis method of implementing the TGP data point to constrain the VTF fits in the vicinity of the glass transition was

applied to the rolling ball data obtained by McDuffie and Kelly and the associated fragilities calculated [95]. Though these fragility values were much more in line with other works implementing dynamic light scattering (Brodin [99]), and dielectric spectroscopy (Reiser [98], Pronin [66], and McDuffie [95]), they all follow the same general trend of constant fragility below 1 GPa. The increasing trend mentioned by Pronin and Reiser is simply not statistically significant. This seems to suggest a strong increase in fragility does not occur until beyond 1 GPa. This may be due to the general expected trend of decreasing fragility with increasing pressure is being approximately balanced by the early onset of hydrogen bond breaking giving rise to a relatively constant fragility in this range.

4.4 Structural Relaxation Surface

Experimentally, it is often difficult to obtain truly isobaric data at high pressure. When a fluid is confined within a pressure vessel or diamond–anvil cell, changes in temperature inevitably induce changes in pressure due to thermal expansion. In his PCS measurements, Cook went to considerable lengths to interpolate between data points in order to construct pseudo–isobaric data runs. Of considerable utility, therefore, would be a framework capable of fitting relaxation data acquired over broad ranges of temperature and pressure (or specific volume) without requiring strict isobaric or, for that matter isothermal, constraints. This motivates the development of a continuous structural–relaxation surface.

To this end, the Andersson–Andersson constrained surface model provided in App. A is employed, extending the classic isobaric Vogel–Tammann–Fulcher (VTF) formalism to account for pressure dependence through polynomial expansions of pressure–dependent parameters. A more rigorous mathematical development of this model is presented in Sec. 5.4.2. Here, only the

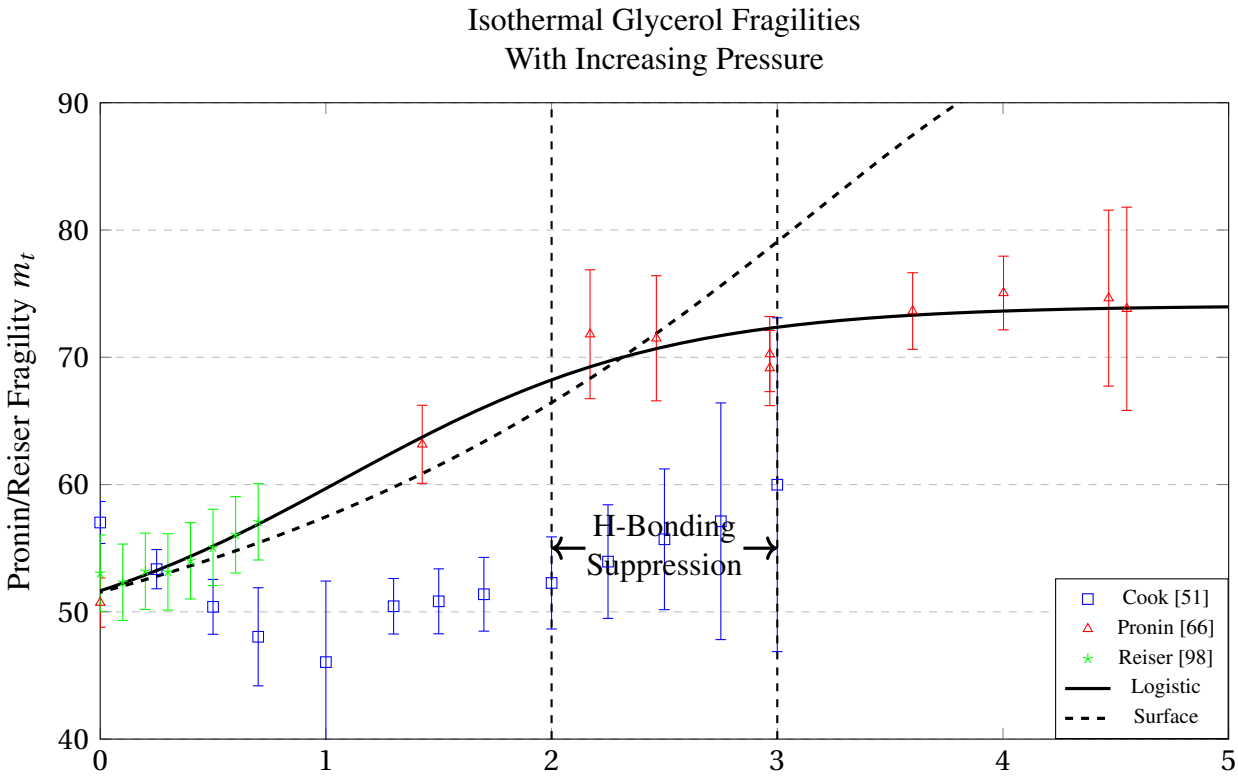


Figure 4.7: The fragility measurements calculated from classic VTF fits to Cook's viscosity data corrected *via* the inclusion of our TGP glass transition point (□) is shown above to be no longer wholly incongruent with more recent work. A Logistic fit to Pronin (△) and Reiser's (⋆) data describes their fragility measurements well with the increase in fragility seeming to saturate between 2 and 3 GPa as originally asserted by Paluch [73] and Pawlus [75]. The Cook fragilities retain relatively large uncertainties, thus cannot reflect accurately the fine behavior of fragility as a function of pressure, yet do seem to reflect this fragility increase between 2 and 3 GPa. A reasonable trend-line is provided that falls within all error bars.

essential structure of the surface and the physical constraints imposed upon it are discussed.

For most liquids, a substantial body of high–quality ambient–pressure data exists in the literature, obtained using a variety of experimental probes of structural relaxation and viscosity, owing to the relative experimental accessibility of this thermodynamic region—glycerol being no exception. Fig. A presents an Angell plot of all atmospheric–pressure data collected for this work, together with a fit to the classic isobaric VTF model. This atmospheric isobar serves as a primary constraint for the relaxation surface.

It should be noted that the Cook data have been excluded from this atmospheric–pressure constraint. The TGP point, shown on the same Angell plot at a location corresponding to $\tau = 220$ s as determined from thermodynamic scaling of the low–pressure TGP data in Fig. 4.5, was likewise not included in the fit, though it exhibits excellent agreement with the remaining data.

At the time the atmospheric data were fit, the degree to which the intercept method of thermodynamic scaling accurately determines the isochronal value associated with the Andersson–Andersson regression of the TGP data was not yet established. One peripheral insight arising from the development of these surface models is the realization that, for liquids whose dynamics conform to thermodynamic scaling, this approach provides a robust means of identifying the precise isochrone represented by a given TGP data run. Even in cases where thermodynamic scaling may not strictly apply, the atmospheric–pressure constraint provides an independent verification of this isochronal value, as the Andersson–Andersson regression through the TGP data must intersect the atmospheric VTF isobar.

The Andersson–Andersson regression through the TGP data provides an additional constraint to be imposed on the surface construction. Specifically, the VTF expression is solved for the zero–mobility temperature T_0 in terms of the pressure–dependent glass–transition temperature $T_g(P)$ and subsequently reinserted into the VTF form, yielding Eq. A.2. Here, $T_g(P)$ is taken from the Andersson–Andersson fit to the TGP data provided in App. B. This procedure results in a relaxation surface constrained both at atmospheric pressure and along an isochrone corresponding to $\tau \approx 220$ s.

The remaining four parameters comprising the coefficient vectors $\vec{d}$ and $\vec{t}$ are treated as free parameters and are determined using a Levenberg–Marquardt optimization algorithm to conform the surface to the remaining data. Fig. 4.8 displays this constrained surface together with the Cook

data, shown for initial parameter estimates. The atmospheric isobaric constraint and the isochronal TGP constraint are indicated by solid black curves on the surface.

It is emphasized that the surface is not fit to the Cook data. Because ~~Cook~~ measured viscosity directly via rolling–ball viscometry, whereas all other data in this chapter probe the structural α–relaxation time, the decision was made to include the Cook data only for qualitative comparison. The Cook viscosity data were scaled using an appropriate high–frequency shear modulus G_∞ to permit comparison with the relaxation–time surface. The color scale in Fig. 4.8 indicates pressure and demonstrates that the overall pressure–dependent trends of the Cook data are well captured by the constrained surface.

**Next we move on an clean what follows up. We add tau data, fit the surface, and again the cook data is NOT included in the fit...just placed on the surface

Glycerol viscosity and relaxation data from several sources are shown, along with the Cook rolling ball data obtain from within a DAC, in Fig. 4.9 [99, 95, 73, 66, 98]. It is clear that data is actually in excellent agreement with other literature data further bolstering the case that the vast majority of the error within the Cook fragility measurements lies in the extreme extrapolation to the glass transition. The inclusion of the TGP isochronal data, though addressing the issue of extrapolation to a significant extent, ultimately provides insufficient accommodation for an accurate determination of the rate of vitrification with increasing pressure in the vicinity of the glass transition, T_g. More data is needed to define the slope near T_g to obtain an isobaric fragility in consensus with other contributors to the literature, *i.e.* Pronin, Paluch, *etc*.

In short, there never was any issue with the validity of Cook's data. As we will see, it is in excellent agreement with other measurements of vitrification or associated alpha relaxation behavior. It is only the lack of data to provide constraint to the VTF fit in key locations, specifically

in the vicinity of T_g, that gives rise to the incongruence of the Cook fragility estimations with literature values, *i.e.* the slope in the vicinity of T_g is poorly constrained. The addition of the TGP isochronal data is not itself sufficient to provide the necessary constraint on this slope for reliable estimations of fragility. However, as we will here demonstrate, only one additional experimentally trivial, constraint is needed for Cook's data to provide fragility values as a function of pressure in excellent agreement with the current literature consensus—an ambient isobar.

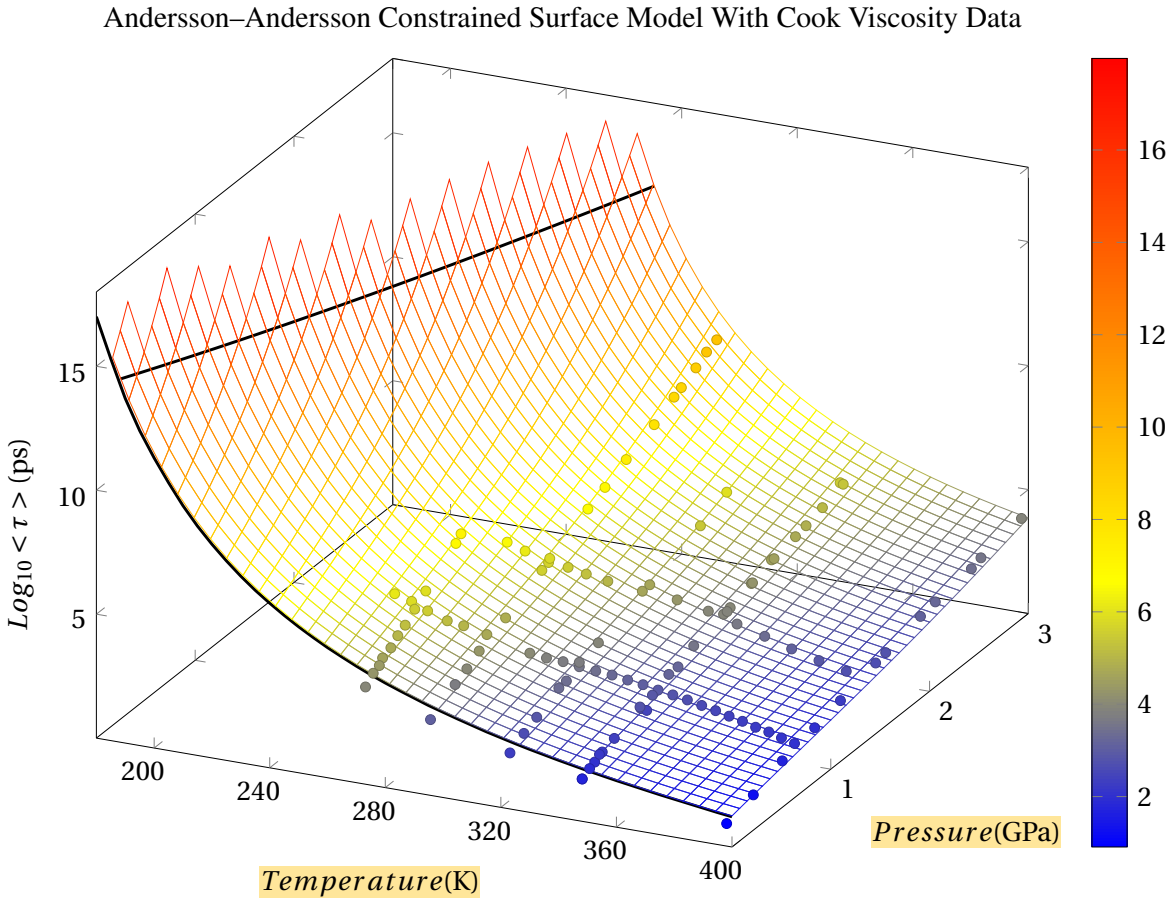


Figure 4.8: Cook's rolling ball viscosity data is here provided along with the Andersson-Andersson constrained surface model, given initial approximations for the fit parameters, described in App. A. The color scale, corresponding to pressure, facilitates visualization of how well the data conforms to the surface. The surface is extended to 3 GPa corresponding to the highest pressures for which Cook obtained data [51].

Fig. 4.8 shows all of Cook's raw viscosity data provided by reference [51] fit to the Andersson-

Andersson constrained surface model described in App. A. The Solid black isobaric line at atmospheric pressure, through which the surface is constrained, is well described in Fig. A, along with the supporting data. This 1-bar regression was also used to define the isochrone for the TGP curve—the second solid black line extending across the top of the surface in Fig. 4.8. The glycerol TGP data fit to a standard Andersson-Andersson model is provided in App. B and corresponds to an isochronal pressure-temperature relationship at $\tau \approx 10^{14}$ s through which the surface is also constrained.

This Andersson-Andersson constrained surface model, fit to the Cook viscosity data, allows for a determination of fragility as a function of pressure in similar fashion to that obtained for cumene in Fig. 5.16 previously. The resulting functional dependence of isobaric fragility is plotted as a solid black line in Fig. 4.7 above and found to be in significantly greater agreement with Pronin, Reiser, and other more recent works up to 2.5 GPa. This result, along with the excellent constrained surface fit, suggest that Cook's data could not be lacking significantly in quality.

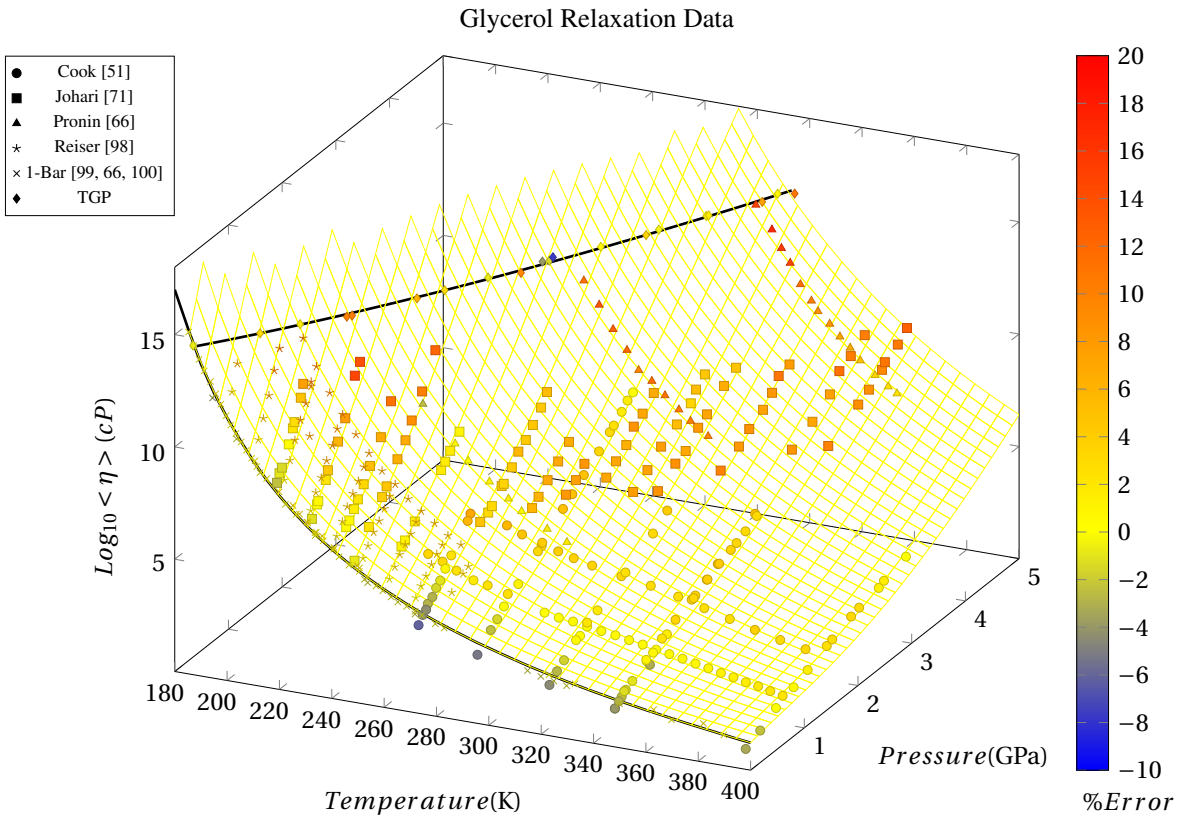


Figure 4.9: The surface shown corresponds to the model described in App. A, fit to a compilation of glycerol relaxation data from Johari and Whalley, Pronin, and Reiser. The Cook, TGP, and atmospheric data were not included in the fit, but are represented on the surface. Data points are colored by their percent deviation from the fitted surface, highlighting systematic trends in the residuals.

~~Here~~ we have the model described in App. A fit to a large selection of glycerol relaxation data from multiple contributers [71, 66, 98]. The same TGP and 1-bar constraint was imposed here as was implemented in Fig. 4.8. The color scale now represents percent deviation of the data from the surface. Note how the high pressure Pronin data begins to diverge from the surface, approaching an error as high as 20%. This data was still included in the fit as it conforms well to the surface at higher temperatures and lower viscosities—only tending away from the surface in a region where the TGP constraint does not allow the data significant influence over the fit parameters.

Fig. 4.8 shows the same Andersson-Andersson constrained surface model fit to a larger selection of literature data including Cook's rolling ball viscosity data. The same TGP and atmospheric constraints are applied. Here the Actual TGP data is provided on the surface yet not included in the fit. It is found that Cook's rolling ball viscosity data is in excellent agreement with other works including Johari, Pronin, Paluch, and Reiser thus we must conclude that the Cook data is very reliable—the issue of increasingly overestimated fragility arising predominately from extreme extrapolation [29, 73, 66, 98]. It should be noted that the four isobaric data sets extracted from Pronin's Fig. 3 in reference [66] do trend increasingly above the fit surface at higher pressures yet agree with Cook and others at lower pressures. The toroidal pressure cell used by Pronin and described in reference [101] may not have provided a hydrostatic sample to the degree that was assumed—a scenario which would certainly explain the data's increasing divergence from the TGP isochrone at higher pressures and lower temperatures as the glass transition is approached. The Pronin isobars are still in reasonable agreement with Cook's data at higher temperatures where hydrostatic conditions are assured, even at the highest pressure. However this data was included as it does not heavily effect the surface at lower pressures where it is in reasonable agreement with the rest of the data yet allows us to extend this fit surface up to 5 GPa. Thus it should be assumed that our calculated fragility above 3 GPa may have reasonable error associated with it and any fine structure is not of high confidence. It should be noted that this Pronin data could be replaced with DPCS measurements at high pressure near the glass transition just as was done with cumene. Molecular simulations may provide high pressure fast dynamics at extreme temperatures.

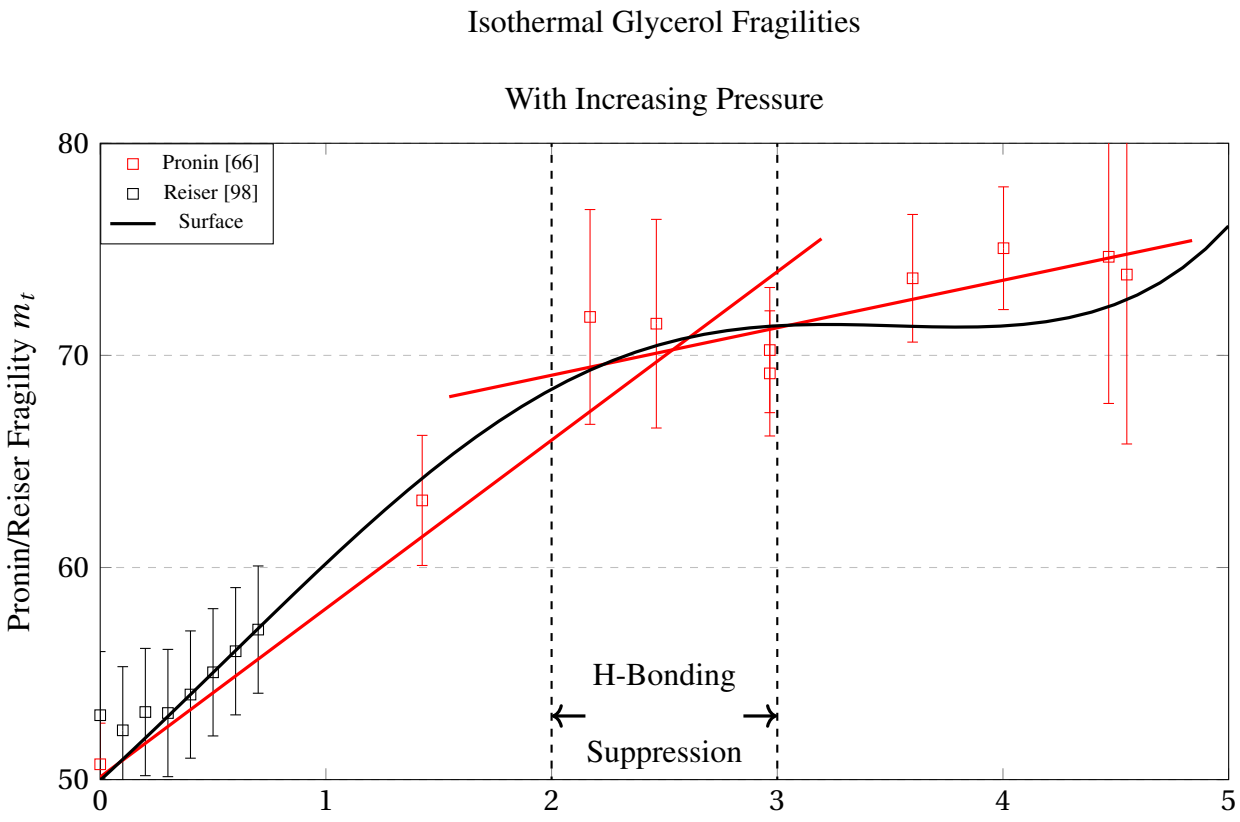


Figure 4.10: Fragility estimations from Pronin (□) and Reiser's (□) data are shown above. The pressure dependence of fragility derived from isobaric temperature derivatives of the Andersson-Andersson surface model described in App. A, fit to the data of Cook, Johari, Pronin, Paluch, and Reiser, is shown as a solid black line. The upward curve in fragility approaching 5 GPa is likely a relic due to the previously discussed issues with the Pronin data at high pressure as well as the surface being generally unconstrained by data beyond this pressure.

The calculated fragility as a function of pressure arising from the Andersson-Andersson surface model from Fig. 4.9 including data from Cook, Johari, Pronin, Paluch and Reiser is provided in Fig. 4.4—extending our fragility measurements out to 5 GPa. The red lines show Pronin's original estimation of the fragility behavior with increasing pressure as was lifted directly from his Fig. 4 in reference [66]. This result shows unequivocally Cook's rolling ball viscosity data to be in excellent agreement with other literature measurements of viscosity, η, and α-relaxation, τ. The Andersson-Andersson constrained surface model relieves us of the experimental burden of

maintaining isobaric conditions during data acquisition, something that is very difficult to accomplish in a diamond anvil cell, as well as avoids the reliance on free-volume models where a reliable equation of state may not be available or more dubious isothermal models such as the modified VTF equation.

4.5 Equation of State and Thermodynamic Scaling

In Ch. 5 Sec. 5.4.1 it is shown that the relaxation dynamics of a simple Van der Waals liquid such as cumene conforms very nicely to a single function of the form $\tau \propto \frac{1}{TV^{\gamma}}$, where γ is a constant scaling factor. This factor gamma has been shown to depend on the intermolecular potential energy landscape, $U(r)\sim\left(\frac{\sigma}{r}\right)^{3\gamma}$, where r is the spatial separation between any two molecules and σ is a material constant [102, 103]. In the case of a simple liquid such as cumene the potential is well approximated by the form of the Lennard-Jones model for which $3\gamma \approx 12$—consistent with the experimentally determined value of $\gamma \cong 4.77$. The dynamics such liquids are strongly driven by simple free-volume considerations and thus we expect thermodynamic scaling to hold with increasing pressure and that is what we have observed to be the case.

However in the case of strongly associated liquids such as glycerol, intra-molecular bonds influence the potential energy landscape. As these bonds are dependent on temperature and pressure, the energy landscape will be temperature and pressure dependent, thus one would not expect the emergence of thermodynamic scaling in these liquids. However several such liquids do exhibit thermodynamic scaling—at least over limited ranges of pressure and temperature. Glycerol is one such example, having a measured scaling parameter within the literature in the range of $1.4 \leq \gamma \leq 1.6$ [98, 104]. This lower value for γ is expected given that free-volume will play a reduced role in the dynamics of associated liquids as molecular mobility is no longer just a mater

of thermal diffusion past nearest neighbors exhibiting Van der Waals repulsion, but also will require energy to break hydrogen bonds with said neighbors. However these values for γ were all determined at relatively low pressures—on the order of 0.1 MPa. Yet we see in Fig. 4.4 a shift in the fragility behavior as a function of pressure occurring between 2 and 3 GPa which has been attributed to a suppression of hydrogen bonding. If hydrogen bonding between glycerol molecules is sufficiently suppressed at very high pressures, one might expect glycerol to behave much more like a simple Van der Waals liquid with a scaling parameter, γ, much closer to 4 as the potential energy landscape no longer is heavily influenced by these suppressed intra-molecular interactions.

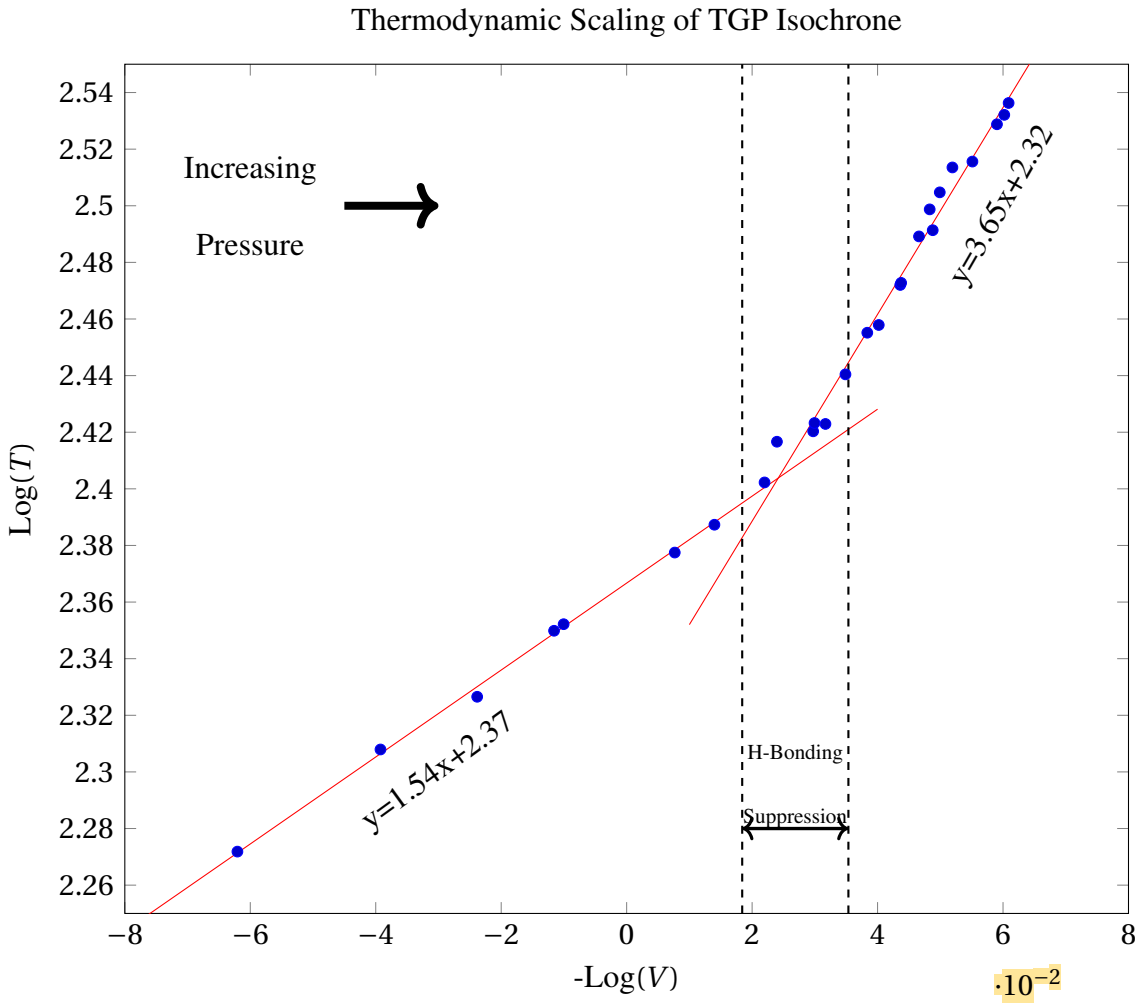


Figure 4.11: The linearized isochronal glycerol TGP data shows a transition from an H-bonded liquid at low pressure with a corresponding scaling factor of $\gamma \approxeq 1.54$ to a liquid which behaves more like one experiencing a simple Van der Waals potential at higher pressures, where H-boding is suppressed, exhibiting a scaling factor of $\gamma \approxeq 3.65$. The region between the dotted vertical lines corresponds the region indicated in Fig. 4.4 previously. The intersection of the two linear regimes is self consistent with the results of Pronin at approximately 2.5 GPa. It is also worthy of note that the average y-intercept of 2.35 implies a TGP isochrone of $\tau \approx 220$ s

In Fig. 4.5, the glycerol TGP data provided in App. B is linearized and plotted to determine the scaling parameter. The region between the dotted lines once again corresponds to that in Fig. 4.4—where pressure suppression of hydrogen bonding becomes prominent. The results are quite compelling. Two distinct regions emerge. At low pressure we find a scaling factor $\gamma = 1.54$ consistent with literature values. At high pressures, well within the region where hydrogen bonding

is assumed to be sufficiently suppressed, we find a significantly greater slope corresponding to $\gamma = 3.65$ consistent with what might be expected of a simpler Van der Waals liquid. A clear transition occurs between 2 and 3 GPa—consistent with our fragility results from our Andersson-Andersson contained surface model as well as Pronin's analysis.

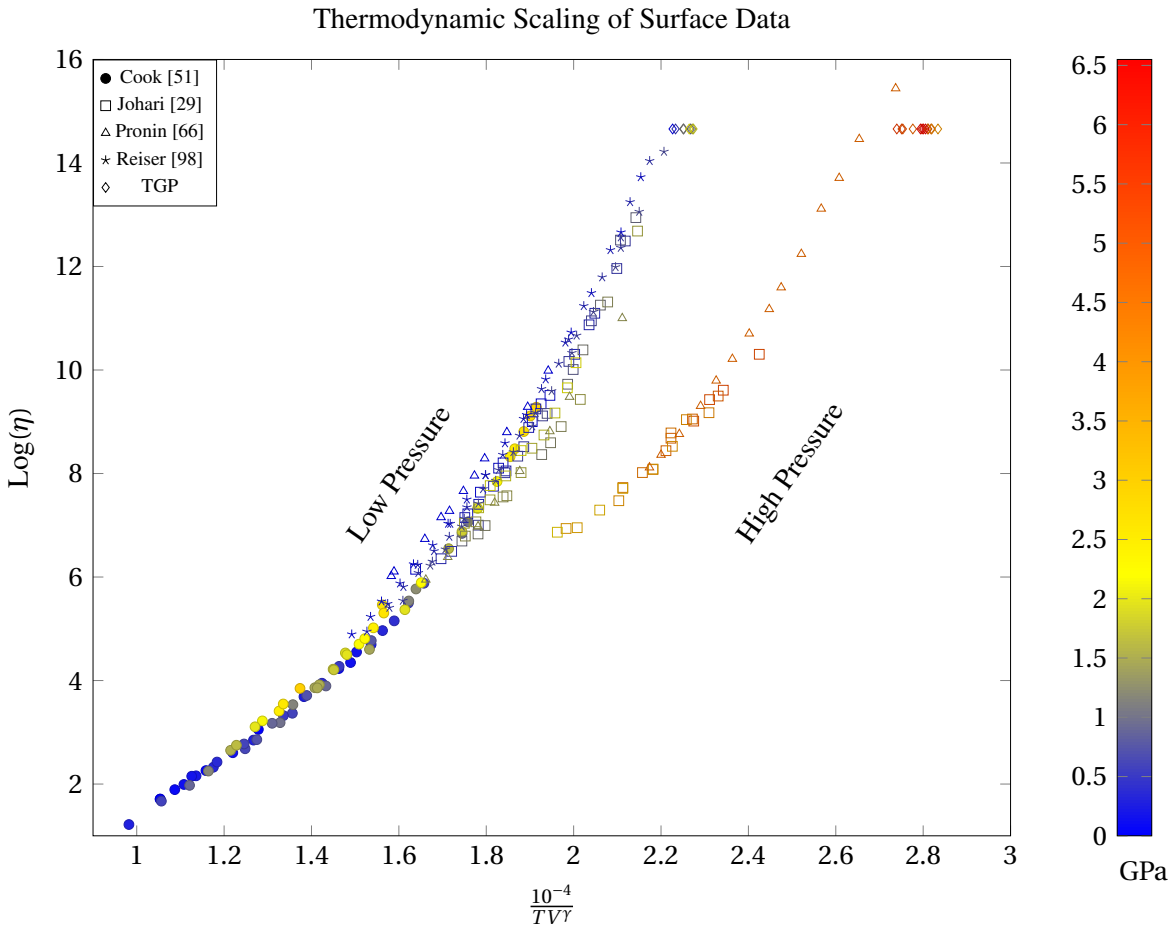


Figure 4.12: The results of thermodynamic scaling of glycerol for all low pressure data (< 2 GPa) with a scaling factor of $\gamma = 1.54$ and high pressure data (> 3 GPa) with a scaling factor of $\gamma = 3.65$.

All of the glycerol data that falls outside of the transition region between 2 and 3 GPa provided by Cook, Johari, Pronin, Reiser, and our TGP data exhibit a reasonable degree of thermodynamic scaling along one of two curves. The results of this scaling are provided in Fig. 4.5. We see that all of the Cook data scales nicely even well into the intermediate pressure regimen-perhaps

due to the relative high temperatures allowing for molecular mobility which facilitates hydrogen bonding at these higher pressures. The Johari data, on the other hand, is all at relatively low to intermediate temperatures and a clear pressure dependent divergence from good thermodynamic scaling is observed as pressure increases and hydrogen bonding begins to be suppressed. All of this still relies on an extreme extrapolation of the Tait equation of state for which Bridgman only provides data up to approximately 1.2 GPa however thermodynamic scaling is possible, albeit in two distinct regimens each with different scaling factor, γ. Where hydrogen bonding is not being suppressed we find a scaling factor $\gamma = 1.54$ that reflects a liquid for which free-volume plays a diminished role in the dynamics as thermal activation is necessary to ionic bonds between neighbors. At very high pressures where hydrogen bonding is completely suppressed we find a scaling factor of $\gamma = 3.65$ that reflects a liquid governed much more strongly by free-volume and Van der Waals interactions.

Once gain we note that the high pressure ~~Pronin~~ data taken in the toroidal pressure cell seems to scale at lower pressures or higher temperatures however, as vitrification sets in at higher pressure the data peels away from the trend expected if thermodynamic scaling were to hold in precisely the same manner that it peeled away from the Andersson-Andersson constrained surface fit previously. It appears once again as if the reported pressures are lower than what the dynamics would imply. As this discrepancy is occurring only at the highest pressure and in the region where vitrification is rapidly increasing, it stands to reason that they have underestimated the prominence of pressure gradients within their toroidal cell. The lower pressure data, even in the vicinity of the glass transition, scales nicely just as it conformed very well to the relaxation surface. However as pressures increase the odds of significant pressure gradients inside the cell increases greatly.

5 Photon Correlation Spectroscopy in a Diamond Anvil Cell

5.1 Justification

Sec. 2.3 describes several probes commonly implemented to measure molecular dynamics in glass-forming systems as the timescale associated with structural relaxation changes by as much as fifteen orders of magnitude while the liquid vitrifies into a glass. Wolynes and Lubchenko assert in their discussion of Dielectric Spectroscopy (DS) that this technique is uniquely suited to track molecular dynamics in glass-forming systems over this large dynamic range. Indeed, when taken as a whole, the multitude of methods that fall under the banner of DS provide an astonishing range of measurement for relaxation times from $\tau \approx 1\,\text{ns}$ up to $\tau > 1000\,\text{s}$[105]. However, many of these methods do not lend themselves well to the geometric constraints and experimental limitations of a diamond anvil cell (DAC). For example, reflection or transmission techniques require that the sample be arranged in a macroscopic coaxial configuration. The methods that may lend themselves well—most notably frequency response and time-domain analysis—are only now being developed in the context of the DAC[106, 107, 108]. Though these methods will certainly contribute greatly to the investigation of glassy dynamics at high pressure moving forward, there is always room for an additional probe.

As shown in Fig. 2.8, depolarized photon correlation spectroscopy (DPCS) covers a dynamic range similar to that of DS frequency-response analysis, extending in principle from approximately 10 ns to beyond one thousand seconds, and provides a direct probe of the structural α-relaxation. Although both DPCS and DS offer comparable windows for measuring molecu-

lar dynamics, they differ fundamentally in how those dynamics are accessed. DPCS is a passive technique that probes spontaneous fluctuations arising from the stochastic thermal evolution of the system, whereas dielectric spectroscopy is inherently a driven measurement, relying on an externally applied electric field to induce dipolar reorientation. While linear-response conditions are typically assumed, the driven nature of DS raises the possibility that the modes most strongly excited by the applied field may not uniquely correspond to the collective structural relaxation of the material.

Additionally, DPCS does not require electrode structures within the sample volume; only optical access is needed, making it a particularly accessible option for extending measurements of structural relaxation into the diamond anvil cell. Despite these differences, both methods are worthy of development, as they possess complementary strengths and limitations.

Dielectric spectroscopy provides a direct probe of structural relaxation only when the dielectric susceptibility faithfully reflects the orientational dynamics of the liquid under study. In such cases, the primary dielectric loss peak may be identified with the bulk α-relaxation. However, this correspondence is not guaranteed. DS is intrinsically sensitive to dipolar reorientation, and when molecular dipoles are associated with specific subunits or segments, the measured response may be dominated by local or segmental dynamics rather than collective structural rearrangement. This situation commonly arises in polymeric systems, where dipole moments oriented transverse to the polymer backbone may reorient independently of the global chain conformation. As noted by the authors, Wrzesińska *et al.*, in the case of poly(dimethylsiloxane), "PDMS is a well-known type B polymer, where the dipole moment is rigidly attached perpendicular to the chain skeleton, and for that reason only segmental motions, called α-processes, are observed" [109].

The physical origin of this limitation was recognized early in dielectric studies of poly-

meric systems. In a seminal investigation of cis-polyisoprene, Yoshida *et al*. demonstrated that dielectric loss spectra may resolve distinct relaxation processes associated with different molecular motions, including a normal mode process arising from end-to-end vector fluctuations of the polymer chain and a segmental mode process that is independent of molecular architecture [110]. The latter reflects local segmental dynamics rather than collective chain rearrangement, illustrating that the dielectric α-process need not uniquely represent bulk structural relaxation when dipolar reorientation is decoupled from collective, global molecular motion.

The study by Tyagi *et al.* provides another clear illustration of the challenges associated with interpreting dielectric spectroscopy in polymeric glass formers, using poly(vinyl acetate) as a representative example. In PVAc, the molecular dipole moment is localized primarily on the acetate side group, causing dielectric spectroscopy to be dominated by local segmental dipolar motion rather than collective structural relaxation. In that work, broadband dielectric spectroscopy was therefore combined with quasielastic neutron scattering in order to access molecular dynamics over a wide range of length and time scales [111].

While dielectric spectroscopy revealed a prominent α-relaxation associated with segmental dipolar motion, comparison with neutron-scattering results demonstrated that the characteristic relaxation times extracted from the dielectric response did not coincide with those governing collective structural dynamics over most probed length scales. The authors further showed that dielectric measurements preferentially weight the slower regions of a heterogeneous distribution of segmental relaxation times arising from polymer microstructure, necessitating the use of an independent probe to obtain a more complete picture of the underlying dynamics. Notably, the same polymer backbone and anisotropic side-group geometry of PVAc that complicate the interpretation of dielectric measurements are expected to introduce significant off-axis components of the

polarizability tensor, making PVAc a particularly attractive candidate for DPCS.

Depolarized photon-correlation spectroscopy couples to fluctuations in the anisotropic component of the molecular polarizability tensor. As a result, DPCS is inherently sensitive to orientational and structural dynamics associated with collective molecular rearrangement, while remaining insensitive to purely isotropic density fluctuations. This selectivity provides a complementary view of structural relaxation dynamics, though it too comes with inherent limitations, perhaps the most significant being that the molecular system must exhibit a sufficiently strong polarizability anisotropy to generate a measurable depolarized scattering signal.

Molecules with extremely high symmetry, such as carbon tetrachloride or sulfur hexafluoride, lack significant anisotropic (off-diagonal) components of the dielectric susceptibility tensor and therefore do not generate an observable depolarized scattering signal. While such highly symmetric systems rarely form molecular glasses due to their strong tendency toward crystallization, they provide a useful conceptual limit that illustrates the symmetry-imposed constraints of DPCS.

Experimentally relevant glass formers typically exhibit some degree of molecular anisotropy. The same geometric features that give rise to orientational frustration and enable glassy dynamics in the first place also tend to generate off-axis components of the dielectric susceptibility tensor. Consequently, glass-forming systems generally lend themselves to producing at least some measurable depolarized scattering signal.

It should not be surprising that, despite the respective limitations of both depolarized photon–correlation spectroscopy (DPCS) and dielectric spectroscopy (DS), many systems may lend themselves equally well to either probe, at least in principle. A clear illustration of this complementarity is provided by *n*-propanol, which possesses a substantial dipole moment of 1.68 Debye and is therefore well suited to dielectric measurements, while also exhibiting sufficient molec-

ular anisotropy to produce a measurable depolarized scattering signal. Although its polarizability anisotropy is smaller than that of aromatic liquids such as cumene, *n*-propanol represents an intermediate case in which both dielectric spectroscopy and DPCS provide access to the bulk α–relaxation.

Importantly, *n*-propanol is a small, comparatively rigid molecule, and therefore does not exhibit the segmental or sub–chain dynamics that can complicate the interpretation of dielectric spectra in polymeric or highly flexible systems. While hydrogen bonding can introduce additional structure into the dielectric response through transient intermolecular associations, these collective dipolar fluctuations remain governed by the underlying structural α–relaxation and do not constitute an independent relaxation mechanism. As a result, DS in *n*-propanol provides a meaningful probe of bulk structural dynamics, albeit one whose amplitude and spectral shape may be influenced by hydrogen–bond correlations.

Cumene (isopropylbenzene) represents a complementary case that is particularly favorable for both probes. Its rigid aromatic structure gives rise to a large polarizability anisotropy, making it exceptionally well suited to DPCS measurements, while the absence of hydrogen bonding or internal molecular flexibility ensures that its dielectric response, though weaker due to the modest permanent dipole moment, remains closely tied to the collective structural α–relaxation. Consequently, although *n*-propanol yields a weaker depolarized scattering signal and cumene a weaker dielectric response, both systems lend themselves naturally to investigation by either technique, with each probe accessing the same underlying structural relaxation dynamics. Even in such systems, DPCS may be a preferable probe simply because it requires only optical access to the sample.

In conclusion, in systems with strong molecular anisotropy but complex or heterogeneous dipolar dynamics, such as polymeric liquids or molecular systems with independently mobile dipo-

Molecule	Glass Former	$\chi_{i\neq j}$	DS Sensitivity	Anisotropic Signal
Cumene (isopropylbenzene)	Yes	Large	Direct α	Very strong
n-Propanol	Yes	Moderate	Direct α	Moderate
Poly(vinyl acetate)	Yes	Large	Segmental	Strong
Glycerol	Yes	Moderate	Mixed	Moderate
Carbon tetrachloride (CCl_4)	No	Vanishing	Weak	Weak
Sulfur hexafluoride (SF_6)	No	Vanishing	Weak	Weak

Table 5.1: Representative molecular systems illustrating the efficacy of dielectric spectroscopy (DS) and depolarized photon-correlation spectroscopy (DPCS) for probing α–relaxation dynamics. Here, off-diagonal $\chi_{i\neq j}$ denotes the relative magnitude of off-diagonal contributions to the dielectric susceptibility tensor that give rise to depolarized scattering. Highly symmetric molecules are included as conceptual reference cases, despite their limited relevance to this work due to their poor glass-forming ability. In such systems, DPCS lacks a measurable depolarized signal due to the near-isotropy of the susceptibility tensor, while DS exhibits no meaningful correlation between the dielectric response and collective structural relaxation.

lar subunits, dielectric spectroscopy may preferentially probe local or segmental relaxations rather than collective structural rearrangement. In these cases, DPCS remains directly sensitive to the bulk α-relaxation, providing a direct probe of structural relaxation dynamics by accessing the stochastic thermal evolution of the system without coupling to driven, local modes [112, 113].

Having established the complementary strengths and limitations of dielectric spectroscopy and depolarized photon-correlation spectroscopy, it is important to emphasize the experimental challenges associated with extending DPCS into the diamond anvil cell. The most significant of these is the exceedingly small sample volume, typically on the order of $\sim 10\,\mathrm{pL}$. In conventional cuvette-based light-scattering experiments, the scattered signal is collected from an extended region illuminated by a focused beam that may span several hundred microns, resulting in effective scattering volumes on the order of a few nanoliters. In the DAC geometry, by contrast, the illuminated region is typically confined to only a few tens of microns.

Under these conditions, it is necessary to ensure that a sufficient number of statistically independent scatterers contribute within the effective scattering volume so that the detected field

obeys the Gaussian statistics assumed by photon correlation spectroscopy. This consideration is particularly relevant in experiments where scattering arises from a low number density of discrete mesoscopic scatterers, such as microspheres undergoing Brownian motion in a host liquid, for which the effective number of scatterers within the probe volume may be limited. For homogeneous liquids, however, light scattering originates from ubiquitous thermal density fluctuations, and even picoliter-scale scattering volumes contain an effectively macroscopic number of dynamically independent fluctuators. As a result, the Gaussian-field assumption remains well satisfied for pure liquids, provided that parasitic scattering from the gasket or diamond anvils is adequately suppressed.

This brings us to another concern regarding DPCS in a DAC—stray spectral reflections from every optical interface, together with unavoidable scattering from the gasket material, are collected alongside the desired signal. As a result, isolating the true scattering volume and suppressing background contributions ~~rapidly becomes~~ a dominant experimental challenge.

Herbst and King addressed many of these difficulties in their effort to extend photon correlation spectroscopy (PCS), identical in application to DPCS but without polarization analysis, into the DAC as a probe of viscosity at high pressures [114]. By analyzing the Brownian motion of polystyrene spheres suspended in the liquid, they obtained reasonable agreement with known values of methanol viscosity at pressures up to 2.9 GPa. Although successful, this approach has not reappeared in the literature since its publication, likely because the experimental complexity is disproportionate to its limited applicability: the method probes viscosities only up to approximately 90 cP.

In principle, the accessible viscosity range could be extended by employing smaller tracer particles and performing longer-timescale autocorrelations. Nevertheless, this approach remains

orders of magnitude removed from the glass-transition regime, where viscosities reach 10^{15} cP. Indeed, in the same year, Cook, working with Herbst and King, demonstrated centrifuge rolling-ball viscometry in a DAC at viscosities exceeding 10^{9} cP [45]. However, rolling-ball techniques are inherently limited to viscosities for which a macroscopic object can traverse the liquid within a practical experimental timescale and therefore cannot be extended to the glass transition.

By contrast, DPCS in a DAC could, at least in principle, access the glass transition and beyond if correlation spectra could be obtained directly from intrinsic density fluctuations of the pure liquid, rather than from nanoparticles suspended within it. The fundamental obstacle to this approach is the extremely weak scattering intensity associated with such fluctuations. In the Rayleigh regime, the intensity of light scattered from spherical particles scales with the sixth power of the particle radius [115]. Consequently, the scattering intensity arising from intrinsic density fluctuations, whose characteristic correlation lengths remain nanometric even in the presence of cooperative dynamics, will be many orders of magnitude weaker than that obtained from the 90 nm polystyrene spheres used by Herbst and King. All of the challenges associated with small sample volumes are therefore dramatically amplified when attempting to extend PCS or DPCS measurements to pure liquids in the DAC.

The goal of this chapter is to ~~outline~~ the process of extending depolarized photon-correlation spectroscopy (DPCS) into the diamond anvil cell, thereby establishing a new probe of slow dynamics at very high pressures, up to and beyond the glass transition. Given the severe experimental constraints outlined above, it is essential to begin with a molecular system that maximizes the depolarized scattering signal. Aromatic liquids such as isopropylbenzene (cumene), with their large anisotropic polarizability arising from the benzene ring, produce exceptionally strong depolarized scattering signals and therefore represent near-optimal candidates for such a proof-of-concept

study.

Cumene, a simple van der Waals–bonded glass-forming liquid, was selected both for this reason and for the availability of independent fast-dynamics data from Brillouin light-scattering measurements. Together, these data allow characterization of relaxation dynamics over an exceptionally broad range of timescales, spanning from tens of picoseconds to hundreds of seconds. In addition, a separate study provided direct measurements of the pressure dependence of the glass-transition temperature via the TGP method described in Ch. 2.3.4. Agreement among these independent probes provides strong evidence for the efficacy of DPCS as a reliable probe of slow dynamics at high pressures in a DAC.

5.2 Experimental Details

As previously mentioned, the light scattered from intrinsic density fluctuations within a pure liquid was expected to be exceedingly weak compared to undesired sources such as specular reflections from the diamond surfaces and gasket interfaces. An initial attempt therefore assumed heterodyne operation, deliberately collecting both the desired scattered signal and accompanying elastic and surface-scattered light, with the latter serving as a local oscillator. The combined light was directed onto the photomultiplier tube (PMT), with neutral-density filters used to attenuate the total intensity to a safe count rate. A Glan–Thompson polarizer was employed primarily to isolate the depolarized component of the scattering signal required for DPCS; as a secondary effect, it also suppresses elastically scattered and specularly reflected light. However, under pressure the diamonds exhibit significant stress-induced birefringence, such that a substantial fraction of laser splash from the diamond tables acquires a rotated polarization and consequently leaks through the polarizer. In practice, the collected light comprised contributions from many coherence areas,

with a local oscillator that overwhelmingly dominated the scattered field arising from density fluctuations. Under these conditions, it is perhaps not surprising that no correlation decay could be resolved even after several hours of data acquisition.

It was therefore clear that spatial filtering was required to both limit the number of coherence areas and to reject much of the excessive local oscillator. This, however, introduced a new problem—physically locating the scattering volume within an exceedingly large region of optical splash. A telescopic viewfinder could be lowered into place to image the interior of the DAC for alignment purposes, but this proved insufficient. The experimenter was still left to blindly raster the collected light across the pinhole in an attempt to isolate a correlation decay signal. Given the uncertainty regarding how long signal averaging would be required before a clear decay emerged, and the tedious nature of blindly rastering over a relatively large area, this approach was impractical.

The solution was to employ the existing Brillouin setup to optimize the spatial filtering before redirecting the light onto the autocorrelator. The complete experimental layout is shown in Fig. 5.2 and is described below along with the process of obtaining an autocorrelation signal.

A Coherent Innova 300 actively locked single-mode Ar^+ medium-frame laser operating at 514 nm provided the light source. To ensure the highest level of power stability, the laser was allowed to warm up at a set current near its operating point for one to two hours. At that stage, manual adjustments were made to the rear mirror to maximize output. Power and mode tracking were then enabled at 1 W output, and the laser was left to warm up for an additional half hour, with occasional alignment adjustments to minimize the operating current. Approximately once per week, the mode-tune algorithm was executed following this warm-up procedure.

A set of reflective neutral-density filters allowed this 1 W output to be incrementally ad-

justed between 1 and approximately 30 mW while maintaining the power stability of the laser. Because the expected scattering signals are exceedingly weak, laser stability was of paramount importance. Future improvements to this system may include the implementation of cross-correlation, which would reduce dependence on power stability and enable analysis of even weaker scattering systems.

As shown in Fig. 5.2, the beam is focused through the DAC located at position (a). The DAC is mounted on a four-axis stage that allows for x, y, z, and θ translations of the cell. What we will refer to as the z-axis is aligned along the incoming focused laser beam and is the primary axis that must be adjusted during signal optimization, as will be discussed shortly. The θ axis lies in the plane of the optical table and allows the DAC to be rotated as shown. In a typical Brillouin experiment, the DAC is rotated away from the collection mirror in order to reduce the amount of spectrally reflected light mixed with the desired signal.

It should be noted that changes to θ do not affect the scattering vector $\vec{q}$, which is geometrically fixed by the collection mirror at location (b). This 1 cm diameter mirror is placed as close as possible to the incoming beam without occultation in order to closely approximate a 180° backscattering geometry. The external angle between the incoming beam and the path from the scattering volume to the center of the mirror is less than 3°. However, due to the indices of refraction of both the diamonds and the liquid sample, the actual scattering angle is significantly smaller—estimated to be less than 1°.

The collimating optic that follows is preceded by an iris, which is used in Brillouin studies to restrict the scattering vector $\vec{q}$ collected. Here, however, the iris is opened reasonably wide but still serves to limit the collection cone, effectively providing a degree of low-resolution spatial filtering by rejecting off-axis light. The collimated light then passes through a Glan–Thompson

polarizer at location (c), which may be removed for alignment purposes and replaced to select the depolarized component of the scattering signal. The beam height is then adjusted by the next two mirrors, placed intentionally after the polarizer, to match the input height of the interferometer.

A 200 µm pinhole provides spatial filtering, and the flip mirror at location (d) is rotated out of the path of the incoming light to allow for spectral analysis. Even at high pressures, near the glass transition, light scattered from lateral acoustic modes gives rise to classic Brillouin peaks, as shown in Fig. 5.1.

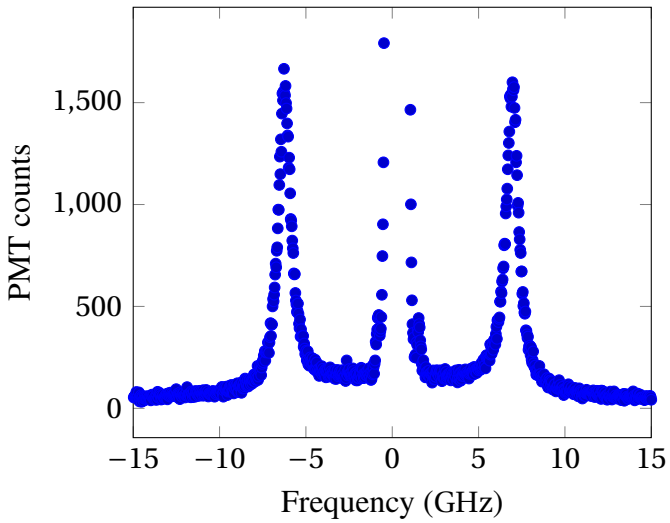


Figure 5.1: Typical Brillouin spectrum with the polarizer removed. The rapidly forming longitudinal acoustic (LA) peaks are used to optimize the DAC position for maximum signal. The polarizer is then reinserted and rotated until the LA peaks are suppressed, ideally leaving only the depolarized signal for DPCS analysis.

Here the spatial filtering may be optimized via maximizing the intensity of these lateral acoustic mode peaks. This is done predominately by adjusting the before-mentioned z-axis which translates the DAC along the path of the incoming laser light such that the collection area being focused onto the pinhole at location (d) corresponds to the liquid sample between the diamond cutlets. The fact that this Brillouin spectra forms significantly more quickly than the much weaker

depolarized spectrum allows for rapid optimization of the spatial filtering. Once this is achieved, the polarizer is replaced and oriented to suppress the LA-modes–passing only the depolarized component of the scattered light. The mirror at location (d) is then flipped up, redirecting the spatially filtered light towards the autocorrelator PMT. The next focusing optic re-collimates the light after the pinhole and is followed by two mirrors to once again adjust the hight of the beam to match the input hight of the autocorrelator optics. The autocorrelator setup itself was previously designed for use with a 1cm cuvette and it was desirable not to strip the system of this capability. As such, the original PCS system was left in tact so that a cuvette and oven could be placed at location (e) or removed at will. To accommodate for DPCS in the DAC, the lens following the hight adjustment mirrors creates an image of the DAC sample at location (e) which the PCS setup then further spatially filters before directing the light to the PMT.

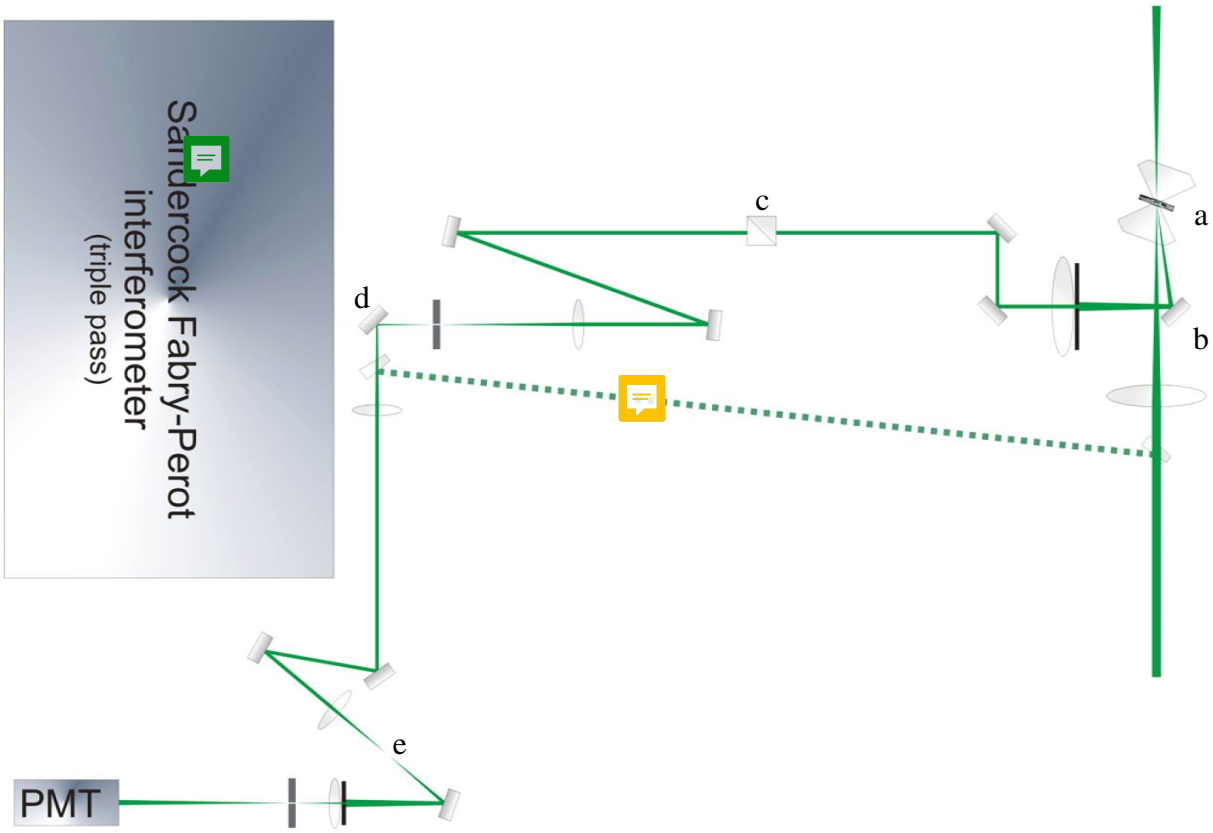


Figure 5.2: Major components of the experimental setup for DPCS: (a) DAC, (b) 1 cm mirror to redirect the scattered signal, (c) Glan–Thompson polarizer, (d) flip mirror to redirect the signal from the interferometer to the autocorrelator. Irises and pinholes are shown as black and gray bars, respectively. The dashed line represents an attempt to provide a local oscillator for heterodyne DPCS analysis.

The present effort to implement DPCS within a DAC can be viewed as an extension of the strategy first demonstrated by Demoulin *et al.*, who showed that time-domain correlation of scattered intensity provides direct access to structural relaxation on timescales spanning microseconds to seconds[62]. Their analysis of undercooled glycerol established the utility of the Cole–Davidson distribution in describing broad, nonexponential relaxation spectra. Later, Lindsey and Patterson demonstrated the correspondence between the Cole–Davidson and Williams–Watts

functions, providing a practical means of translating between frequency- and time-domain analyses[63]. Taken together, these works justify the approach pursued here: by developing the experimental capability to isolate weak depolarized scattering from liquids under extreme confinement in a DAC, one gains a direct probe of structural relaxation that is theoretically continuous with both ultrasonic/frequency-domain results and with the broad body of photon correlation data analyzed in the time domain.

5.2.1 Homodyne *vs*. Heterodyne

As mentioned in Sec. 5.1, DPCS may be performed in either a homodyne (self-beating) or a heterodyne (local oscillator) arrangement. What was not previously discussed is the requirement that one must be squarely within one of these two regimes in order to avoid prohibitively complex analysis.

Elimination of the heterodyne component in the collected light signal is a significant hurdle when attempting PCS within a DAC. The surrounding structure, the edges of the gasket, every surface interface, and even inevitable dust on the diamond table scatter light with appreciable intensity. This problem is further exacerbated by the exceedingly small scattering volume (on the order of tens of picoliters), making it easy for the heterodyne contribution to be comparable to—or even significantly stronger than—the light scattered from the sample volume.

In depolarized studies, much of the heterodyne signal is immediately rejected by the Glan–Thompson prism. However, due predominantly to birefringence in the diamonds at high pressures, a substantial component of the local oscillator inevitably bleeds through. Herbst and King devoted considerable effort to addressing this challenge when first achieving PCS in a DAC[114]. They were able to approach near-homodyne conditions through proper orientation of the DAC to deflect

spectral reflections away from the collection optics, combined with spatial filtering of the collected light.

Their method, however, is insufficient if one wishes to extend this technique to directly probe density fluctuations in a pure liquid. Despite similar sample volumes, Herbst and King measured the thermal diffusion of 90 nm polystyrene spheres and thus benefited from orders-of-magnitude greater scattering intensity. Since their method did not fully eliminate the heterodyne component, it is unreasonable to assume that, when scattering from density fluctuations in a pure liquid, near-homodyne conditions could be confidently preserved. Some amount of spectrally reflected light from every gasket surface will inevitably be collected along with the desired signal, despite the best attempts at spatial filtering. In fact, given the exceedingly weak signal expected from density fluctuations in a few ~~pico~~liters of pure liquid, it is reasonable to suspect that pure heterodyne conditions may be unavoidable.

A simple means to test whether conditions are reasonably within the heterodyne regime is to increase the intensity of the local oscillator and observe whether the extracted α-relaxation times are affected. If the ratio of idler to signal can be increased significantly while the correlation function remains unchanged, then it is reasonable to assume heterodyne conditions.

To implement this test, we inverted the procedure of Herbst and King: rather than rotating the DAC to reduce specular reflections, we incrementally rotated it to increase the amount of reflected light impinging on the collection optics, thereby providing a local oscillator of increasing intensity. Although the simplified optical schematic shown in Fig. 5.2 does not explicitly reflect this constraint, successful heterodyne detection requires that the local oscillator and the scattered field traverse optical paths that are as nearly identical as practicable prior to mixing at the detector. Utilizing spectrally reflected light from the diamond culets and gasket interfaces as the local oscil-

lator naturally satisfies this requirement, as both the scattered and reflected fields originate within the same confined DAC geometry and propagate through essentially the same optical elements. Consequently, any difference in optical path length is limited to at most a small fraction of the culet separation, ensuring adequate phase coherence at the detector. Correlation functions were obtained at a fixed pressure and temperature while incrementally increasing the angle subtended between the DAC and the incoming laser beam. The results are summarized in Tbl. 5.2.

Angle (degrees)	Counts (thousands)	$\langle\tau\rangle$ (μs)	$\Delta\langle\tau\rangle$ (μs)
0	200	1360	54
5	164	1188	57
10	156	1296	73
15	120	1232	33
20	250	1449	57

Table 5.2: PMT count rates and associated α-relaxation times $\langle\tau\rangle$ at various orientations of the DAC. Zero degrees corresponds to the incoming laser at normal incidence to the diamond table; the DAC was then rotated *away* from the collection mirror.

What is of primary interest here is how the expectation value of the relaxation time changes with increasing intensity of any heterodyne component. It is reasonable to assume that the intensity of light scattered directly from the sample is not substantially increased simply by rotating the DAC twenty degrees. Thus, the observed increase in PMT counts at certain angular positions must arise predominantly from additional scatter and specular reflections originating from the sources previously discussed.

Although Fig. 5.3 does appear to suggest an increasing trend in $\langle\tau\rangle$ as the heterodyne com-

ponent is enhanced, it should be emphasized that the PMT counts more than double. As a bare minimum, then, the intensity of the local oscillator has been doubled, yet this corresponds to only a ~17% increase in $\langle\tau\rangle$. This outcome implies that the experiment is operating near, though perhaps not fully within, the heterodyne regime. We will later find this assessment consistent with the agreement of these results with dynamic data obtained via other probes, including Brillouin scattering, TGP, *etc*.

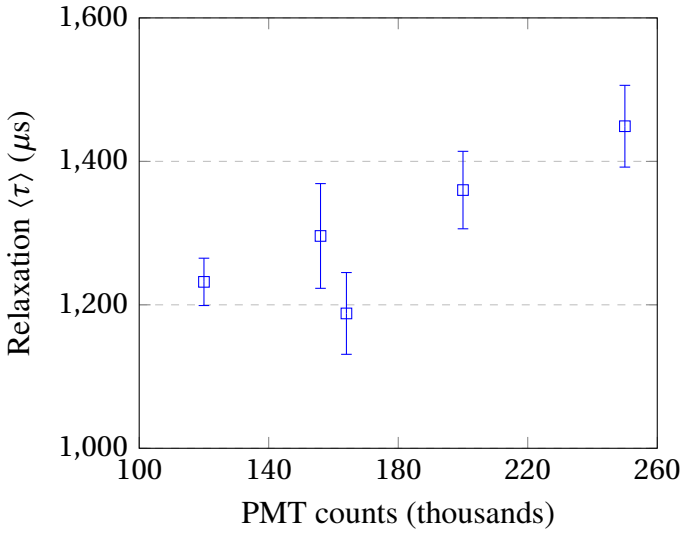


Figure 5.3: Structural relaxation times $\langle\tau\rangle$ as a function of increasing scatter and specular reflections serving as a local oscillator. Error bars represent 95% confidence intervals derived from the mean square error (MSE) of the fit parameter τ.

5.2.2 Time-Correlation Functions

The correlation spectra arising from the Brownian motion of monodisperse spheres in a liquid will exhibit a single activation energy associated with a particle breaking free of the cage formed by its nearest neighbors. Accordingly, one expects, and indeed observes, a simple single-exponential decay in the correlation spectra, with a characteristic time τ corresponding to this structural relaxation.

By contrast, an average glass-forming system is significantly more complex, exhibiting a distribution of relaxation times. This distribution may be interpreted either as arising from spatial heterogeneity—different local environments—or from distinct relaxation modes.

Historically, these stretched exponential time-correlation spectra have been fit using the Kohlrausch–Williams–Watts (KWW) function, which takes the form of Eq. 5.1:

$$C(t) = e^{-(t/\tau)^{\beta}} \tag{5.1}$$

The use of the KWW function to describe glassy systems is theoretically compelling for many reasons, not the least of which is that the Laplace transform of the time-correlation function is directly related to the complex dielectric constant ε^* as follows[116, 117]:

$$\frac{\varepsilon^* - \varepsilon_\infty}{\varepsilon_0 - \varepsilon_\infty} = \mathscr{L}\left[-\frac{\mathrm{d}C(t)}{\mathrm{d}t}\right] \tag{5.2}$$

Importantly, Eq. 5.2 does not rely on the assumption of a single exponential decay; it remains valid for arbitrary normalized relaxation functions, including stretched exponentials with $0 < \beta < 1$, in which case the resulting dielectric response reflects a distribution of relaxation times. However, in the simple case where $\beta = 1$, Eq. 5.1 reduces to a single exponential decay, corresponding to a system governed by a single dominant relaxation process. This situation is exemplified by the Brownian motion of monodisperse spherical particles in a liquid, for which the dynamics are controlled by a single characteristic energy scale associated with viscous dissipation and local solvent rearrangement. As a result, the correlation function is well described by a single relaxation time τ. When transformed via Eq. 5.2, this yields a simple Debye process with characteristic relaxation time τ[67].

$$\mathscr{L}\left[-\frac{\mathrm{d}C(t)}{\mathrm{d}t}\right]_{\beta=1} = \frac{1}{1+i\omega\tau} \tag{5.3}$$

When working in frequency space, such as with Brillouin light scattering or dielectric frequency-response analysis, the relaxation dynamics of glass-forming systems are often described in a form closely analogous to Eq. 5.1, namely via the Cole–Davidson distribution:

$$\frac{\varepsilon^* - \varepsilon_\infty}{\varepsilon_0 - \varepsilon_\infty} \approx \left(\frac{1}{1+i\omega\tau_{\mathrm{CD}}}\right)^{\beta_{\mathrm{CD}}} \tag{5.4}$$

The symmetry between Eq. 5.4 and Eq. 5.1, together with the fact that the simple Debye process described by Eq. 5.3 corresponds directly to a pure exponential decay in the time domain, might suggest that the Cole–Davidson function is the Laplace transform of the KWW stretched exponential. It is not. Both functions may be interpreted as arising from a distribution of elementary relaxation processes, each independently describable either by a simple exponential decay in the time domain or by the corresponding Debye relaxation in the frequency domain. However, because they assume fundamentally different distribution functions, the extracted values of τ and β are not identical.

This discrepancy is not inconsequential, since multiple probes—operating in both the frequency and time domains—are required to track molecular dynamics in glass-forming systems across an enormous dynamic range. In this study, we will relate data fit to a Cole–Davidson distribution to our time-domain DPCS measurements. Lindsey and Patterson addressed this issue by explicitly assuming that the KWW and Cole–Davidson functions arise from distributions of simple exponential decays and Debye processes, respectively, as follows[63]:

$$e^{-(t/\tau_{\mathrm{KWW}})^{\beta}} = \int_0^{\infty} e^{-t/\tau}\, \rho_{\mathrm{KWW}}(\tau)\, \mathrm{d}\tau \tag{5.5}$$

$$\left(\frac{1}{1+i\omega\tau_{\mathrm{CD}}}\right)^{\beta_{\mathrm{CD}}} = \int_0^{\infty} \frac{1}{\tau}\left(\frac{1}{1+i\omega\tau}\right) \rho_{\mathrm{CD}}(\tau)\, \mathrm{d}\tau \tag{5.6}$$

A great deal of effort was devoted to solving for these distributions, ρ_{CD} and ρ_{KWW}, analytically and via series expansion, respectively. The differences between them were then assessed numerically, and a means of converting between the τ and β parameters of the two distributions was established. Consequently, throughout the literature it has been common practice to fit frequency-domain data to a Cole–Davidson distribution and time-domain data to a KWW stretched exponential, converting one to the other via the relationships described by Lindsey and Patterson when direct comparison is required.

An alternative approach—arguably more physically meaningful—is to compare the expectation values of the relaxation times. If the assumed distributions are valid, then although the τ parameters may differ, the expectation values $\langle\tau\rangle$ should agree. For a known distribution $\rho(\tau)$, the expectation value is defined as

$$\langle\tau\rangle = \int_0^{\infty} \tau\, \rho(\tau)\, \mathrm{d}\tau \tag{5.7}$$

For the KWW stretched exponential and Cole–Davidson distributions, this yields

$$\langle\tau\rangle_{\mathrm{KWW}} = \frac{\tau_{\mathrm{KWW}}}{\beta_{\mathrm{KWW}}}\Gamma\left(\frac{1}{\beta_{\mathrm{KWW}}}\right) \tag{5.8}$$

$$\langle\tau\rangle_{\mathrm{CD}} = \beta_{\mathrm{CD}}\,\tau_{\mathrm{CD}} \tag{5.9}$$

However, further results provided by Lindsey and Patterson show that for $\beta_{\mathrm{CD}} < 0.5$, the ratio

$$\frac{\langle\tau\rangle_{\mathrm{KWW}}}{\langle\tau\rangle_{\mathrm{CD}}}$$

quickly becomes significantly greater than unity—*i.e.*–, the difference in the underlying distributions invalidates the assumption of equivalence. For cumene, $\beta_{\mathrm{CD}} \approx 0.5$; thus, it is prudent to explicitly convert between parameters rather than relying on expectation-value equivalence.

Lindsey and Patterson also provided an explicit form for $\rho_{\mathrm{CD}}(\tau)$, given by

$$\rho_{\mathrm{CD}}(\tau) = \frac{\sin(\pi\beta_{\mathrm{CD}})}{\pi}\left(\frac{\tau}{\tau_{\mathrm{CD}} - \tau}\right)^{\beta_{\mathrm{CD}}} \tag{5.10}$$

This distribution, obtained from Eq. 5.6 together with the normalization condition

$$\int_0^\infty \frac{1}{\tau}\,\rho_{\mathrm{CD}}(\tau)\,\mathrm{d}\tau = 1,$$

allows one to define the Cole–Davidson function in the time domain. Recognizing the validity of Eq. 5.3, and recalling that $\rho_{\mathrm{CD}}(\tau)$ was derived under the assumption that the Cole–Davidson form corresponds to a distribution of simple Debye processes, the Cole–Davidson function can be written in the time domain as

$$\left(\frac{1}{1 + i\omega\tau_{\mathrm{CD}}}\right)^{\beta_{\mathrm{CD}}} = \mathscr{L}\left[-\frac{\mathrm{d}}{\mathrm{d}t}\int_0^{\tau_{\mathrm{CD}}} \frac{1}{\tau}e^{-t/\tau}\,\rho_{\mathrm{CD}}(\tau)\,\mathrm{d}\tau\right] \tag{5.11}$$

The work of Lindsey and Patterson has been available for almost four decades, yet it is clear why it became common practice to fit time-domain data to a KWW stretched exponential and then convert the parameters, rather than fitting directly to the Cole–Davidson distribution. Requiring a nonlinear fit routine to perform this integration would have been prohibitively resource-intensive on 1980s hardware. It remains nontrivial even today, though it can be recast into a form involving one- and two-parameter Gamma functions which, with the aid of modern lookup tables, can be evaluated relatively efficiently:

$$\int_0^{\tau_{\mathrm{CD}}} \frac{1}{\tau} e^{-t/\tau} \rho_{\mathrm{CD}}(\tau)\,\mathrm{d}\tau = \frac{1}{\pi}\Gamma\big(1-\beta_{\mathrm{CD}}\big)\,\Gamma\Big(\beta_{\mathrm{CD}}, \tfrac{t}{\tau_{\mathrm{CD}}}\Big)\sin\big(\pi\beta_{\mathrm{CD}}\big) \tag{5.12}$$

Thus, in order to obtain values of τ that are directly comparable to those derived from frequency-domain Brillouin studies, all DPCS time-domain correlation decays will be fit in *Mathematica* to a distribution of single-exponential decays corresponding to a Cole–Davidson function in the frequency domain, as expressed in Eq. 5.13:

$$C_{\mathrm{CD}}(t) = \frac{1}{\pi}\Gamma\big(1-\beta_{\mathrm{CD}}\big)\,\Gamma\Big(\beta_{\mathrm{CD}}, \tfrac{t}{\tau_{\mathrm{CD}}}\Big)\sin\big(\pi\beta_{\mathrm{CD}}\big) \tag{5.13}$$

5.3 Representative DAC Autocorrelation at 75 °C and Regression Strategy

The sequence presented in Fig. 5.6 is taken from the 75 °C isotherm of cumene over a wide pressure range. As noted previously in Sec. 5.2, autocorrelations obtained in a DAC are not expected to be as tidy as those from conventional cuvette geometries. Two experimental realities dominate the appearance of these data. First, microscopic impurities, gasket scatter, and dust furnish a strong local oscillator. At lower temperatures these larger scatterers are effectively

immobile on our timescales and provide a stable heterodyne reference. At higher temperatures, however, some of these particles become sluggishly mobile. When such a particle enters or leaves the scattering volume mid-acquisition, the baseline of the autocorrelation can swing substantially and may take impractically long to average back to a flat baseline.

This fluctuating baseline is shown in Fig. 5.4 for one of our higher-pressure datapoints—26.8 kbar. It should be noted that these are not taken at set time intervals but are instead the result of periodically saving the acquisitions during long runs; one never knows when a severe baseline fluctuation may render the data unfittable, so earlier saved autocorrelations can preserve a data point that often required hours, if not days, to obtain.

It may also be seen from Fig. 5.4 that the first autocorrelation was saved relatively early, before the decay had fully converged to the actual relaxation—*i.e.* it lies well above the regressions saved at later times. However, all of the subsequent regressions quickly converge, as was typical for all acquisitions, even though the baseline at high pressures and temperatures very rarely does so for the aforementioned reasons.

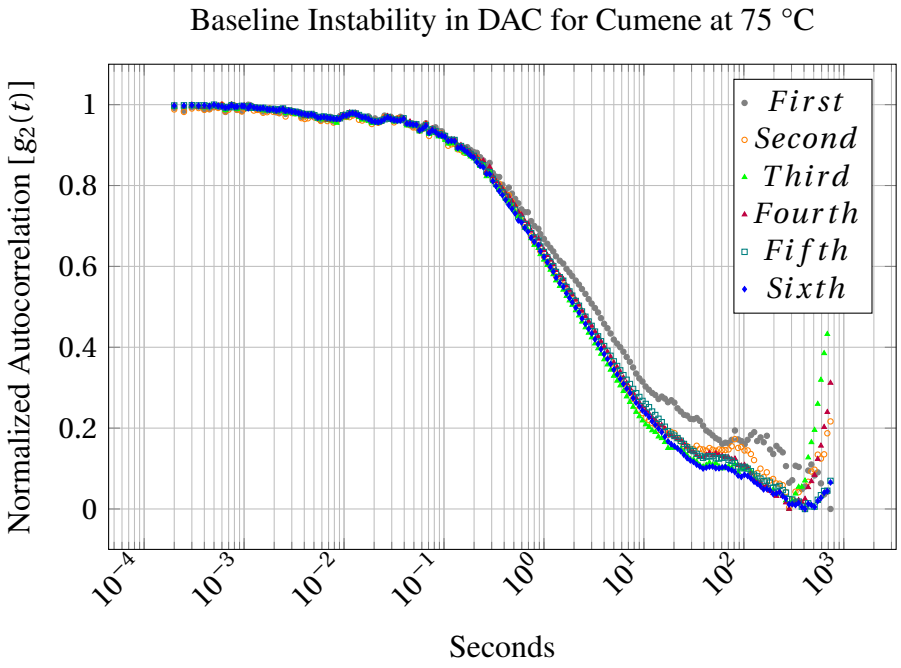


Figure 5.4: Autocorrelations of cumene at 75 °C and 26.8 kbar illustrating the fluctuating baselines characteristic of high-temperature DAC measurements. Each curve represents a saved acquisition taken at a different time during the run. The earliest save ("First") was recorded before the decay had fully converged to the true relaxation and therefore lies well above the later regressions. Subsequent saves converge rapidly to a consistent relaxation profile, while the overall baseline remains unstable likely due to mobile impurities entering or leaving the scattering volume mid-acquisition.

An additional consideration is that the DAC severely limits the illuminated/scattering volume. Scattering volumes typical for PCS are often on the order of tens to hundreds of nanoliters. However, because the culets of the DAC constrain the length of the scattering column—often to ten microns or less—the effective scattering volume may be reduced by two orders of magnitude or more. For the fastest autocorrelation channels this often implies $\ll 1$ photon per scan on average, so the short-lag "head" of the normalized autocorrelation function $g_2(t)$ is shot-noise dominated. This gives rise to the poorly defined low-lag region shown in Fig. 5.5. Unlike baseline drift, this head noise persists even at relatively low temperatures, as it originates from photon statistics within a tiny scattering volume. At lower temperatures or higher pressures, however, the intrinsic dynamics

are much slower, and the shot noise inherent to these fast channels becomes of little consequence. In fact, it is quite common for the autocorrelation curve to lack a head entirely, as the liquid's relaxation time increases beyond the range of the correlator's fastest channels. This behavior is clearly visible in the low-pressure data presented in Fig. 5.6.

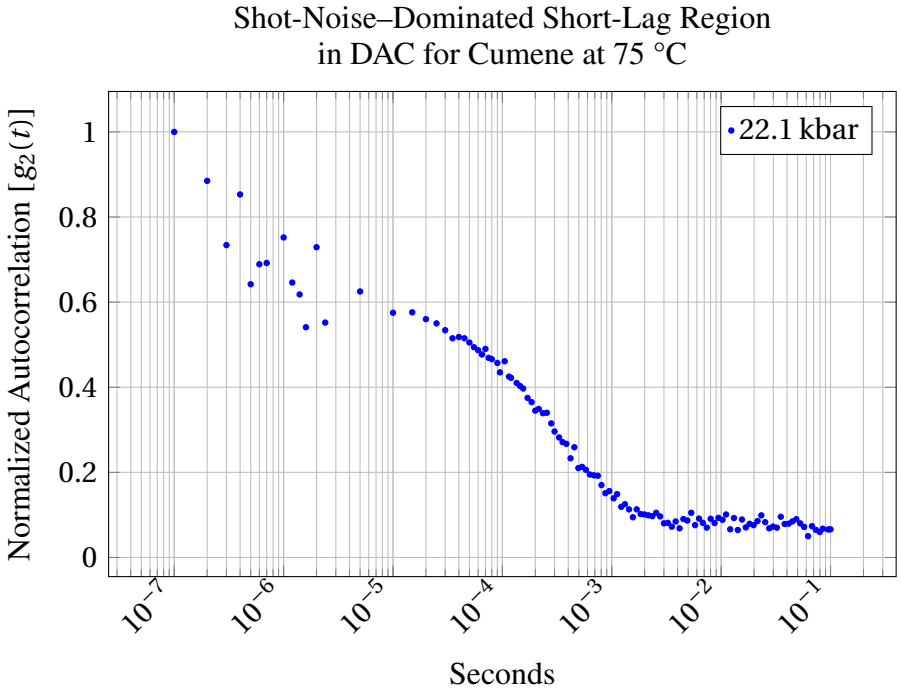


Figure 5.5: Autocorrelation of cumene at 75°C and 22.1 kbar illustrating the shot-noise–dominated short-lag region that arises from the severely reduced scattering volume in the DAC.

Despite these complications being more pronounced at higher temperatures, the 75 °C data series is included here as it represents the "worst-case" data quality we encountered and thus motivates the regression fit strategy outlined below which has been implemented here to extract reliable relaxation times.

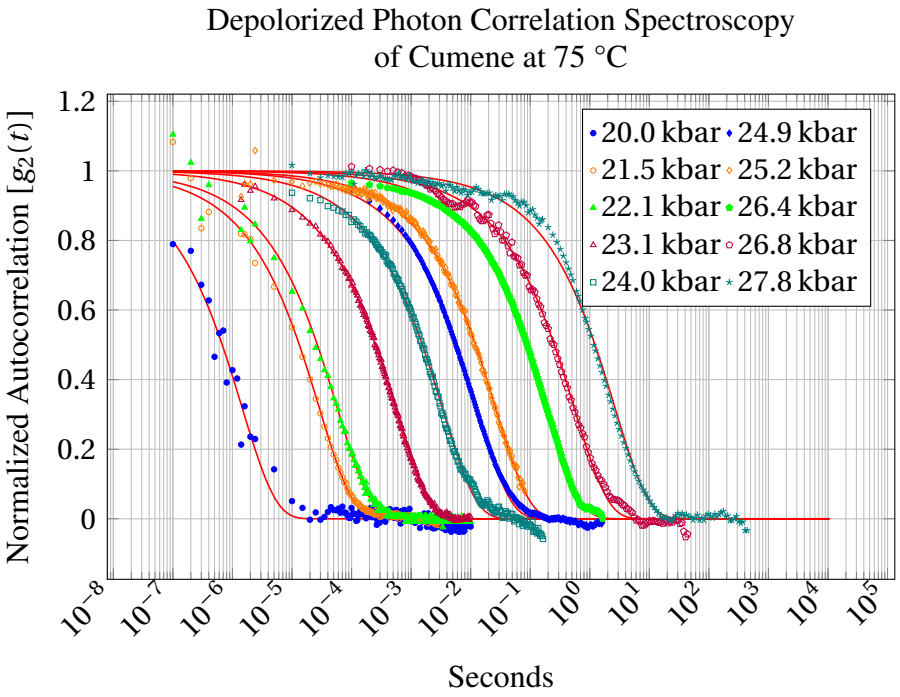


Figure 5.6: Depolarized PCS autocorrelations of cumene in a DAC at 75 °C across a range of pressures. The sequence highlights two dominant elevated-temperature artifacts: slowly varying baseline offsets (heterodyne local-oscillator drift) and photon-limited short-lag noise. Mitigation and fitting strategy are discussed in the text.

Obtaining robust Cole–Davidson fits in the time domain ideally requires well-defined heads and baselines. Of these, the baseline is often most critical given that, for the Cole-Davidson distribution, τ_{CD} is the discrete upper cutoff of the structural-relaxation mode distribution. The expectation value satisfies $\langle\tau\rangle_{\mathrm{CD}} = \beta_{\mathrm{CD}}\,\tau_{\mathrm{CD}}$—see Eq. 5.9. In the DAC, especially at higher temperatures, a clean baseline can be unattainable due to the before-mentioned local-oscillator drift, while at the fastest lags the head can be too noisy to anchor the early-time curvature due to shot noise. The practical remedy is therefore to constrain the shape parameter, β_{CD}, or at least to establish a functional dependence of β_{CD} on pressure and temperature. Without such a constraint, the fit becomes underdetermined when either the head or baseline is unreliable. When a portion of the autocorrelation function must be excluded, whether the early-time head or the late-time baseline, fixing β_{CD}

removes the principal degeneracy between amplitude and shape in the fit. This constraint allows τ_{CD} to be determined primarily by the curvature of the mid-lag region. In practice, most datasets furnish at least one clean side, so constraining β_{CD} empirically or through a pressure-temperature-dependent model would ensure consistent τ_{CD} values and hence consistent mean relaxation times $\langle\tau\rangle = \beta_{CD}\tau_{CD}$.

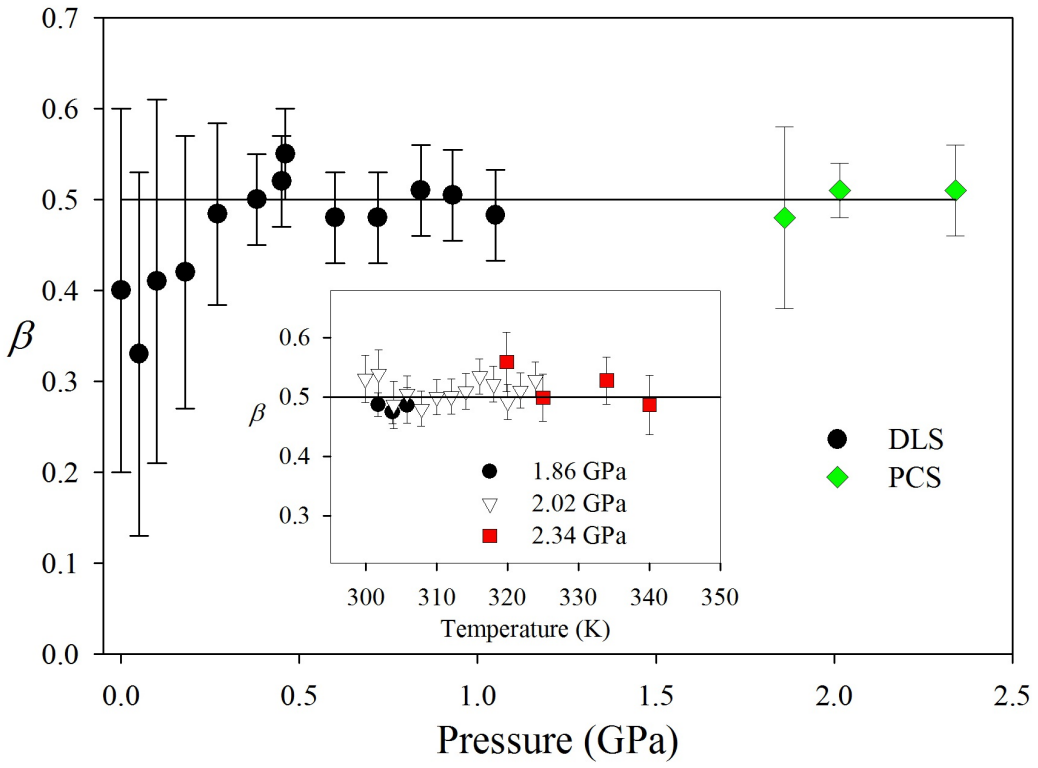


Figure 5.7: Pressure dependence of the Cole–Davidson shape parameter, β_{CD}, obtained from Brillouin (black circles) and DPCS (green diamonds) measurements. Error bars represent 1σ uncertainties derived from the nonlinear fit covariance matrices. The inset shows the temperature dependence of β_{CD} at selected pressures, demonstrating its statistical invariance across a large experimental domain.

As it happens our lab has plenty of relaxation data at various pressures and temperatures for cumene in the form of Brillouin light-scattering data as well as DPCS. The values for β_{CD} along with 1σ uncertainties derived from the covariance matrix of the nonlinear fits, is shown in Fig. 5.7. It is notable that the error bars for β_{CD} become excessively large near ambient pressure.

The larger uncertainties in the Brillouin fits at low pressure arise due to the structural–relaxation (Mountain) component becoming less distinct from the quasi-elastic background near the Bose peak, increasing parameter covariance in the Cole–Davidson fits. However, as there was no evident trend in the value of β_{CD} approaching the Bose peak, there is no reason to believe such a trend would suddenly appear as we approach ambient pressure. To extend this analysis, DPCS data was also included under conditions where an autocorrolation could be obtained having a solid low-lag head and baseline thus allowing for a nonlinear fit from which β_{CD} could be confidently obtained. This is particularly compelling as it not only extends our analysis of the pressure dependence of β_{CD} up to nearly 2.5 GPa, but also further assures us that our time-domain Cole-Davidson fits to our autocorrolations are returning the appropriate values for comparison with our frequency-domain data. It is very evident that there is no significant pressure dependence in the value of β_{CD} from ambient pressure upto nearly 2.5 GPa.

The inset plot of Fig. 5.7 is comprised entirely of DPCS autocorrolation fits where solid low-lag heads and baselines were obtainable. It is clear that over the range of 40 °C for which this data was obtained there is no significant temperature dependence in the value of β_{CD}. Thus from here we make the assumption that, for the glass former cumene, over the ranges of pressure and temperature here investigated, the value of $\beta_{CD} = 0.509$. This value is used consistently for all Cole-Davidson fits to cumene data here presented.

For internal validation, we applied this procedure to runs where both a clean head and a flat baseline are present. The full autocorrolation is first fit with $\{\tau_{CD}, \beta_{CD}\}$ free to obtain $\tau_{CD}^{(full)}$ and $\beta_{CD}^{(full)}$. The analysis is then repeated twice: once with the short-lag head removed and $\beta_{CD} = 0.509$, and once with the baseline removed and the same fixed β_{CD}. In both cases the recovered τ_{CD} agrees with $\tau_{CD}^{(full)}$ within the 95% confidence interval, indicating that a single reliable side of the

decay suffices when β_{CD} is constrained.

5.4 Results and Discussion

5.4.1 Thermodynamic Scaling

Another means of validating the DPCS data—and of confirming that the measurements were obtained under heterodyne or homodyne experimental conditions—is to examine how well they conform to the established thermodynamic scaling behavior of cumene. The cumene TGP data, provided here and in App. C, allows us to determine the appropriate scaling factor.

Liquids that obey thermodynamic scaling exhibit a relationship between temperature, volume, and relaxation time of the form:

$$\tau \propto \frac{1}{TV^{\gamma}} \tag{5.14}$$

Thus, since the TGP data comprise an isochronous curve at $\tau \approx 100\,\mathrm{s}$, taking the logarithm of both sides of Eq. 5.14 yields the linear expression in Eq. 5.15. The slope of this relation gives the scaling factor γ, while the y-intercept provides an estimate of the relaxation time associated with the TGP data.

Experimentally, the TGP process provides the glass-transition temperature as a function of pressure. To proceed, it is necessary to convert these pressures to the corresponding specific volumes. For cumene, this conversion is carried out using the modified Tait equation of state provided in App. D.

$$\log_{10}(T) = -\gamma \log_{10}(V) - \log_{10}(\tau_G) \tag{5.15}$$

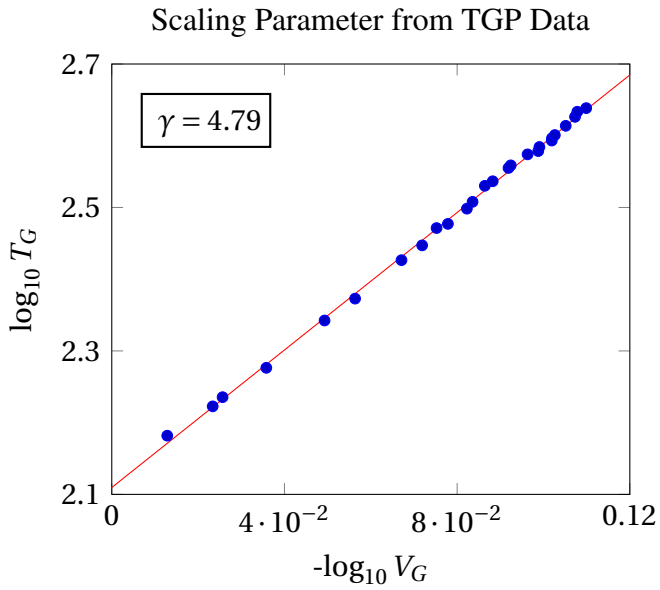


Figure 5.8: Linearized cumene TGP data from App. C fit to the regression model $\log_{10}(T) = -\gamma \log_{10}(V) - \log_{10}(\tau_G)$. The slope gives $\gamma = 4.79$, and the y-intercept of 2.11 corresponds to an average TGP relaxation time of $\tau \approx 130\,\text{s}$.

The value of the scaling factor having been obtained in this fashion, the DPCS data provided in App. F may be checked to determine whether it scales consistently with the Brillouin fast-dynamics data and the TGP data for $\gamma = 4.77$. The corresponding specific volumes are again obtained via Eq. D.1. The results of this thermodynamic scaling are presented in Fig. 5.9.

With respect to the heterodyne/homodyne considerations addressed in Sec. 5.2.1, the majority of the data were collected with the DAC oriented such that a significant amount of local oscillator light from the surfaces was captured along with the desired signal, thereby ensuring that we were well within the heterodyne regime. These data were fit to Eq. 5.13 to obtain the α-relaxation parameter, τ_{CD}. The scaled values are labeled "DPCS" in Fig. 5.9 and show excellent agreement with both the scaled Brillouin and TGP data.

Because the intensity of light scattered from the sample volume is so much weaker than that of the local oscillator, it was often necessary to signal average for several hours in the heterodyne

configuration. As noted in Sec. 5.2.1, it was generally assumed that the experiment always operated well within the heterodyne regime due to the exceedingly small sample volume. Thus, minimizing scatter from the DAC surfaces in order to reduce acquisition time seemed reasonable.

With this in mind, when the 75° isotherm was measured, the DAC was oriented to minimize PMT counts. This resulted in relatively rapid acquisition—in some cases as little as ten minutes. However, when these data were fit to Eq. 5.13 and scaled, the resulting points systematically lay above the rest of the data known to be solidly heterodyne. This suggested that the measurement was perhaps closer to purely homodyne conditions than originally assumed.

Lacking a reliable means of quantifying the degree of homodynicity, the data were instead fit to Eq. 5.16, which assumes a Cole–Davidson distribution of purely heterodyne correlation decays:

$$\int_0^{\tau_{\mathrm{CD}}} \frac{1}{\tau} e^{-2t/\tau} \rho_{\mathrm{CD}}(\tau)\, d\tau = \frac{1}{\pi} \Gamma\big(1-\beta_{\mathrm{CD}}\big)\, \Gamma\Big(\beta_{\mathrm{CD}}, \tfrac{2t}{\tau_{\mathrm{CD}}}\Big) \sin\big(\pi\beta_{\mathrm{CD}}\big) \tag{5.16}$$

When refit under the assumption of homodynicity, the scaled values were found to agree with the remainder of the scaled data. This indicates that by orienting the DAC to reject non-target light as aggressively as possible, the experiment operated under nearly pure homodyne conditions. With further development—particularly for liquids with strong depolarized scattering—it may be feasible to run confidently in a homodyne configuration, thereby accelerating data acquisition. However, because the only reliable way to confirm reasonably homodyne conditions is to compare with data known to be heterodyne, we elected to intentionally orient the DAC so that sufficient splash from the surfaces was collected to ensure the data were solidly within the heterodyne regime during acquisition. This procedure was used for all data except the 75°C isotherm.

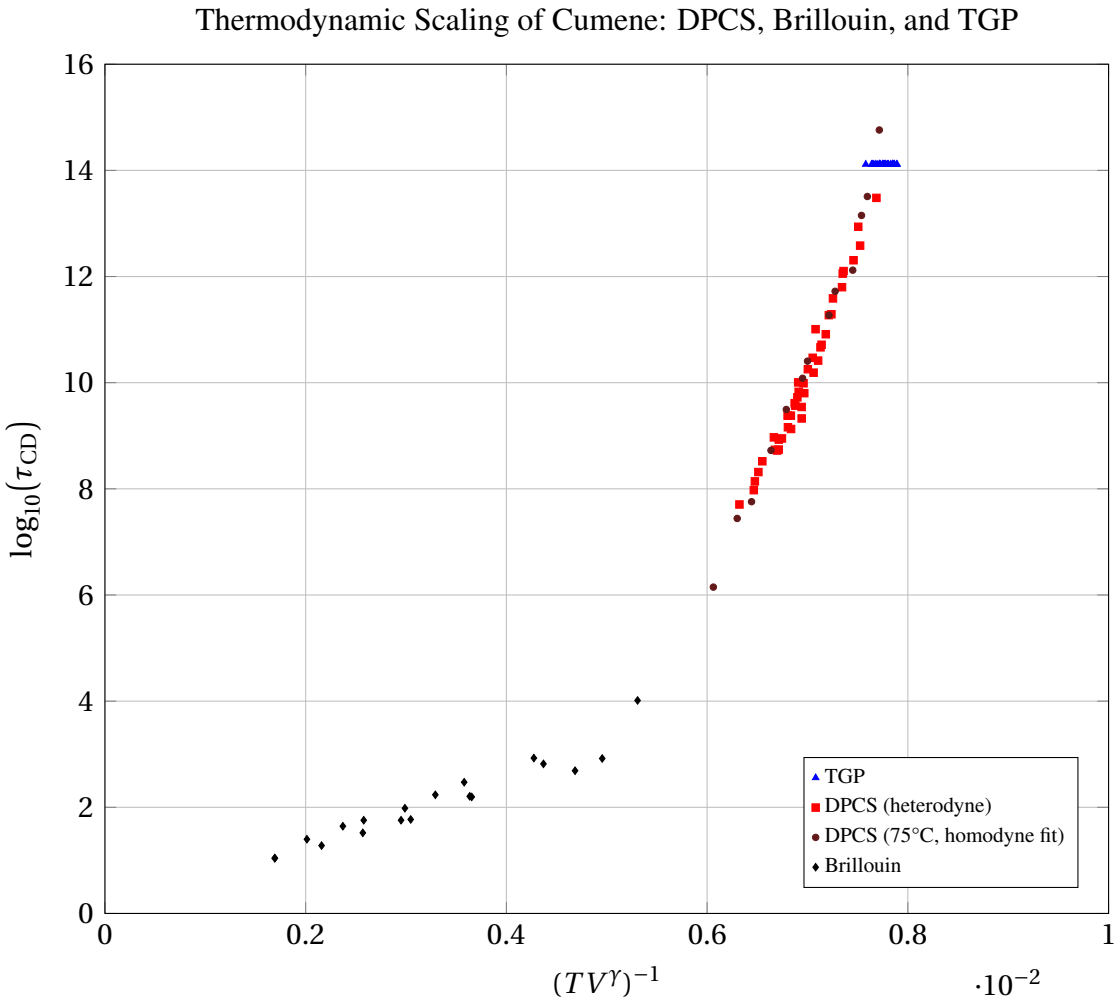


Figure 5.9: Thermodynamic scaling of cumene showing $\log_{10}(\tau_{CD})$ *vs.* $(TV^{\gamma})^{-1}$ using γ determined from TGP. DPCS data acquired under intentionally heterodyne conditions (squares) align with scaled Brillouin fast-dynamics data (diamonds) and TGP (triangles). The 75°C isotherm, acquired with the DAC oriented to minimize PMT counts, is consistent with the heterodyne data when refit assuming homodyne correlation decays (circles).

5.4.2 Alpha Relaxation Dynamics

As discussed in Sec. 5.2.1 and Sec. 5.2.2, the broader goal of this chapter is to establish DPCS within a DAC as a reliable probe of the structural α-relaxation. The utility of time-domain correlation methods for glass-forming liquids was first demonstrated by Demoulin *et al.*, who showed that the Cole–Davidson distribution provides a robust description of broad, nonexponential

relaxation in undercooled glycerol[62]. Later, Lindsey and Patterson established the correspondence between the Williams–Watts stretched exponential and the Cole–Davidson form, thereby providing a practical framework for translating between time-domain and frequency-domain probes[63]. Building on this foundation, we pool together DPCS, DLS, and TGP data to provide a comprehensive picture of the α-relaxation dynamics of cumene, spanning timescales from picoseconds to hundreds of seconds, and extending to pressures in excess of 3 GPa

To obtain a full description of the relaxation dynamics of cumene over the entire experimental range, data obtained from DPCS, as well as DLS and TGP methods, have been pooled together to fill in as much of the dynamic space as possible. For an initial attempt, a surface fit of this dynamic data was performed using the Vogel–Tammann-Fulcher (VTF) equation,

$$\ln(\tau) = \ln(\tau_\infty) + \frac{D T_0}{T - T_0}, \tag{5.17}$$

where τ_∞ is the infinite-temperature relaxation time (assumed constant for all pressures), T_0 is the zero-mobility temperature at which the α-relaxation time asymptotically diverges, and D is the fragility parameter. A simple polynomial expansion as used by Ransom and Oliver to extend the Tate equation of state into high pressures can be analogously implemented to the VTF equation[46]. Such an expansion of D and T_0 allows this classically temperature-dependent model to capture dynamics over a wide pressure range—from ambient conditions up to approximately 4.5 GPa in this study. The resulting pressure-extended VTF model, and the values for these parameters obtained via a Levenberg–Marquardt fit, are summarized in Eq. 5.18.

$$H(T,P) = \log(\tau_\infty) + \frac{D(P)\,T_0(P)}{T - T_0(P)}$$

with:

$$D(P) = \sum_{j=0}^{2} d_j P^j, \qquad T_0(P) = \sum_{j=0}^{3} t_j P^j, \tag{5.18}$$

$$\tau_\infty \equiv 1.118179\,\mathrm{ps}, \qquad \vec{t} = \begin{pmatrix} 115.201\,\mathrm{K} \\ 1.17558\,\mathrm{K\,kbar^{-1}} \\ 0.141404\,\mathrm{K\,kbar^{-2}} \\ -0.00176523\,\mathrm{K\,kbar^{-3}} \end{pmatrix}, \quad \vec{d} = \begin{pmatrix} 1.59655 \\ 1.29546 \\ -0.03236 \end{pmatrix}.$$

It should be noted that the values for τ_∞, d_0, and t_0, obtained from a separate VTF fit to the ambient-pressure isobaric data from Fig. 5.10, were fixed as constants for the surface fit. This constraint forced the surface to pass through the well-established ambient-pressure values and significantly improved the fidelity of the fit in the cold low-pressure regime, where the data are relatively sparse.

To further constrain the pressure-extended VTF surface, we enforce both the ambient-pressure ($P = 1\,\mathrm{bar}$) VTF anchor of Fig. 5.10 and the Anderson–Anderson (AA) isochrone obtained from the TGP data provided in App. C. The AA isochrone appears in Fig. 5.11 as the solid line traversing the 130s isochrone along the top of the plot.

Let $P_{\mathrm{ref}} = 1\,\mathrm{bar}$. The AA isochrone supplies a reference level

$$y^* \equiv H(T_t, P_{\mathrm{ref}}) = \mathrm{Int} + \frac{D(P_{\mathrm{ref}})\,T_0(P_{\mathrm{ref}})}{T_t - T_0(P_{\mathrm{ref}})},$$

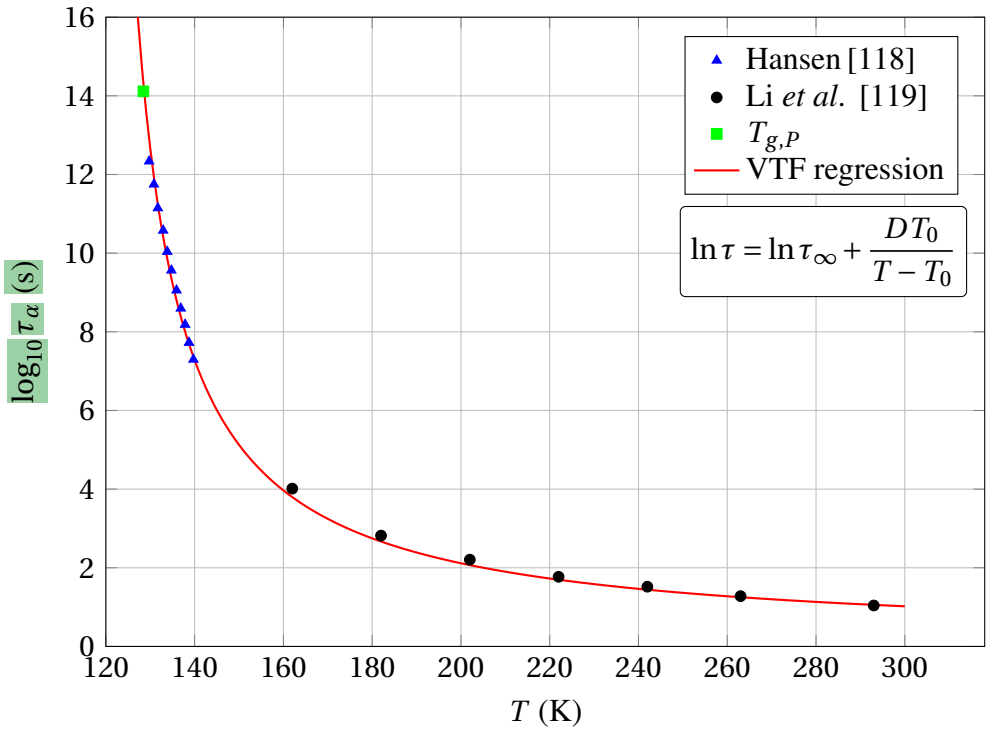


Figure 5.10: Atmospheric-pressure structural-relaxation data for cumene, plotted as $\log_{10}\tau_\alpha$ versus temperature at atmospheric pressure. The solid red curve is a Vogel–Tammann–Fulcher (VTF) regression fit to this 1-bar data that returns the specific values of τ_∞, t_0, and d_0 used in the construction of the surface model described by Eq. 5.18.

where T_t is the AA intercept at P_{ref}. We then take the pressure-dependent AA curve $T_{\mathrm{AA}}(P)$ (from the AA fit to the TGP data) as the locus on which the surface must equal y^* for all P.

With the surface written as

$$H(T,P) = \mathrm{Int} + \frac{D(P)\,T_0(P)}{T - T_0(P)} \quad \text{(cf. (5.18))},$$

imposing the isochronal condition $H(T_{\mathrm{AA}}(P), P) = y^*$ yields the constrained representation

$$H_{\mathrm{constr}}(T,P) = y^* - \frac{D(P)\,T_0(P)}{T_{\mathrm{AA}}(P) - T_0(P)} + \frac{D(P)\,T_0(P)}{T - T_0(P)}. \tag{5.19}$$

By construction, this satisfies $H_{\mathrm{constr}}(T, P_{\mathrm{ref}}) = H(T, P_{\mathrm{ref}})$ as it recovers the 1-bar VTF of Fig. 5.10 exactly and $H_{\mathrm{constr}}\big(T_{\mathrm{AA}}(P), P\big) = y^*$ thus it threads the AA isochrone.

In the global regression we hold the ambient-pressure triplet $\big(\mathrm{Int},\, t_0,\, d_0\big)$ fixed to the 1-bar fit (so that $T_0(P_{\mathrm{ref}}) = t_0$ and $D(P_{\mathrm{ref}}) = d_0$), adopt $T_{\mathrm{AA}}(P)$ from the AA analysis of the TGP data, and fit the remaining pressure coefficients (*e.g.* d_1-d_2 and t_1–t_3) to all (T, P) data.

What results is a surface model for the alpha-relaxation dynamics of cumene where the 1-bar anchor pins the absolute level and curvature at P_{ref}, while the AA isochrone adds an orthogonal geometric constraint along an isochronal contour as pressure increases. The aim here is to force $T_0(P)$ to follow the observed bending of the TGP isochrone and leave $D(P)$ to encode the residual shape (fragility). The fitted mesh in Fig. 5.11 is thus "stitched" to the 1-bar VTF along the front edge and to the AA line across the top, enabling stable interpolation in the interior where direct relaxation data are sparse.

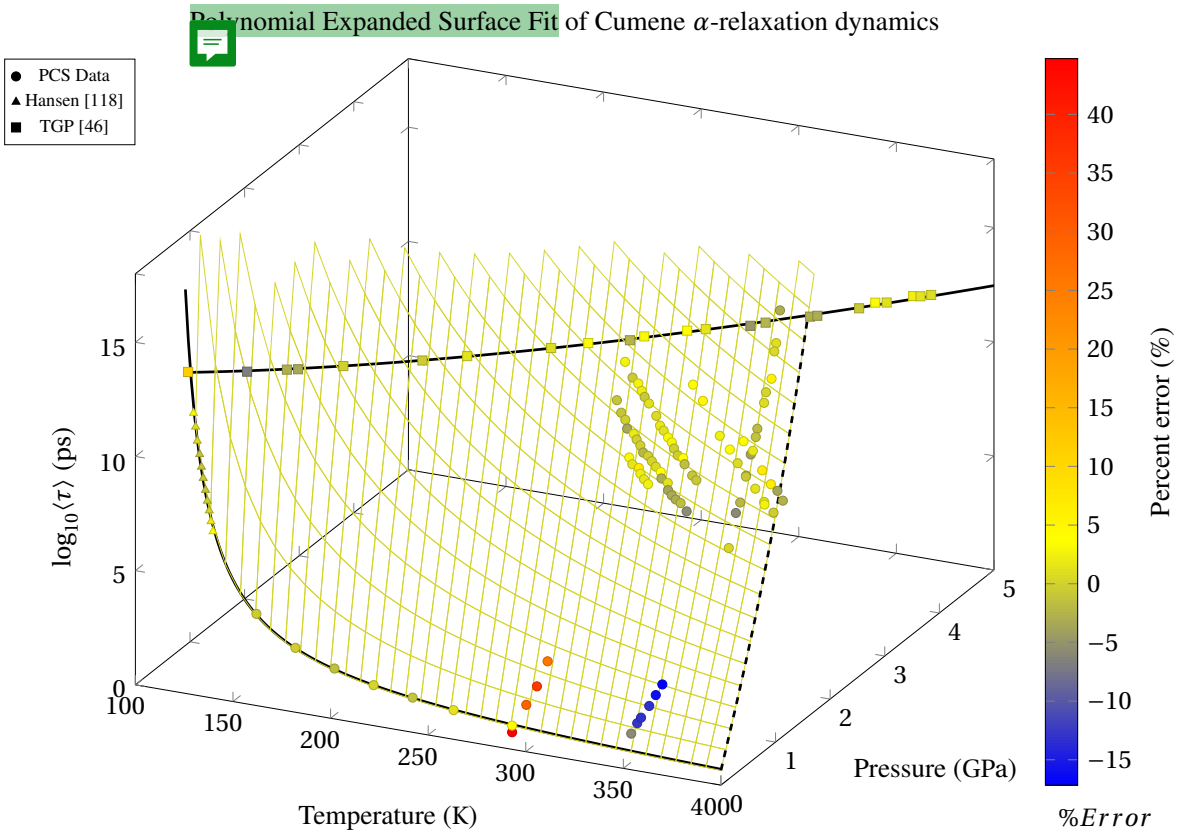


Figure 5.11: Surface fit of relaxation times for cumene as a function of temperature and pressure. Experimental data points (symbols) are overlaid with the fitted Vogel–Fulcher–Tammann (VTF) surface (mesh). The fit incorporates constraints from ambient-pressure data to improve fidelity in the low-pressure regime.

It is evident from the error bars in Fig. 5.11 that the model provides excellent agreement with the α-relaxation dynamics of this simple van der Waals system. However, our low-pressure DPCS runs exhibit noticeably higher error—both relative to the fitted surface and to well-established atmospheric-pressure data. In particular, the room-temperature measurements substantially underestimate the relaxation times as the pressure approaches ambient, while the 75°C data tend to overestimate them, albeit to a more modest extent.

The origin of this discrepancy is not yet clear, though several observations are noteworthy. The room-temperature run was the first successful data set obtained on this system, and a

reasonable hypothesis is that, at relatively low pressures, any impurities present in the DAC sample are significantly more mobile than they would be near the glass transition. Under such conditions, the assumption of purely heterodyne scattering may be unwarranted. Furthermore, the room-temperature and 75°C data arise from different DAC loadings, separated by several years of increased loading experience. It is therefore plausible that the 75°C sample was simply a cleaner, better-prepared load; in that case, despite the higher temperature, its improved adherence to the model would naturally follow from reduced impurity-induced artifacts.

Whatever the case, the remainder of the surface conforms well to our data set. This agreement leaves little doubt that the model adequately describes the relaxation dynamics as a function of pressure and temperature, even in regions where data are absent, provided we remain within the triangular domain bounded by the 1-bar constraint, the AA isochrone, and the dashed line at 400 K. However, once we move beyond the region where data constrain the surface, significantly less confidence can be placed in the model's fidelity.

To further assess fidelity of the surface model, Fig. 5.12 compares the model-predicted and experimental $\log_{10}\langle\tau\rangle$ values for all (T, P) conditions, with each point color-coded by the local percent error (the same metric used on the surface). A linear regression constrained through the origin gives a slope of $m = 0.9523$, close to unity, while the point cloud remains symmetrically distributed about the $y = x$ line. Together, the near-unity slope and balanced scatter indicate that the surface reproduces the experimental α-relaxation times with little systematic bias across the full thermodynamic range.

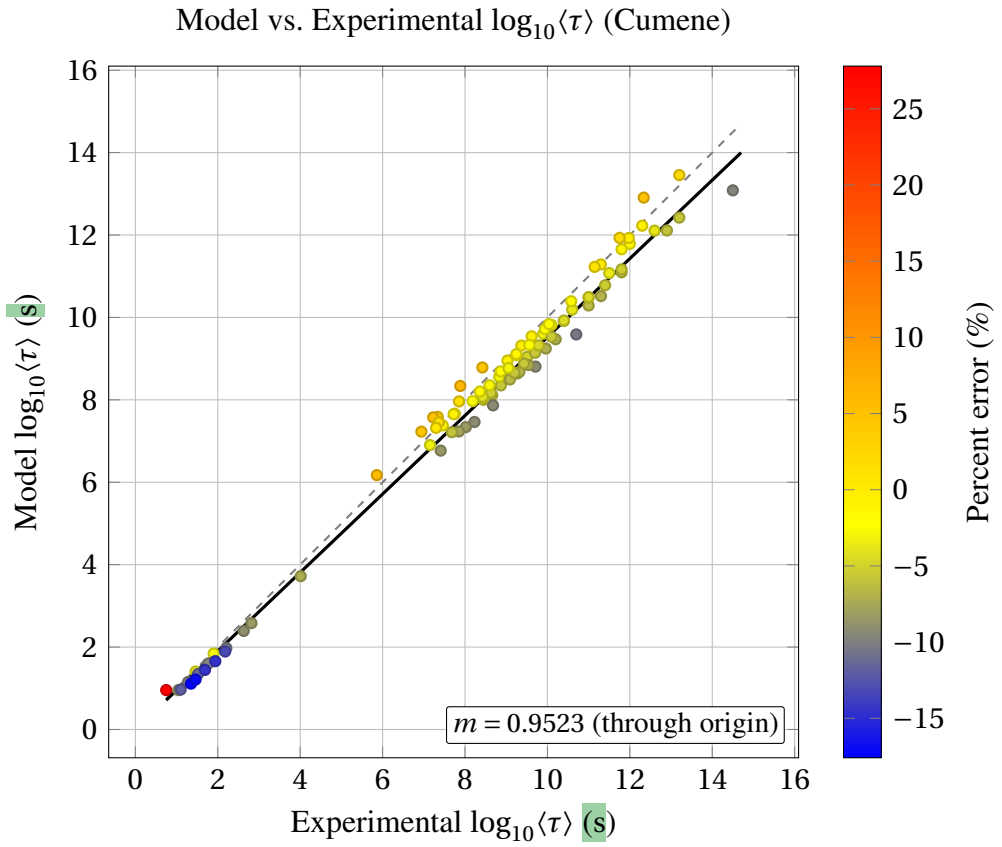


Figure 5.12: Model versus experimental $\log_{10}\langle\tau\rangle$ values for cumene, with points color-coded by the local percent error. The dashed line ($y = x$) denotes perfect agreement between model and data, while the solid black line shows the through-origin regression fit.

Although this model provides confidence when interpolating within the domain constrained by experimental data, there is decidedly less assurance when extrapolating a surface based on an arbitrarily imposed polynomial expansion of the VTF parameters. This lack of robustness under extrapolation represents a significant limitation to the model's broader utility. One of the primary motivations for developing a pressure–temperature dependent surface is to enable investigation of the pressure dependence of fragility, among other derived dynamical quantities. In this context, it is worth emphasizing that the highest–order terms in Eq. 5.18 are negative. As a consequence, the model becomes inherently nonphysical at sufficiently high pressures: rather than asymptotically approaching a regime of zero mobility with decreasing temperature, the surface may instead predict

a nonphysical *decrease* in the α–relaxation time upon cooling. This breakdown is illustrated in Fig. 5.13, which shows the surface extended into regions where it no longer preserves the expected asymptotic behavior.

While the behavior of the surface along isobars—where classical fragility may be assessed as a function of pressure—appears well behaved within the triangular region in which experimental data constrain the fit, caution is warranted when considering dynamics along isotherms. When extrapolated to higher pressures, the model fails to predict a physically meaningful zero–mobility pressure at which relaxation modes should be progressively suppressed. One might reasonably suggest, particularly given that the highest–order coefficients t_3 and d_2 are small, that the surface has been over–parameterized and that a reduced–order polynomial expansion would remedy this issue. However, such reduced–order fits retain negative highest–order terms in both the $\vec{t}$ and $\vec{d}$ parameter vectors, thereby exacerbating the nonphysical behavior under extrapolation rather than correcting it.

These limitations are especially relevant when considering additional derived quantities, such as the activation volume, which depends explicitly on pressure derivatives of the relaxation time at fixed temperature. At elevated temperatures, the nonphysical downturn of the fitted surface renders any calculation that relies on the surface gradient along an isotherm dubious at best. Moreover, beyond the bulk of the relaxation–time data, where only the glass–transition constraint provided by TGP remains, there is no meaningful basis for confidence in assessments of free volume or other pressure–derivative quantities. This consideration further motivates the development of a surface construction that embeds the correct asymptotic approach to zero mobility at low temperatures and high pressures implicitly.

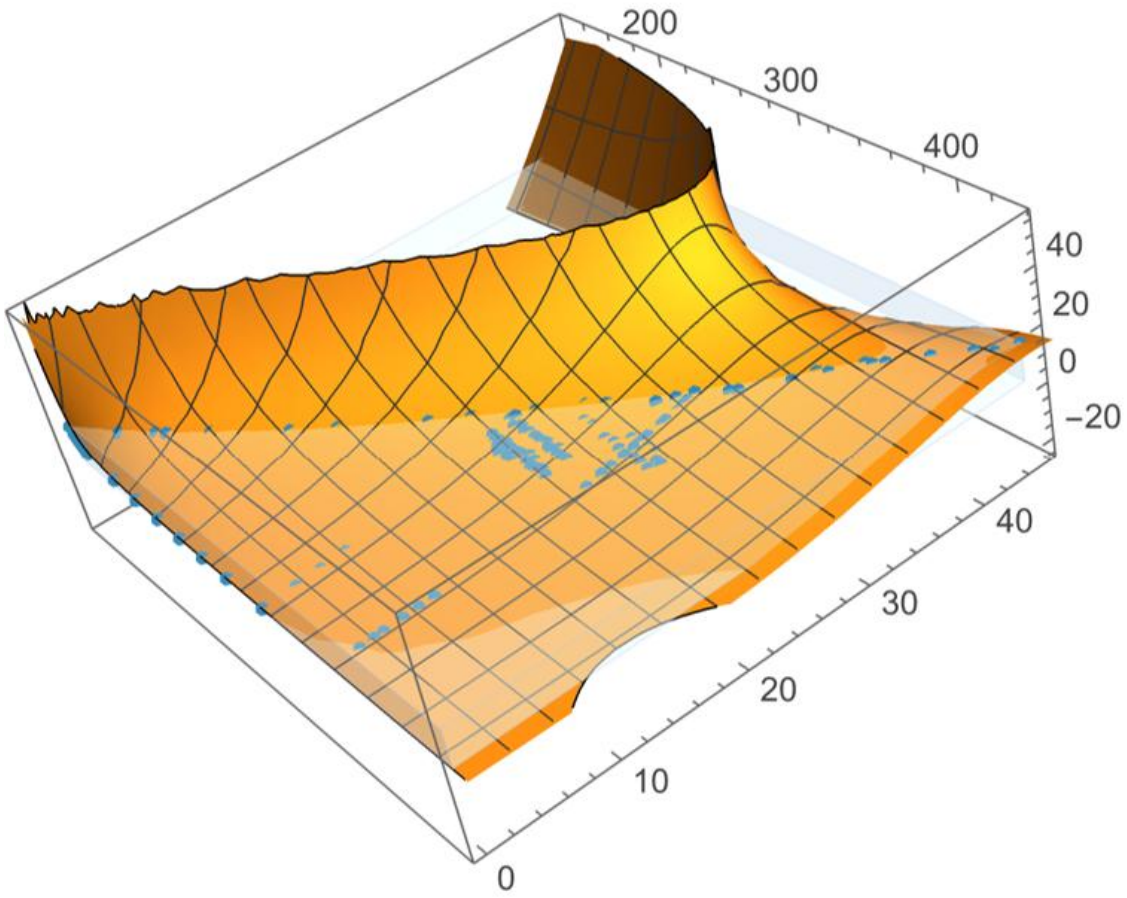


Figure 5.13: Visualization of the non–physical behavior inherent to the polynomial VTF surface expansion. At sufficiently high pressures the fitted surface fails to asymptotically approach zero mobility with decreasing temperature, instead exhibiting an nonphysical downturn in the predicted α-relaxation time. This illustrates the limitations of the polynomial form used in Eq. 5.18.

Even were this not the case, and it may not be for all liquids, the underlying issue remains—a polynomial expansion for the zero-mobility temperature and fragility functions, $T_0(P)$ and $D(P)$, is inherently non-physical. A model that better constrains the surface behavior, in particular its tendency to asymptotically approach a regime in which relaxation modes are suppressed (both for decreasing temperatures and increasing pressures), would be preferable to this simple polynomial expansion. For this reason, attention was turned toward creating a pressure–temperature dependent model of the α-relaxation for which, regardless of the best-fit parameters, the physical

requirement of an asymptotic approach to regions of zero mobility is maintained along any isobar or isotherm. For anyone familiar with developments in the glassy-dynamics community over the last few decades, the "modified" VTF (Eq. 5.20) first suggested by Paluch *et al*. should come to mind—a natural analogue of the classic VTF but expressed as a function of pressure rather than temperature[120].

$$\ln\langle\tau(P)\rangle = \ln\tau_\infty + \frac{D_P\,P_0}{P_0 - P}, \tag{5.20}$$

Historically, the modified VTF was introduced as a direct analogue of the classic VTF relation but reformulated for the pressure path. Paluch and co-workers demonstrated that, along an isotherm, the dramatic slowing down of molecular mobility with increasing pressure follows a divergence form that mirrors the temperature-driven Vogel–Fulcher–Tammann behavior. In this formulation, P_0 plays the role of a zero-mobility pressure, and D_P quantifies the steepness of the divergence, thereby providing a pressure-based analogue of fragility. Because the modified VTF guarantees an asymptotic divergence of the relaxation time as $P \to P_0$, it naturally enforces physically reasonable behavior along every isotherm, in sharp contrast to the polynomial expansions used in Eq. 5.18. This makes it particularly attractive for systems in which pressure-induced vitrification or dynamic arrest is expected, or for cases where extrapolation beyond the directly measured pressure range is required.

5.4.3 Anderson–Anderson-constrained zero-mobility surface

In Eq.5.18 we introduced a Vogel–Tammann–Fulcher (VTF) surface whose temperature–pressure curvature was constrained by the Anderson–Anderson isochrone model fit to our TGP

data from App. C near the glass-transition at approximately $\tau \approx 130$s. There, the AA model was used to enforce the physically required shape of the boundary $T_g(P)$, while the VTF divergence retained its usual temperature dependence.

Here we apply the same Anderson–Anderson model, but with a different interpretation. Instead of modeling the glass-transition isochrone, we assume that the *zero-mobility* isochrone, the locus along which the relaxation time diverges, shares the same AA-type curvature as $T_g(P)$. The rationale is that both the glass transition and the zero-mobility boundary reflect the same limiting physics of suppressed structural relaxation at low temperature and high pressure.

Anderson–Anderson form for the zero-mobility isochrone

The original AA expression for the glass-transition isochrone is as follows:

$$T_g(P) = k_1 \left(1 + \frac{k_2}{k_3} P\right)^{1/k_2} \tag{5.21}$$

To repurpose this shape for the zero-mobility surface, we replace k_1 by the zero-mobility temperature for the VTF model at 1-bar—t_0 from Eq. 5.18.

$$T_0(P) = T_\infty \left(1 + \frac{k_2}{k_3} P\right)^{1/k_2} \tag{5.22}$$

Here $T_\infty = 115.2$K . This ensures that the zero mobility temperature provided by our AA constraint is in agreement with the 1-bar constraint from Fig. 5.10 as with our previous model.

The zero-mobility isochrone can equally well be expressed in terms of the pressure required to reach divergence at a fixed temperature. To obtain this form, we simply solve the Anderson–

Anderson expression for P rather than for T.

$$P_0(T) = \frac{k_3}{k_2}\left[\left(\frac{T}{T_\infty}\right)^{k_2} - 1\right] \tag{5.23}$$

The AA-constrained two-variable VTF surface

Constraining the zero-mobility isochrone introduces only two fit parameters when used in conjunction with the 1-bar VTF constraint. It is suggested here that a sufficient model for the relaxation dynamics of a simple van-der-Waals liquid such as cumene, taking the form of Eq. 5.24, requires only one additional degree of freedom—a constant D_p analogous to the fragility parameter D_0 in the classical VTF expression.

$$H(T,P) = \log(\tau_{\text{inf}}) + \frac{D_0\, T_0(P)}{T - T_0(P)} + \frac{D_p\, P}{P_0(T) - P} \tag{5.24}$$

Substituting Eqs.5.22 and5.23 into Eq. 5.24 gives the explicit form of Eq. 5.25,

$$H(T,P) = \log(\tau_{\text{inf}}) + \frac{D_0\, T_\infty \left(1 + \frac{k_2}{k_3} P\right)^{1/k_2}}{T - T_\infty \left(1 + \frac{k_2}{k_3} P\right)^{1/k_2}} + \frac{D_p\, P}{\frac{k_3}{k_2}\left[\left(\frac{T}{T_\infty}\right)^{k_2} - 1\right] - P}, \tag{5.25}$$

where $\tau_\infty = 1.118179$ ps, $T_\infty = 115.201$ K, and $D_0 = 1.59655$ are fixed from the 1-bar constraint, leaving only D_p, k_2, and k_3 as degrees of freedom in the nonlinear surface fit.

The resulting parameter values are listed in Tbl. 5.3. The corresponding fitted surface is shown in Fig. 5.14, where the rear vertical isochrone surface represents the zero-mobility asymptote, and a second vertical isochrone surface marks the 130s TGP condition. The increasing separation between these two vertical isochrone surfaces with rising pressure reflects the expected

decrease in fragility at higher pressures.

It is important to note that this model constitutes a zero-mobility–constrained surface. Both the TGP and zero-mobility isochrones are generated using the same Anderson–Anderson regression; however, it is the additional modified-VTF-shaped constraint imposed along the isotherms that forces the relaxation dynamics to asymptotically approach a physically meaningful region of zero mobility for *all* temperatures and pressures. This stands in sharp contrast to the earlier polynomial-expansion model, whose form allowed nonphysical behavior away from the data domain and did not enforce divergence along any thermodynamic path. By embedding the correct asymptotic structure directly into the surface, rather than attempting to approximate it with independent polynomial terms, the present model permits reliable extrapolation beyond the measured (T, P) range. Such extrapolation is essential for assessing the pressure and temperature-dependent fragility, particularly in regions where direct measurements are not experimentally accessible.

Parameter	**Estimate**	**Standard Error**	**t-Statistic / P-Value**
D_p	3.91293	0.228946	17.0911 / 3.60118×10^{-34}
k_2	1.71628	0.0125992	136.222 / 3.76332×10^{-135}
k_3	13.7721	0.213287	64.5708 / 3.49584×10^{-96}

Table 5.3: Fit parameters for the three-degree-of-freedom relaxation-surface model of Eq. 5.24. The table lists the parameter estimates together with their standard errors, t-statistics, and associated p-values derived from the nonlinear regression. These parameters define the Anderson–Anderson–constrained VTF surface for the zero-mobility isochrone.

It is evident from inspection of the relaxation surface in Fig. 5.14 that the fragility of cumene decreases with increasing pressure. In the present framework, fragility is defined not through the classical VTF parameters of the earlier polynomial model, but directly from the geom-

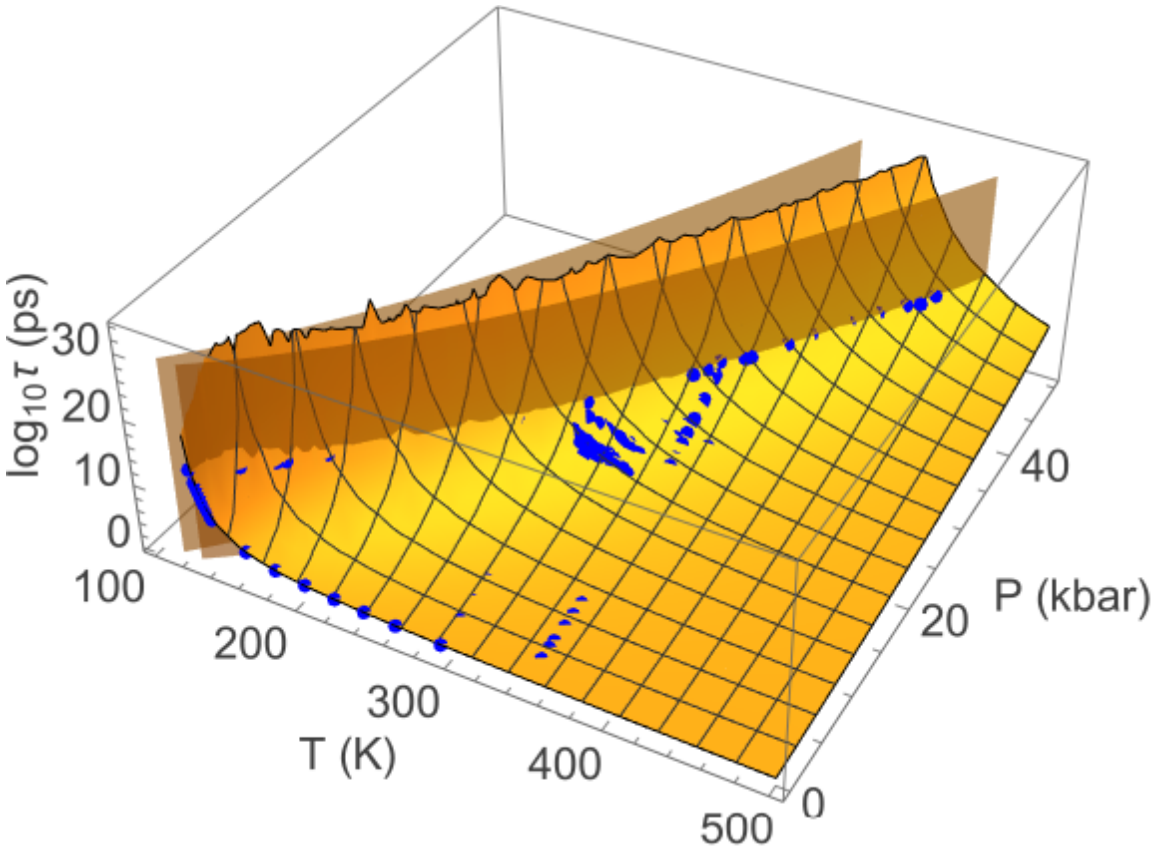


Figure 5.14: Fit surface resulting from the three-degree-of-freedom model of Eq. 5.24, constrained by the Anderson–Anderson representation of the zero-mobility isochrone. The rear vertical sheet represents this zero-mobility asymptote $T = T_0(P)$ (equivalently, $P = P_0(T)$), while the curved surface shows the fitted relaxation landscape over (T, P) space. Experimental data are overlaid as blue points.

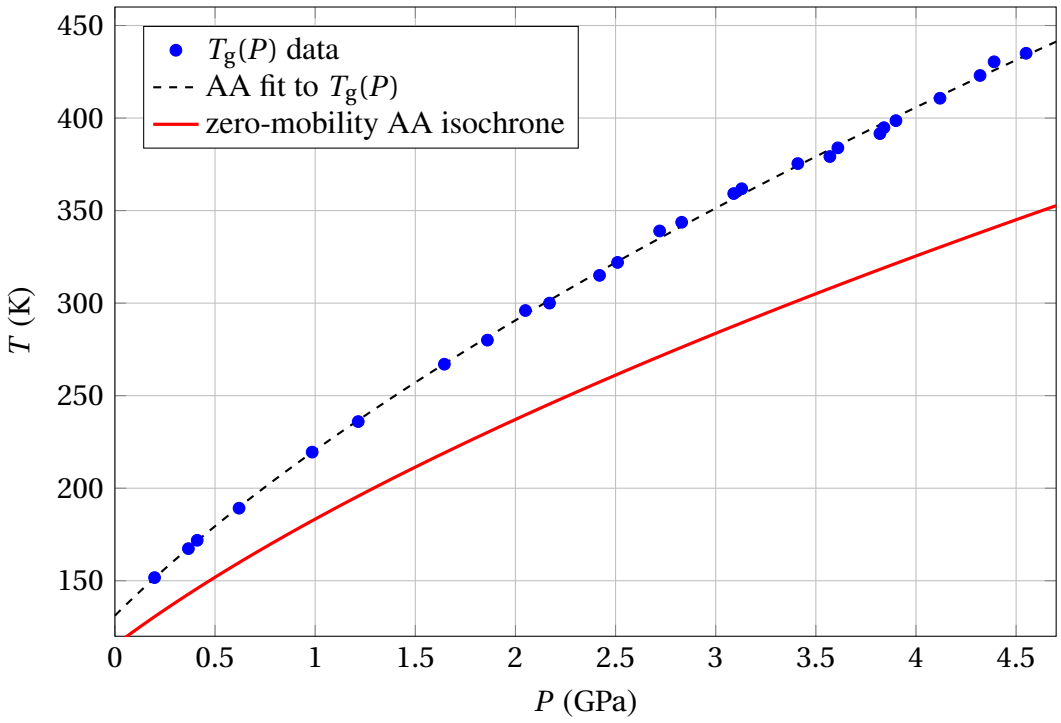


Figure 5.15: Cumene glass-transition temperatures $T_g(P)$ with the Anderson–Anderson fit from App. C, together with the AA zero-mobility isochrone for comparison. The decrease in the classical isobaric fragility with increasing pressure is evident in the growing separation between these two isochronal curves.

etry of the logarithmic relaxation surface

$$H(T,P) \equiv \log_{10} \tau_\alpha(T,P), \tag{5.26}$$

given by the Anderson–Anderson–constrained VTF form of Eq. 5.18. This formulation naturally allows fragility to be expressed in terms of the gradient of the surface, thereby providing a more physical interpretation of how relaxation modes are suppressed along isobars and isotherms.

Temperature–based scaled fragility

Following the scaled definition of fragility introduced in Sec. 3.1 (Eq. 3.2), the isobaric scaled fragility is obtained by differentiating $H(T,P)$ with respect to the reduced temperature $T_g(P)/T$ at fixed pressure:

$$m^*(P) \equiv \frac{1}{\Delta} \frac{\partial H}{\partial\big(T_g(P)/T\big)}\bigg|_{T=T_g(P)} \tag{5.27}$$

where $\Delta = \log_{10}(\tau_g/\tau_\infty)$ is the scaled dynamic range between the infinite–temperature and glass–transition relaxation times. Because $T_g(P)$ is constant along an isobar,

$$\frac{\partial}{\partial(T_g/T)} = -\frac{T^2}{T_g}\frac{\partial}{\partial T}$$

and therefore the scaled fragility can be written compactly in terms of the gradient of the relaxation surface:

$$m^*(P) = -\frac{T_g(P)}{\Delta}\,(\nabla H)_T|_{T=T_g(P)} \tag{5.28}$$

Physically, this quantity reflects how rapidly the dynamics slow when the temperature is decreased slightly below $T_g(P)$ at fixed pressure. Because small variations in temperature do not strongly modify the underlying potential–energy landscape near the glass transition, $m^*(P)$ primarily characterizes the *probe response* of the system—how sharply the available relaxation pathways become dynamically inaccessible as thermal energy is withdrawn. A strong argument was made previously for the adoption of the scaled fragility definition given in Eq. 3.2. Nevertheless, to facilitate direct comparison between fragility values derived from the present surface and those reported elsewhere in the literature, it is appropriate to continue working with the classical Angell definition of fragility. Accordingly, results will primarily be discussed in terms of the conventional

fragility, although, where easily accommodated, both definitions will be presented.

Before proceeding to an analysis of the pressure dependence of isobaric fragility, it is first necessary to address the fragility of cumene at atmospheric pressure. Reported values of this parameter exhibit a notable degree of variability within the literature, even after accounting for differences in how the glass transition temperature is defined, with most studies adopting the standard criterion of $\tau = 100\,\mathrm{s}$. Understanding the origin of this discrepancy serves not only to validate the surface model developed here, which relies critically on a fixed isobaric regression at atmospheric pressure, but also to reinforce a conclusion that is already well established in the literature. Specifically, light scattering experiments, as a non-invasive probe, provide a more faithful measure of the intrinsic α-relaxation associated with the stochastic thermal evolution of the system, even in a comparatively simple liquid such as cumene, which might otherwise be expected to lend itself equally well to dielectric spectroscopy and DPCS.

Our surface model, anchored by a 1–bar VTF regression, relies on data collected by Hansen and Li *et al*., together with the TGP point derived from the Anderson–Anderson fit described in App. C. Fig. 5.10 demonstrates remarkably good agreement between the model and the data over the full dynamic range. A slight tendency for the VTF fit to underestimate the data is observed as temperatures approach 160 °C; however, this region lies near the upper operational limit of Brillouin light scattering, where the extracted relaxation times begin to approach the Bose peak. Consequently, confidence in these intermediate–temperature points is reduced.

Importantly, a substantial body of ambient–pressure data exists for cumene, and the 1–bar datasets employed here were selected specifically because they were obtained using light–scattering probes—DPCS in the case of Hansen and Brillouin light scattering in the work of Li *et al*.. The TGP point would likewise be expected to provide a direct measure of the structural

relaxation, as few experimental observations are more direct than the physical vitrification of the material. However, as discussed previously in the description of the surface model, the precise relaxation time τ_i associated with this isochrone is not directly experimentally measurable and is instead constrained by where it crosses this atmospheric VTF isobar. The corresponding isochronal value is nonetheless independently verifiable in the case of cumene via the intercept obtained from the thermodynamic scaling of the data shown in Fig. 5.8.

Pressure dependence of fragility

The surface model defined in Eq. 5.25 provides a robust and internally consistent framework for examining the pressure dependence of isobaric fragility. In the discussion that follows, we focus primarily on the conventional Angell fragility in order to facilitate direct comparison with previously reported literature values. For completeness, the pressure dependence of both the conventional fragility and the scaled fragility $m^*(P)$ defined in Eq. 5.28 are shown in Fig. 5.16.

At ambient pressure, the fragility extracted from the surface lies slightly below the commonly cited values for cumene discussed previously. This difference is attributable in part to the use of light-scattering data in contrast to DS, as well as to the manner in which fragility is evaluated within the present framework. Specifically, the fragility reported here is obtained directly from the local gradient of the relaxation-time surface, rather than from fits restricted exclusively to the viscous regime. As a consequence, the resulting value is sensitive to the overall goodness of fit rather than to enforced low-temperature curvature alone.

Despite this modest reduction in magnitude at low pressure, the qualitative pressure evolution of the fragility is clear and robust. The fragility exhibits a smooth but distinctly nonlinear decline with increasing pressure, consistent with the growing separation between the glass-transition

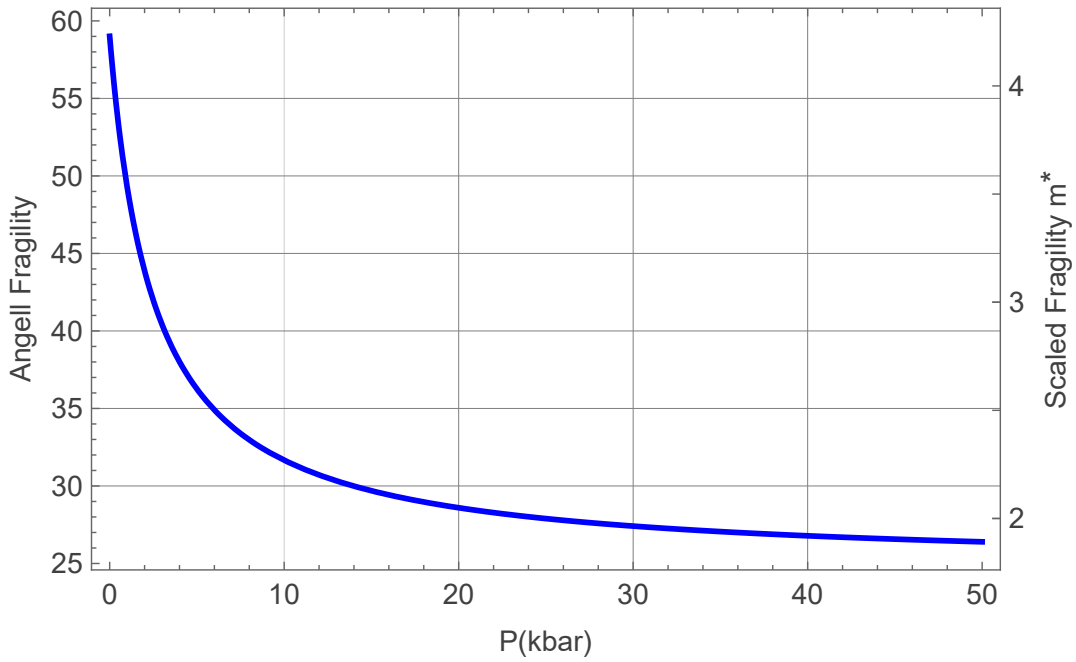


Figure 5.16: Pressure dependence of the isobaric fragility of cumene obtained from the relaxation-time surface. The left axis shows the conventional Angell fragility m, while the right axis displays the scaled fragility m^* defined in Eq. 5.28. When viewed in terms of the scaled fragility, the data clearly suggest an asymptotic approach toward a finite high-pressure limit that retains more fragility than would arise from purely Arrhenius behavior.

isochrone $T_g(P)$ and the zero-mobility isochrones shown in Fig. 5.11. With increasing pressure, the system becomes progressively stronger, approaching Arrhenius-like behavior, consistent with a stiffening of the underlying relaxation landscape.

One particularly striking feature, especially evident when viewed in terms of the scaled fragility, is the emergence of a high-pressure asymptote. It is often implicitly assumed that increasing pressure yields a stronger liquid, with the limiting case being purely Arrhenius behavior. Although the fragility of cumene decreases rapidly as cooperative dynamics are suppressed, these dynamics do not appear to be fully eliminated. The limiting fragility obtained from the surface model of Eq. 5.25 as pressure approaches infinity is approximately $m^* = 1.8$, corresponding to $m = 25$ in terms of the conventional Angell fragility. This represents a clear instance in which the

utility of the scaled fragility defined in Eq. 3.2 becomes evident. As shown in Fig. 5.16, even without extrapolation beyond the experimentally accessible data, the scaled fragility is clearly converging toward a value significantly greater than unity. This limiting behavior remains distinctly more fragile than that of a purely Arrhenius system, indicating that increasing pressure does not drive the liquid into a truly strong glass-forming regime. Instead, the surface model suggests the persistence of residual non-Arrhenius character even at elevated pressures, consistent with an energy landscape that becomes increasingly stiff while retaining a finite degree of dynamical heterogeneity.

The existence of this asymptotic fragility is not imposed by the model but rather it emerges naturally from the global structure of the relaxation-time surface. The rapid approach toward this asymptote well within the domain of the available relaxation data lends confidence to its physical significance. At least in the case of cumene, it appears that the system cannot be driven fully into Arrhenicity by the application of pressure alone.

Isothermal pressure–modified fragility

An entirely analogous expression follows for compression along isotherms. Defining the glass–transition pressure $P_g(T)$ via $H\big(T, P_g(T)\big) = \log_{10}\tau_g$, the pressure–modified fragility is defined with respect to the reduced pressure and may be written as

$$m_P^*(T) \equiv \frac{1}{\Delta}\frac{\partial H}{\partial (P/P_g(T))}\bigg|_{P=P_g(T)} = \frac{P_g(T)}{\Delta}\,(\nabla H)_P|_{P=P_g(T)} \tag{5.29}$$

In contrast to temperature variation, increasing pressure directly alters the intermolecular structure: local cages tighten, barrier heights increase, and the potential–energy landscape itself is reshaped. Thus $m_P^*(T)$ represents a *susceptibility of the landscape to mechanical compression*,

rather than a probe of dynamical accessibility.

For this reason, it is the opinion of the author that the isothermal pressure–modified fragility should be understood as fundamentally *not* a measure of landscape fragility; rather, it quantifies how readily the landscape itself deforms under pressure—no more a "fragility" than EMF is a force. Even so, it remains a useful parameter when investigating vitrification under compression, particularly as one approaches pressures for which all relaxation modes are suppressed and the structural entropy becomes effectively locked in. This "pressure quenching" is, in many cases, a more practical route to rapid vitrification when thermal quenching is too slow to avoid crystallization.

Connection to activation volume

The same isothermal pressure derivative that defines the pressure–modified fragility also appears naturally in the conventional definition of the activation volume. For a relaxation process described by $\tau(T,P)$, the molar activation volume is defined as

$$\Delta V^{\ddagger}(T,P) \equiv RT \left.\frac{\partial \ln \tau(T,P)}{\partial P}\right|_{T}, \tag{5.30}$$

where R is the universal gas constant. Expressed in terms of the surface $H = \log_{10}\tau$ implemented here, Eq. 5.30 becomes

$$\Delta V^{\ddagger}(T,P) = (\ln 10)\, RT\, (\nabla H)_P|_T\,. \tag{5.31}$$

Evaluated along the glass–transition isochrone $P = P_g(T)$, the activation volume is therefore directly proportional to the same local surface derivative that enters the definition of the

pressure–modified fragility:

$$\Delta V_g^{\ddagger}\big(T, P_g(T)\big) = (\ln 10)\, R\, T_g(T)\, \frac{\Delta}{P_g(T)}\, m_P^*(T). \tag{5.32}$$

This relation makes explicit that $m_P^*(T)$ provides a scaled measure of the pressure sensitivity of the dynamics, normalized by the glass–transition pressure and the accessible dynamic range. When combined with the appropriate thermodynamic prefactors, it yields the conventional activation volume $\Delta V_g^{\ddagger}(T,P)$, which retains its usual interpretation as a molar volume characterizing pressure–induced slowing of the structural dynamics.

Within the surface–based framework developed here, activation volume and modified fragility are thus revealed to be complementary projections of the same relaxation–surface gradient. The distinction lies not in the derivative itself, but in the physical interpretation imposed by the choice of scaling: multiplication by $P_g(T)$ yields a scaled susceptibility describing how readily the relaxation landscape deforms under compression, whereas multiplication by $RT_g(T)$ recovers the conventional activation volume familiar from high–pressure relaxation studies. This unification clarifies the relationship between pressure fragility and activation volume without requiring separate fits or auxiliary models, and highlights the utility of the constrained surface in providing physically consistent derivatives across the full (T,P) domain.

Isochronic fragility and surface geometry

The preceding measures of fragility emphasize either thermal variation at fixed pressure or mechanical compression at fixed temperature. A more neutral characterization follows by considering variations taken *along an isochrone* of the relaxation surface itself. Within the present

framework, this quantity arises naturally from the local geometry of the surface $H(T,P)$ and requires no additional fitting assumptions.

Along an isochrone defined by $H(T,P) = \log_{10}\tau = \text{const}$, the total differential vanishes,

$$\left.\frac{\partial H}{\partial T}\right|_P dT + \left.\frac{\partial H}{\partial P}\right|_T dP = 0, \tag{5.33}$$

so that the slope of the isochrone in the (T,P) plane is given by

$$\left.\frac{dP}{dT}\right|_H = -\frac{(\partial H/\partial T)_P}{(\partial H/\partial P)_T}. \tag{5.34}$$

This ratio of surface–gradient components provides a direct measure of the relative influence of temperature and pressure on the dynamics at fixed relaxation time and may be identified as an isochronic fragility. Unlike isobaric or isothermal measures, this quantity does not privilege a particular thermodynamic path, but instead reflects the intrinsic geometry of the relaxation surface.

If an equation of state is invoked to transform the surface from (T,P) to (T,V) coordinates, an equivalent isochronic fragility may be obtained in terms of the slope of constant–τ contours in (T,V) space. In this representation, the separation between thermal and volumetric contributions to the dynamics becomes explicit, and the conventional isochoric fragility discussed in the literature is recovered as a particular projection of the same surface geometry. The agreement between these equivalent constructions emphasizes that isochronic, isobaric, and isochoric fragilities are not independent material properties, but rather complementary manifestations of the same underlying relaxation landscape.

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A Andersson-Andersson Constrained Surface

Equation 5.18 in Sec. 5.4.2 extends the classic isobaric VTF model to account for variation in pressure simply by allowing the constants $\ln(\tau_\infty)$, D, and T_0 to be polynomial functions of pressure. In light of our TGP process providing an extremely well defined isochronal curve corrosponding to the glass transition temperature, T_g, via the Andersson-Andersson equation, it makes sense to constrain this surface model to this isochronal curve[48]. The standard isobaric VTF equation may be solved for the zero mobility temperature, T_0, in terms of the glass transition temperature, T_g, as follows:

$$\begin{aligned} \ln(\tau_g) &= \ln(\tau_\infty) + \frac{DT_0}{T - T_0} \\ \Rightarrow T_0 &= \frac{\left(\ln(\tau_\infty) - \ln(\tau_g)\right) T_g}{\ln(\tau_\infty) - \ln(\tau_g) - D} \end{aligned} \tag{A.1}$$

Substituting this back into the original VTF model for an arbitrary temperature, T, yields the following model in which all parameters are again functions of pressure and the glass transition temperature T_g is provided by the Andersson-Andersson equation from appendix B. This model was fit to both alpha relaxation data, τ, all measured in picoseconds and viscosity data, η, all measured in centipoise extracted from the work of Cook, McDuffie, Paluch, Pronin, Reiser, Broden, and Johari[51, 95, 73, 66, 98, 99, 29].

$$\ln(\tau) = \ln(\tau_\infty) + \frac{\left(\ln(\tau_\infty) - \ln(\tau_g)\right) D T_g}{\left(\ln(\tau_\infty) - \ln(\tau_g) - D\right) T - \left(\ln(\tau_\infty) - \ln(\tau_g)\right) T_g}$$

FOR:

$$D = \sum_{j=0}^{3} d_j P^j \quad \textit{AND} \quad \ln(\tau_\infty) = \sum_{j=0}^{1} t_j P^j$$

WHERE $\ln(\tau_g) \equiv 32.489$ *AND* *VECTORS*: (A.2)

$$\vec{d} = \begin{pmatrix} 19.0557 \\ -0.690394 \\ 0.0115522 \\ -0.0000719137 \end{pmatrix} \quad \vec{t} = \begin{pmatrix} -6.73079 \\ 0.258835 \end{pmatrix}$$

To further constrain the surface, the values of d_0 and t_0 were not included in the Levenberg-Marquardt surface fit routine, but rather obtained from a standard VTF fit to ambient pressure isobaric data from Brodin, Pronin, and Nagai shown below[**Nagai**, 66, 99]. This atmospheric isobar was also used to determine the characteristic time/viscosity for the glycerol TGP isochronal curve via assuring that both curves cross on the surface at zero pressure.

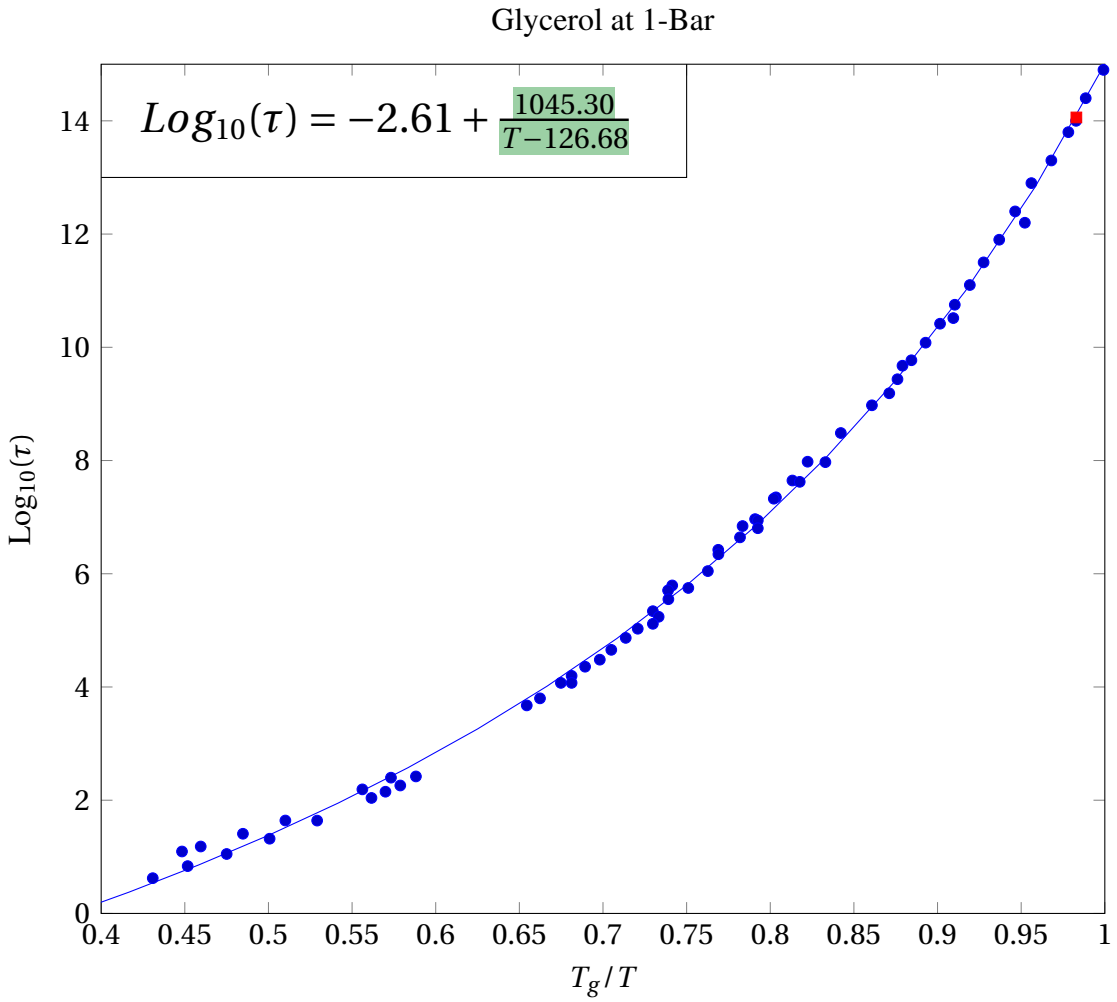


Figure A.1: Shown is the classic Angel Plot for glycerol at atmospheric pressure resulting in a glass transition temperature of T_g = 185.14 K defined at $\tau_g \equiv 100$ s. From the associated VTF model fit we may define the zero order coefficients of the Andersson-Andersson surface model as $d_0 = 19.06$ and $t_0 = -6.73$ leaving the nonlinear surface fit with only four remaining degrees of freedom. The average relaxation time associated with the isochrone obtained from the glycerol TGP experiments at atmospheric pressure is also shown above (■) corresponding to an α-relaxation time or viscosity of $\approx 10^{14}$ seconds or centipoise respectively. This TGP isochrone was latter determined to greater precision via the linearizion of the TGP data as reported in figure 4.5. This more precise value of $\tau = 220$ s was implemented when providing constraint to the Andersson-Andersson surface model.

B TGP Data (Glycerol)

Table B.1: Glycerol TGP Data

Pressure GPa	Temperature K		Volume $cm^3\,g^{-1}$
0.000	187.000		1.154
0.460	203.000		1.095
0.770	212.000		1.056
1.070	224.000		1.027
1.110	225.000		1.023
1.660	239.000	*	0.982
1.910	244.000		0.968
2.300	253.000		0.951
2.500	261.000		
2.790	263.000		
2.830	265.000		
2.910	265.000		
3.270	276.000		
3.650	285.000	**	0.915
3.800	287.000		0.912
4.220	297.000		0.904
4.240	297.000		0.904
4.680	308.000		0.898
4.890	310.000		0.894
4.950	315.000		0.895
5.170	320.000		0.891
5.470	326.000		0.887
5.800	328.000		0.881
6.380	338.000		0.873
6.550	341.000		0.871
6.680	344.000		0.869

* Specific volumes corresponding to this pressure and above were obtained with moderate extrapolation of the Tait EOS beyond Bridgman's data

$$T_G(P) = 188.376\left(1 + 2.046\frac{P}{5.596}\right)^{\frac{1}{2.046}} \tag{B.1}$$

C TGP Data (Cumene)

Table C.1: Cumene TGP Data

Pressure GPa	Temperature K	Volume $cm^3\,g^{-1}$
0.198	152.000	0.971
0.367	167.000	0.948
0.411	172.000	0.943
0.620	189.000	0.921
0.985	220.000	0.893
1.215	236.000	0.878
1.645	267.000	0.857
1.860	280.000	0.847
2.050	296.000	0.841
2.170	300.000	0.836
2.420	315.000	0.827
2.510	322.000	0.825
2.720	339.000	0.820
2.830	344.000	0.816
3.090	359.000	0.809
3.130	362.000	0.808
3.410	375.000	0.801
3.570	379.000	0.797
3.610	384.000	0.796
3.820	392.000	0.791
3.840	395.000	0.791
3.900	399.000	0.790
4.120	411.000	0.785
4.320	423.000	0.781
4.390	430.000	0.780
4.550	435.000	0.776

$$T_G(P) = 131.145\left(1 + 1.64181\frac{P}{1.21852}\right)^{\frac{1}{1.64181}} \tag{C.1}$$

D Equation of State (cumene)

All conversions from pressure to density or specific volume for cumene in this work were obtained via the modified Tait equation of state developed by Tim Ransom as shown in equation D.1 below [47]. The density as a function of temperature at atmospheric pressure, $\rho_0(T)$, was provided by Cibulka [121].

$$\rho(T,P) = \frac{\rho_0(T)}{1 - C(T)\,ln\left(\frac{B(T)+P}{B(T)+0.0001GPa}\right)}$$

$$FOR \quad VECTORS:$$

$$B(T) = \sum_{j=0}^{4} b_j \left[\frac{T - T_0}{100}\right]^j \quad AND \quad C(T) = \sum_{j=0}^{1} c_j \left[\frac{T - T_0}{100}\right]^j$$

$$WHERE \quad T_0 \equiv 298.15\ \mathrm{K} \quad AND: \tag{D.1}$$

$$\vec{b} = \begin{pmatrix} 0.11110\ \mathrm{GPa} \\ -0.080954\ \mathrm{GPa\,K^{-1}} \\ 0.0226\ \mathrm{GPa\,K^{-2}} \\ -0.0034\ \mathrm{GPa\,K^{-3}} \\ 0.00028\ \mathrm{GPa\,K^{-4}} \end{pmatrix} \quad \vec{c} = \begin{pmatrix} 0.093736 \\ -0.0040032\ \mathrm{K^{-1}} \end{pmatrix}$$

E Equation of State (glycerol)

To facilitate the conversion of glycerol pressure data (GPa) to specific volume ($Å^{-3}$) the Tait equation of state, fit to Bridgman's glycerol data, was implemented [97]. It should be noted that Bridgman only obtained data up to just under 1.2 GPa yet Cook *et al*. extrapolates this equation of state up to 3 GPa [51]. This work extrapolates this equation of state up to 6.68 GPa to allow for the thermodynamic scaling assessment of Sec. 4.5. Clearly, experimental measurement of the glycerol equation of state at these elevated pressures is desperately needed yet remains absent within the literature. However, despite dependence upon these extreme extrapolations, thermodynamic scaling of glycerol data implementing this equation of state yields self-consistent results as was demonstrated in Fig. 4.5. The isothermal Tait equation of state is given as follows:

$$\frac{V}{V_0} = 1 - c \times Ln\left(\frac{b+P}{b+P_0}\right), \tag{E.1}$$

where P_0 and V_0 are simply arbitrarily chosen reference values of pressure and volume respectively at a specific temperature. For this work P_0 was set to atmospheric pressure and V_0 was found to be sufficiently well modeled as a linear function of temperature as follows:

$$V_0(T) = 0.065T + 119.7, \tag{E.2}$$

where temperature, T, is expected in (°C) and the resulting specific volume, V_0, is returned in units of ($Å^3/molocule$). The density data for glycerol to which Eq. E.2 was fit was taken from

Cheng [122]. The remaining fit parameters c and b may be trivially related to the bulk modulus K_0 and its pressure derivative $K_0^{'}$ via Eq. E.3.

$$K_0 = \frac{P_0}{c} + \frac{b}{c} \quad and \quad K_0^{'} = \frac{1}{c} - 1 - Log\left(\frac{b}{b+P_0}\right) \tag{E.3}$$

The reported values of the bulk modulus, K_0, and its pressure derivative, $K_0^{'}$, calculated from five separate isothermal Tait equations of state by Cook *et al*. was fit to a third order polynomial model shown in Fig. E. This allows for the accurate interpolation of K_0 and $K_0^{'}$ at any arbitrary temperature within this range.

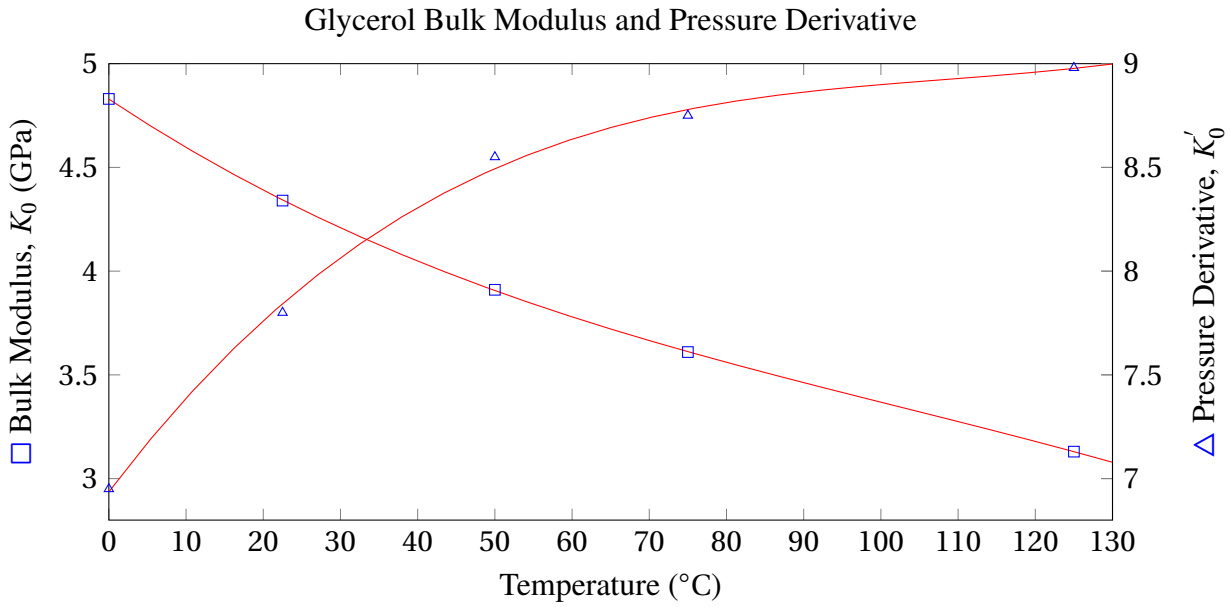


Figure E.1: The glycerol bulk modulus, K_0, and its pressure derivative, $K_0^{'}$, as extracted from reference [51] are provided above. Each are fit to a third order polynomial to allow for interpolation to arbitrary temperatures within the range of 0 °C to 130 °C.

From the interpolated values of the glycerol bulk modulus and its first pressure derivative, the c and b constants of the Tait model provided in Eq. E.1 can be obtained via the simultaneous solution of Eq. E.3.

F Cumene Data

Table F.1: Cumene α-relaxation data

Pressure	Temperature	Volume	$Log_{10}(\tau_{cd})$
GPa	K	$cm^3\,g^{-1}$	psec
1.809	302.41	0.855269301	9.501225002
1.796	304.34	0.856450603	9.265885723
1.851	306.16	0.853731579	9.024928767
1.814	307.98	0.856237884	8.859162925
1.813	309.87	0.856721661	8.648566862
1.823	311.65	0.856549883	8.438877631
1.864	294.55	0.850429432	11.75429022
1.857	296.95	0.851347869	11.28683581
1.866	298.8	0.851259543	10.97157456
1.834	300.65	0.853456629	10.70641649
1.888	302.5	0.850865834	10.41382764
1.875	304.35	0.851991538	10.16823223
1.879	306.15	0.852169269	9.952990421
1.862	308.05	0.853536996	9.704193429
1.889	309.77	0.852418889	9.52735403

Pressure	Temperature	Volume	$Log_{10}(\tau_{cd})$
GPa	K	$cm^3 g^{-1}$	psec
1.896	311.55	0.852426944	9.313236403
1.904	313.35	0.852384843	9.078420565
1.931	315.15	0.851305467	8.826515356
1.876	317.05	0.854761127	8.66865517
1.92	319.05	0.852766801	8.421121658
1.87	320.59	0.855889794	8.218191399
1.875	322.33	0.855997927	8.014881601
1.883	324.15	0.855955509	7.838815731
1.904	326.05	0.855205469	7.674126902
1.883	329.73	0.857206242	7.404269109
2.013	294.57	0.842514964	13.180914
1.998	298.81	0.844196189	12.63691797
2.036	300.7	0.842646706	12.28122205
2.034	302.51	0.843134939	12.00560945
2.018	304.45	0.844372076	11.80165749
2.037	306.15	0.843758895	11.49803058
2.042	309.73	0.844268132	10.98816149
2.044	311.61	0.844567843	10.61174454
2.046	313.41	0.844850584	10.36503661
2.062	315.22	0.844418201	10.11494108

Pressure	Temperature	Volume	$Log_{10}(\tau_{cd})$
GPa	K	$cm^3 g^{-1}$	psec
2.067	317.01	0.844545332	9.887492957
2.044	318.77	0.846103751	9.687611888
2.043	320.73	0.84657707	9.423214735
2.087	322.26	0.844647502	9.241429424
2.045	324.15	0.847210031	9.0776077
2.054	327.81	0.847532	8.626835981
2.062	329.7	0.84752475	8.434821105

G Serial Specifications

Baud rate:	9600 bps
Data bits:	8
Stop bits:	1
Parity:	None
Flow control:	None
Termination:	Line feed (\n)
Command Set:	

atd	Returns last analog to digital conversion from 777 rate-meter as number between 0 and 1024 (technically between 0 and 615 due to limitations of the LM358N)
max	Returns atod value corresponding to max output of the LM358N amplifier as it is not capable of swinging to the positive rail. This value is set in firmware and can only be updated by re-flashing the firmware. Its current value is 615 corresponding to approximately 3 volts.
pmt?	Returns current setting for maximum PMT counts as number between 0 and 1024 (technically between 0 and 615 due to limitations of the LM358N)
c#r	Channel (# number between 0 and 4) read. Returns both the stored channel position and width as follows: Channel # Position: 0450 Width: 050
c#s#	Channel (# number between 0 and 4) set (# number between 0 and 65536). Sets channel start position at which shutter will be closed. Channel is disabled by setting this to location 0, thus it is not possible to shutter on the 0th channel.
c#w#	Channel (# number between 0 and 4) width (# number between 0 and 255). As these are only stored in an 8-bit location, it is only possible to shutter a maximum width of 255 with a single channel. However, if a larger shutter is needed, multiple channels may be used sequentially.
pmt#	pmt (# number between 0 and 1024) This will set the value of the pmt max counts at which point the shutter will be held closed to protect from excessive counts. Again, technically this value must be less than 615 as per the firmware set max value corresponding to the upper limit of the LM358N. To offer the highest possible efficacy of this protection, this value should be set as low as possible without effecting data collection.

Table G.1: RS232 serial specifications and command list

H Schematics

H.1 SPEX500M Grating Drive

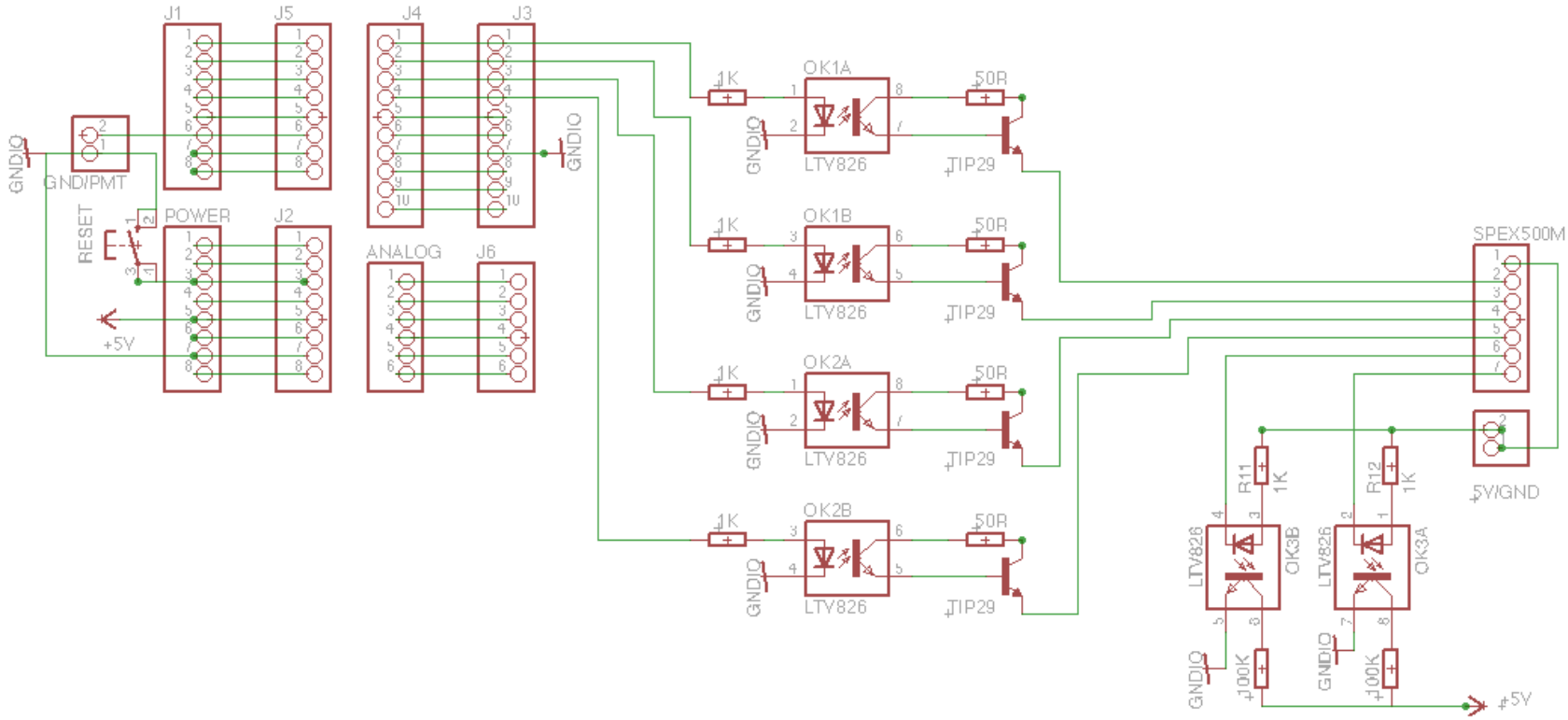


Figure H.1: Spex500M Grating Drive Schematic

H.2 Shutter Control Schematic

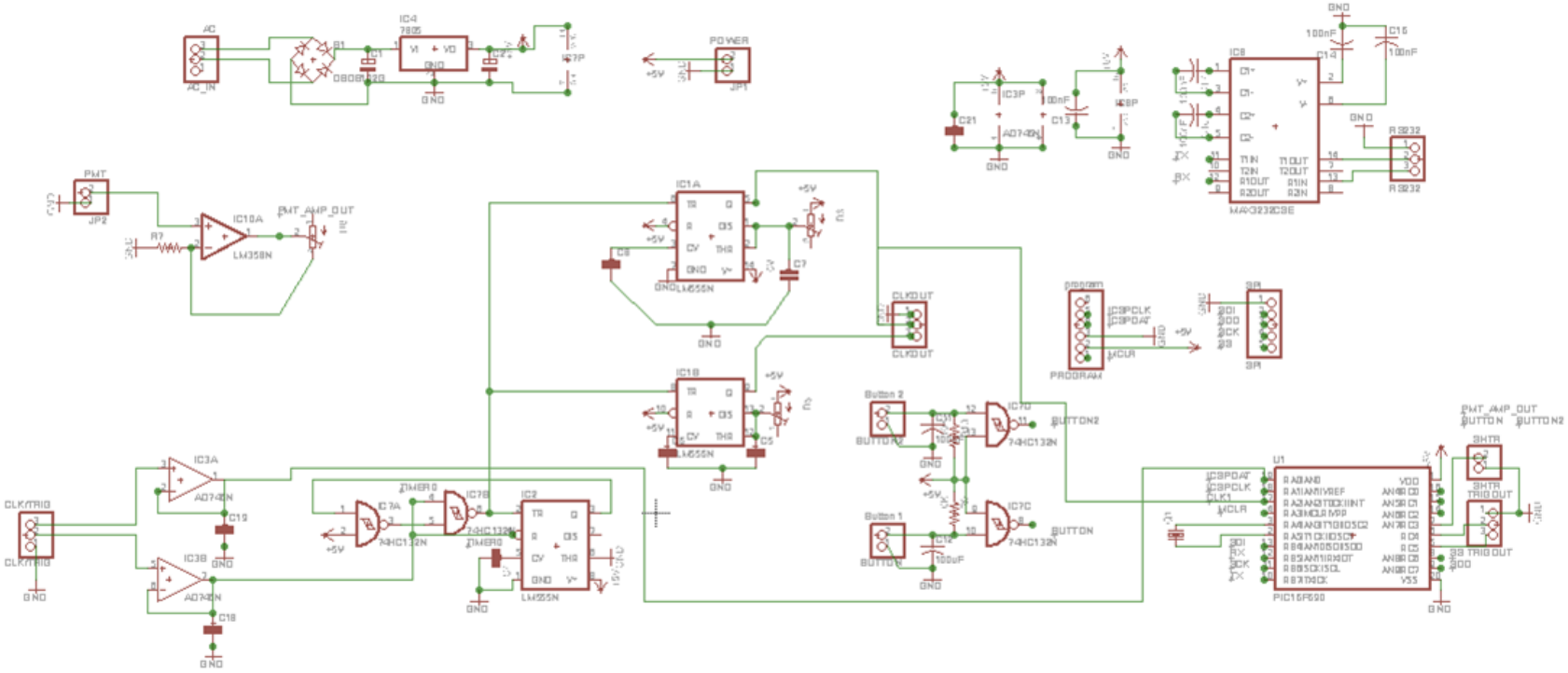


Figure H.2: Shutter Controller Schematic

I Embedded Code

I.1 Shutter Controller Code

File: main.c
Created: June 30th, 2013
Compiler: Microchip Hi-Tech C
Description: Microchip PIC16F690 shutter control main program (v5.2)

```
#include <stdio.h>
#include <stdlib.h>
#include <xc.h>
#include <htc.h>
#include "usart.h"
#include "main.h"

// PIC16F690 Configuration Bit Settings
#pragma config FOSC = HS
#pragma config WDTE = OFF
#pragma config PWRTE = OFF
#pragma config MCLRE = ON
#pragma config CP = OFF
#pragma config CPD = OFF
#pragma config BOREN = ON
#pragma config IESO = ON
#pragma config FCMEN = ON

//IO definitions
```

```
#define trigger PORTAbits.RA0          //define trigger pin
#define button PORTCbits.RC1           //define button pin
#define shutter PORTCbits.RC3          //define shutter pin
#define trigger_out_1 PORTCbits.RC4    //define trigger output
#define trigger_out_2 PORTCbits.RC5    //define trigger output
#define PMT_channel 4                  //Analog pin
#define PMT_AMP_MAX 615                //Max value of PMT op-amp

//Definitions to make the code that follows more readable.
#define time TMR0
#define time_reset TMR0=0; count=0;
#define CMD_Length 10                  //Max length of USART commands
#define debounce 200                   //debounce wait
#define close shutter=1                //define shutter close
#define open shutter=0                 //define shutter open
#define button_pressed button==1
#define button_released button==0
#define trigger_high trigger==1
#define trigger_low trigger==0

//Global Variables to handle RS232 Communications
char usartdata;                        //Last Received Byte of RS232 data
char usartcount=0;                     //Number of received bytes
char usartarray[CMD_Length];           //String received via RS232
unsigned int count=0;                  // value of the MCP Clock
char button_time=0x0;

// Global variables for channel positions, widths, and PMT max value
unsigned int max_ADC=0xFE;
unsigned int Last_ADC=0;
unsigned int Channel_0_Position;
unsigned int Channel_0_Width;
unsigned int Channel_1_Position;
unsigned int Channel_1_Width;
unsigned int Channel_2_Position;
unsigned int Channel_2_Width;
unsigned int Channel_3_Position;
unsigned int Channel_3_Width;
```

```
unsigned int Channel_4_Position;
unsigned int Channel_4_Width;

//Initialize Port I/O directions
void port_init() {
    TRISB = 0b11111111;                 // configure port B as input
    TRISA = 0b11111111;                 // configure port A as input
    TRISC = 0b00000111;                 //C0 through C2 for PMT and buttons.
}

//Load saved values from EEPROM
void init() {
    Channel_0_Position = eeprom_read(0x0)*256+eeprom_read(0x1);
    Channel_0_Width = eeprom_read(0x2);
    Channel_1_Position = eeprom_read(0x3)*256+eeprom_read(0x4);
    Channel_1_Width = eeprom_read(0x5);
    Channel_2_Position = eeprom_read(0x6)*256+eeprom_read(0x7);
    Channel_2_Width = eeprom_read(0x8);
    Channel_3_Position = eeprom_read(0x9)*256+eeprom_read(0xA);
    Channel_3_Width = eeprom_read(0xB);
    Channel_4_Position = eeprom_read(0xC)*256+eeprom_read(0xD);
    Channel_4_Width = eeprom_read(0xE);
    max_ADC=eeprom_read(0xF)*256+eeprom_read(0x10);
}

//Returns power of 10
unsigned int power_10(int power) {
    unsigned int value=1;
    for(int i=1;i<power+1;++i) value=value*10;
    return value;
}
```

```
//Error handeling for the program
void error_handel(int error_code) {
    switch (error_code) {
        case 0:
            T0IE=0;                         //disable the button pressed timer interrupt
            break;
        case 2:
            close;
            USARTWriteString("PMT Overflow Error\n");
            break;
        case 3:
            USARTWriteString("Unknown Command\n");
            usartcount=0;
            break;
        default:
            close;
            USARTWriteString("Unknown Error\n");
            break;
    }
    if(error_code!=3) {                     //UnShuttered Mode
        while(button_released) {
            CLRWDT(); //Wait for reset button to be pressed
            if(shutter == 0) Last_ADC = ADCRead(PMT_channel);
            if(Last_ADC > max_ADC) close;
        }
        for(char i=0;i<debounce;) {         //Software debounce for button
            CLRWDT();
            if (button_released) ++i;
            else i=0;
        }
        USARTWriteString("Shutter Enabled\n");
        while (trigger_high) CLRWDT();      //Resume shuttering at next trigger
        open; time_reset;
    }
}
```

```
void usart_parse() {                            //Parse out RS232 Commands
    usartcount=0; //reset bytes received
    char arraysize=0; //size of received string
    int j; //the test parameter
    for(arraysize=0;usartarray[arraysize]!='\n';++arraysize);    //determine size of string

    if(usartarray[0]=='c' && usartarray[2]=='s') {                  //Syntax c#s####
        unsigned int Channel_Set_Value=0;
        for(int i=0;i<arraysize-3;++i) {
            Channel_Set_Value=Channel_Set_Value+(usartarray[arraysize-1-i]-0x30)*power_10(i);
        }
        eeprom_write(0x0+(usartarray[1]-0x30)*0x3,Channel_Set_Value » 8);
        eeprom_write(0x1+(usartarray[1]-0x30)*0x3,Channel_Set_Value & 0x00ff);
        init();
    }

    if(usartarray[0]=='m' && usartarray[1]=='a' && usartarray[2]=='x') {
        USARTWriteInt(PMT_AMP_MAX,4);
        USARTWriteByte('\n');
    }

    if(usartarray[0]=='a' && usartarray[1]=='t' && usartarray[2]=='d'){
        if(usartarray[3]=='+') USARTWriteInt(Last_ADC,4);
        else USARTWriteInt(ADCRead(PMT_channel),4);
        USARTWriteByte('\n');
    }

    if(usartarray[0]=='p' && usartarray[1]=='m' && usartarray[2]=='t') {
        if(usartarray[3]=='?') {
            USARTWriteInt(max_ADC,4);
            USARTWriteByte('\n');
        }
        else {
            unsigned int pmt_value=0;
            for(int i=0;i<arraysize-3;++i) {
                pmt_value=pmt_value+(usartarray[arraysize-1-i]-0x30)*power_10(i);
            }
```

```
        if(pmt_value > PMT_AMP_MAX) pmt_value = PMT_AMP_MAX;
        if(eeprom_read(0xF) != (pmt_value»8)) eeprom_write(0xF,pmt_value»8);
        if(eeprom_read(0x10) != (pmt_value & 0x00ff)) {
            eeprom_write(0x10,pmt_value & 0x00ff);
        }
        init();
    }
}

if(usartarray[0]=='c' && usartarray[2]=='w') {
    unsigned int Channel_Width=0;          //syntax: c#w###
    for(int i=0;i<arraysize-3;++i) {
        Channel_Width=Channel_Width+(usartarray[arraysize-1-i]-0x30)*power_10(i);
    }
    eeprom_write(0x2+(usartarray[1]-0x30)*0x3,Channel_Width);
    init();
}

if(usartarray[0]=='c' && usartarray[2]=='r') {
    unsigned int Channel_Set_Value=0;    //syntax c#r
    unsigned int Channel_Width_Value=0;
    switch (usartarray[1]){
        case '0':
            Channel_Set_Value=eeprom_read(0x0)*256+eeprom_read(0x1);
            Channel_Width_Value=eeprom_read(0x2);
            break;
        case '1':
            Channel_Set_Value=eeprom_read(0x3)*256+eeprom_read(0x4);
            Channel_Width_Value=eeprom_read(0x5);
            break;
        case '2':
            Channel_Set_Value=eeprom_read(0x6)*256+eeprom_read(0x7);
            Channel_Width_Value=eeprom_read(0x8);
            break;
        case '3':
            Channel_Set_Value=eeprom_read(0x9)*256+eeprom_read(0xA);
            Channel_Width_Value=eeprom_read(0xB);
            break;
```

```
            case ’4’:
                Channel_Set_Value=eeprom_read(0xC)*256+eeprom_read(0xD);
                Channel_Width_Value=eeprom_read(0xE);
                break;
            default:
                USARTWriteString("Select Channel 1 - 5\n");
                break;
        }
        USARTWriteString("Channel ");
        USARTWriteByte(usartarray[1]);
        USARTWriteString(" Position: ");
        USARTWriteInt(Channel_Set_Value,4);
        USARTWriteString(" Width: ");
        USARTWriteInt(Channel_Width_Value,3);
        USARTWriteByte(’\n’);
    }
}
unsigned int ADCRead(unsigned char ch) {
    unsigned int ADC_result=0;
    if(ch>13) return 0;                     //Invalid Channel
    ADCON0=ADCON0 & 0b11000011;
    ADCON0=ADCON0 | (ch«2);                 //Select ADC Channel
    GO_DONE=1;                              //Start conversion
    while(GO_DONE);                         //wait for the conversion to finish
    ADC_result = ADRESH;
    ADC_result = (ADC_result«8)|ADRESL;
    return ADC_result;
}
void interrupt tc_int(void) {
    if (RCIE && RCIF) {
        usartdata=USARTReadByte();
        if(usartcount==CMD_Length) error_handel(3);
        else {
            usartarray[usartcount]=usartdata;
            ++usartcount;
            if(usartdata==’\n’) usart_parse();
        }
```

```
        RCIF=0;
        return;
    }
    if(RABIF && RABIE) {
    trigger_out_1=trigger;
    trigger_out_2=trigger;
    time_reset;
    RABIF=0;
    return;
    }
    if(T2IF && T2IE) {                  //Enter open shutter mode
    if(button_time!=0xFA) ++button_time;
    T2IF=0;
    return;
    }
}

void main(int argc, char** argv) {
    USARTInit();                        //Initialize RS232
    ADON=1;                             //switch on the ADC module
    RCIE=1;                             //Enable USART Rx Interrupt
    port_init();
    ANSEL=0b00010000;                   //AN4 analog pin
    ANSELH=0b00000000;                  //ADConverter options
    ADCON0=0b10010001;
    ADCON1=0b00100000;                  //TAD for minimum for 20Mhz
    IOCA=0b00000001;                    //enable A0 interrupt for trigger
    IOCB=0b00000000;                    //disable port B interrupt
    RABIE=1;                            //Enable RB6 interrupt for trigger
    INTCONbits.GIE=1;                   //enable global interrupts
    INTCONbits.INTE=0;
    INTCONbits.PEIE=1;
    TMR1CS=1;                           //Set TMR1 as counter
    T1OSCEN=0;
    T1CON=0b00000111;
    time_reset;
    T0IE=0;                             //Timer 0 interrupt disabled
    T0CS=1;                             //External Pin Counter Mode
```

```
OPTION_REGbits.T0SE=0;
T2CON=0b01111011;                   //Prescaler 1:16 Postscaler 1:16 Timer OFF
T2IE=1;                             //Timer 2 interrupt enabled
init();                             //read eeprom values

close;                              //Shutter Closed on Startup
while(button_released) CLRWDT();
for(char i=0;i<debounce;) {
   CLRWDT();
   if (button_released) ++i;
   else i=0;
}
while (trigger_high) CLRWDT();

for(;;) {                           //Begin Main Loop
   CLRWDT();
   count = count + TMR0; TMR0=0;
   Last_ADC = ADCRead(PMT_channel);
   if(Last_ADC>max_ADC) error_handel(2);
   if(button_pressed) {             //for unshuttered mode
      close;                        //Close Shutter
      T2CONbits.TMR2ON=1;
      button_time=0x0;
      for(char i=0;i<debounce;) {
         CLRWDT();
         if(button_time==0xFA) open;    //reopen shutter
         if (button_released) {
            ++i;
            button_time=0;
         }
         else i=0;
      }
      T2CONbits.TMR2ON=0;
      error_handel(0);              //runs unshuttered
   }
```

```
        if(count>=Channel_0_Position && count<=Channel_0_Position+Channel_0_Width
        && Channel_0_Position!=1) close;
        else if (count>=Channel_1_Position && count<Channel_1_Position+Channel_1_Width
        && Channel_1_Position!=1) close;
        else if (count>=Channel_2_Position && count<Channel_2_Position+Channel_2_Width
        && Channel_2_Position!=1) close;
        else if (count>=Channel_3_Position && count<Channel_3_Position+Channel_3_Width
        && Channel_3_Position!=1) close;
        else if (count>=Channel_4_Position && count<Channel_4_Position+Channel_4_Width
        && Channel_4_Position!=1) close;
        else open;
    }
}
```

I.2 TGP Stage Driver Code

File: TGP_Stage_Driver.ino
Created: July 6th, 2016
Compiler: Arduino IDE 1.0.5-r2
Description: Arduino-Uno embedded code for stage driver

```
#include <LiquidCrystal.h>
#include <MSTimer2.h>

class motor {
    public:
    void initialize(int d, int s);          //d = direction pin and s = step
    int location();                         //return current location
    void zero();                            //Set zero at current location
    void go(long int);                      //goto specified location
    int dir();                              //return direction pin
    void setStepDelay(int);                 //set the delay between steps in ms
    int stepDelay();                        //return current step delay
    void setSlack(long int);                //set motor slack

    private:
    long int x;                             //Stepper Location
    int dirPin;                             //Direction selection input pin
    int stepPin;                            //Step input pin
    int step_Delay;                         //delay between steps
    int slack;                              //backslash overshoot
    void over();                            //overshoot to adjust for backlash
    void oneStep();                         //motor takes one step
    void slackAdjust();                     //adjust for mechanical slack
};

void motor::setSlack(long int x) {
    slack = x;
}

int motor::stepDelay() {
```

```
    return step_Delay;
}

void motor::initialize(int d, int s) {
    dirPin = d;
    stepPin = s;
    x = 0;
    step_Delay = 1000;
    slack = 0;
    pinMode(dirPin, OUTPUT);
    pinMode(stepPin, OUTPUT);
    digitalWrite(dir(), LOW);
    digitalWrite(stepPin, LOW);
}

long int motor::location() {
    return x;
}

void motor::zero() {
    x = 0;
}

int motor::dir() {
    return dirPin;
}

void motor::over() {
    if(slack > 1) digitalWrite(dir(), HIGH);
    else digitalWrite(dir(), LOW);
    for(long int y; y < slack; ++y) {
        oneStep();
    }
}

void motor::go(long int y) {
```

```
    bool slackflag = 0;
    if(x < y) {
        digitalWrite(dir(), HIGH);
        if(slack > 0) slackflag = 1;
    }
    else {
        digitalWrite(dir(), LOW);
        if(slack < 0) slackflag = 1;
    }
    while(x != y) {
        if(x > y) ++x;
        else --x;
        oneStep();
    }
    if(slackflag) slackAdjust();
}

void motor::setStepDelay(int del) {
    digitalWrite(stepPin, HIGH);
    digitalWrite(stepPin, LOW);
    delayMicroseconds(step_Delay);
}

//**************************************
// Main Program Section
//**************************************
//Define Buttons
#define xup 8
#define xdown 10
#define yup 6
#define ydown 7
#define button 9
#define reset A5
#define motor_speed A4

motor xmotor;
motor ymotor;
```

```
LiquidCrystal lcd(13, 11, A0, A1, A2, A3);  //lcd(RS,E,D4,D5,D6,D7) RW-Gnd

//store location of up to 10 rubies in 3x10 array
//0th row 0 for unused or 1 for used
//1st row x location
//2nd row y location
long int rubys[3][10];
int stepDelay = 200 + 1024;                 //min step delay for motor 532 added to slow motor
int setFlag=0;                              //flag 1 to print "SET" on LCD or 2 print current ruby
const int motorSlack = 1000;                //motor slack
char currentRuby=0;

void setRuby(long int x, long int y) {
int i = 0;
while(rubys[0][i] == 1 && i < 10) ++i;      //find non-activated ruby location
while(digitalRead(button) == 0);
rubys[0][i] = 1;                            //ruby set for use
rubys[1][i] = x;                            //set x location
rubys[2][i] = y;                            //set y location
}

void set() {
    setFlag = 1;
    xmotor.setSlack(0);
    ymotor.setSlack(0);
```

```
    while(digitalRead(button) == 0);            //wait for button release
    delay(500);                                 //debounce
    while(digitalRead(reset) == 1 && rubys[0][9] == 0) {
        //Directional Control from Joystick
        if(digitalRead(xup) == 0) {
            xmotor.go(xmotor.location() + 1);
        }
        if(digitalRead(xdown) == 0) {
            xmotor.go(xmotor.location() - 1);
        }
        if(digitalRead(yup) == 0) {
            ymotor.go(ymotor.location() + 1);
        }
        if(digitalRead(ydown) == 0) {
            ymotor.go(ymotor.location() - 1);
        }
        //Button to set Ruby position
        if(digitalRead(button) == 0) {
            setRuby(xmotor.location(), ymotor.location());
            while(button==0);
            delay(500);                         //debounce
        }
    }
    setFlag = 0;
    xmotor.setSlack(motorSlack);
    ymotor.setSlack(motorSlack);
}

void motorLCD() {
    int percentMotor = 100 - (int)((double)(xmotor.stepDelay() - stepDelay)/1024*100);
    lcd.clear();
    lcd.print("Motor Speed: ");
    lcd.print(percentMotor);
}

void stepTo(char x) {
```

```
    if(x > 47 && x < 58) {                  //character between 0 and 9
        x -= 48;                            //convert from ASCII to numerical value binary
        if(rubys[0][x] == 1) {              //check to see if location is set
            xmotor.go(rubys[1][x]);
            ymotor.go(rubys[2][x]);
            currentRuby = x + 48;
            setFlag = 2;
        }
    }
}

void zeroRubys() {
    for(int i = 0; i < 10; ++i) {
        for(int k = 0; k<3; ++k) {
            rubys[k][i] = 0;
        }
    }
}

void motorSpeed() {
    int currentDelay = xmotor.stepDelay();
    int newDelay = analogRead(motor_speed) + stepDelay;
    if(abs(currentDelay-newDelay) > 10) {
        xmotor.setStepDelay(newDelay);
        ymotor.setStepDelay(newDelay);
        motorLCD();
        delay(1000);
    }
    else
        lcd.clear();
        lcd.print("X: ");
        lcd.print(xmotor.location());
        lcd.setCursor(0,1);
        lcd.print("Y: ");
        lcd.print(ymotor.location());
```

```
        if(setFlag == 1) {
            lcd.setCursor(11,0);
            lcd.print("*SET*");
            lcd.setCursor(11,1);
            int i = 0;
            while(rubys[0][i] == 1 && i < 10) ++i;
            lcd.print(i);
        }

        if(setFlag == 2) {
            lcd.setCursor(15,0);
            lcd.print(currentRuby);                 //setflag set to ruby location
        }
    }
}

void setup() {
    //Initialize Serial and LCD;
    Serial.begin(9600);
    pinMode(12,OUTPUT);
    digitalWrite(12,LOW); //ground R/W
    lcd.begin(16, 2);

    //TIMER2 USED FOR MOTOR SPEED CONTROL ANALOG READ
    MsTimer2::set(200, motorSpeed);
    MsTimer2::start();

    //intialize ruby locations
    zeroRubys();

    //Initialize Motor pin9 Direction pin8 Step for x-axis
    xmotor.initialize(2,3);
    xmotor.setSlack(motorSlack);
    ymotor.initialize(4,5);
    ymotor.setSlack(motorSlack);
```

```
    //Set Button Pin modes
    pinMode(reset,INPUT);                   //A0 Set Button
    pinMode(button,INPUT);                  //A1 Button on Controller
    pinMode(xup,INPUT);                     //A2 X-axis Positive
    pinMode(xdown,INPUT);                   //A3 X-axis Negative
    pinMode(yup,INPUT);                     //A4 Y-axis Positive
    pinMode(ydown,INPUT);                   //A5 Y-asix Negative
    pinMode(A5,INPUT);                      //A5 analog input for speed control
}

void printIDN() {
    Serial.println("TGP microscope stage driver");
    Serial.println("————————————————");
    Serial.println("0-9 Moves to Ruby Location");
    Serial.println("Z Erases stored Ruby Locations");
    Serial.println("S Returns 0-9 for ruby last selected");
    Serial.println("N Returns Number of Set rubies");
}

void returnstate() {
    Serial.print(currentRuby);
}

void returnNumber() {
    int i = 0;
    for(int k = 0; k < 10; ++k) {
        i += rubys[0][i];
    }
    Serial.print(i);
}

void loop() {
```

```
    //serial handler
    if(Serial.available()) {
        byte character=Serial.read();
        if(character==’I’ || character==’i’) printIDN();
        else if(character == ’Z’ || character == ’z’) zeroRubys();
        else if(character == ’S’ || character == ’s’) returnstate();
        else if(character == ’N’ || character == ’n’) returnNumber();
        else stepTo(character);
    }

    //Button Handler
    if(digitalRead(button) == 0) set();

    if(digitalRead(reset) == 0) {
        MsTimer2::stop();
        lcd.clear();
        lcd.print(" Hold Button");
        lcd.setCursor(0,1);
        lcd.print(" to Clear");

        //time button for reset
        long int i=0;
        while(digitalRead(reset) == 0 && i < 1000000) ++i;
        if(i == 1000000) {
            lcd.clear();
            lcd.print("Ruby Locations");
            lcd.setCursor(0,1);
            lcd.print(" Cleared");
            zeroRubys();
        }
        while(digitalRead(reset) == 0);          //hold till release
        lcd.clear();
        MsTimer2::start();
    }
}
```

I.3 SPEX 500M Controller Code

File: Spex500M_v2.ino
Created: October 31st, 2014
Compiler: Arduino IDE 1.0.5-r2
Description: Grating motor control and PMT counter

```
#include <EEPROMex.h>
#include <EEPROMVar.h>
#include <EEPROMex.h>                    //EEPROM extended functionality

#define serial_speed 9600                //RS232 Serial Communication Speed

//Arduino Uno digital ports for Travel Limit Switches
#define high_limit 6
#define low_limit 7

//Port Values for Stepper Positions
#define motor_port PORTB
char position_1=B00000101;
char position_2=B00000001;
char position_3=B00001001;
char position_4=B00001000;
char position_5=B00001010;
char position_6=B00000010;
char position_7=B00000110;
char position_8=B00000100;
```

```
char step_delay_ms=2;                   //2ms delay between motor steps
boolean echo=false;                     //Initialize with Echo Off for Communication
boolean limit=true;                     //Initialize with limit switches enabled
boolean stringComplete = false;         //indicate line feed for end of command
String serialRecieved = "";             //a string to hold incoming commands
long int location;                      //Current Location of Motor in steps
long int destination;                   //Current Destination in steps
char current_step = 0;                  //Value 0-7 holds current step location
char slk = 0;                           // 1 if moving in negative direction for slack adjust
long int slk_steps = 1000;              //Number of steps for slack adjust (About 15 Angstroms)
unsigned long int counts=0;             //PMT Counts
String inputString= "";                 //received serial commands
char incomingByte=0;                    //length of inputString

void SetupTimer1(){
   TCCR1A = 0;                          // normal counting mode
   TCNT1 = 0;                           // clear the timer count
   TCCR1B = TCCR1B | B00000111;         //Set for external clock
   TIMSK1=1;                            //Enable Timer1 interrupt for overflow handling
}

//Print Instruction and ID when prompted or on startup
void idn() {
```

```
    Serial.println("");
    Serial.println("SPEX 500M");
    Serial.println("Stepper Control Board");
    Serial.println("Original Design by:");
    Serial.println("THE Jennifer Stabach");
    Serial.println("");
    Serial.println("Command Summary");
    Serial.println("———————");
    Serial.println("IDN?: Menu");
    Serial.println("?: poll current location");
    Serial.println("STP?: poll motion status");
    Serial.println("STP: stop motion");
    Serial.println("S,#: move to #");
    Serial.println("DLY,#: step delay # ms");
    Serial.println("E: toggle echo");
    Serial.println("0: release motor");
    Serial.println("+#: move # steps");
    Serial.println("-#: move # steps");
    Serial.println("ZR: Set Location Zero");
    Serial.println("ZR,#: Set Location #");
    Serial.println("EP: Save Location to EPROM");
    Serial.println("V?: Read Analog Voltage");
    Serial.println("LMT: Enable/Dissable Limit Switches");
    Serial.write(0x4);                          //Labview Termination (EOT)
}

void set_step(char current_step) {
```

```
    switch(current_step) {
        case 0:
            motor_port=position_1;
            break;
        case 1:
            motor_port=position_2;
            break;
        case 2:
            motor_port=position_3;
            break;
        case 3:
            motor_port=position_4;
            break;
        case 4:
            motor_port=position_5;
            break;
        case 5:
            motor_port=position_6;
            break;
        case 6:
            motor_port=position_7;
            break;
        case 7:
            motor_port=position_8;
            break;
    }
    delay(step_delay_ms);
}

void setup() {
```

```
    SetupTimer1();
    pinMode(high_limit,INPUT);              //limit switch pin set to input
    pinMode(low_limit,INPUT);               //limit switch pin set to input
    if(position_1 = B00010100) DDRD = B00111110;    //Old Board
    else DDRD = B00000010;                  //Board updated for TMR1 Counter
    DDRB = DDRB | B00001111;                //Motor drive pins for board with TMR1 Counter
    Serial.begin(serial_speed);             //init serial port
    serialRecieved.reserve(20);             //reserve 20 bytes for the serialRecieved

    //Load Last Known Location from EEPROM
    int i = EEPROM.read(0);                 //Location of first good EEPROM block
    location = (signed long int) EEPROM.readLong((i+1)*sizeof(long));
    destination = location;
    current_step = location % 8;            //set current step
    set_step(current_step);                 //Grab Motor on current step
}

void respond()
                                            //degbug operations
    if(serialRecieved=="1\n") motor_port=B00000101;
    if(serialRecieved=="2\n") motor_port=B00001001;
    if(serialRecieved=="3\n") motor_port=B00001010;
    if(serialRecieved=="4\n") motor_port=B00000110;
    if(serialRecieved=="0\n") motor_port=B00000000;

    //return Counts
    if(serialRecieved=="k\n") Serial.println(counts + TCNT1);
```

```
//Returns count rate in counts/second
if(serialRecieved=="RT\n") {
    unsigned long int rate;
    TCNT1=0;
    counts=0;
    delay(100);
    rate = (counts + TCNT1)/.1;
    Serial.println(rate);
    Serial.write(0x4);                  //Labview Termination
}


//Respond IDN?
if(serialRecieved=="IDN?\n" || serialRecieved=="idn?\n") idn();

if(serialRecieved=="0\n") motor_port=0;

//Enable/Dissable Limit Switches
if(serialRecieved=="LMT\n") {
    Serial.print("Limit Switches ");
    if (limit) {
        limit=false;
        Serial.println("Dissabled");
        Serial.write(0x4);              //Labview Termination
    }
    else {
        limit=true;
        Serial.println("Enabled");
        Serial.write(0x4);              //Labview Termination
    }
}

//PMT Counts over provided time in ms
if(serialRecieved[0]=='C' && serialRecieved[1]==',') {
    serialRecieved[0]='0';
    serialRecieved[1]='0';
    int x = serialRecieved.toInt();
    TCNT1=0;
```

```
    counts=0;
    delay(x);
    Serial.println(counts + TCNT1);
    Serial.write(0x4);                  //Labview Termination
}

//Analog Voltage Return from pin 0
if(serialRecieved=="V?\n") {
    Serial.println(analogRead(0));
    Serial.write(0x4);                  //Labview Termination
}

if(serialRecieved=="VP\n") {
    Serial.print(analogRead(0));
    Serial.print(",");
    Serial.println(location);
    Serial.write(0x4);                  //Labview Termination
}

//Respond with Current Position
if(serialRecieved[0]=='?') {
    Serial.println(location);
    Serial.write(0x4);                  //Labview Termination
}

//Set Current Location to Zero
if(serialRecieved=="ZR\n") {
    while(current_step != 0) {
        ++destination;
        step_motor();
    }

    destination=0;
    location=0;
}

//Set Location to Zero
```

```
if(serialRecieved[0]=='Z' && serialRecieved[1]=='R' && serialRecieved[2]==',') {
    long int x;
    for(char i=0;i<3;++i) serialRecieved[i]='0';
    //Prep string for conversion to integer
    if(serialRecieved[3]=='-') {
        x=-1;
        serialRecieved[3]='0';
    }
    else x = 1;
    x = x*serialRecieved.toInt();
    location = x;
    destination = x;
}

//Save Location to EEPROM
if(serialRecieved=="EP\n") {
    int i=EEPROM.read(0);                   //starting memory location used
        if(i > 240) {                       //warning if EEPROM gets old
        Serial.println("EEPROM APPROACHING FAILER");
        Serial.println("Replace ATmega328");
        Serial.write(0x4); //Labview Termination
    }
    for(char k=0;k==0;i++) {                //This will store long in EEPROM
        if(i < 255) {
            EEPROM.updateLong((i+1)*sizeof(long),location);
            if(EEPROM.readLong((i+1)*sizeof(long))==location) ++k;
        }
        else {
            Serial.println("Memory Write Error");
            Serial.println("!!!REPLACE EEPROM!!!");
            Serial.write(0x4);              //Labview Termination
            ++k;
        }
    }
    –i;                                     //i updated in loop thus decrement back

    //Return Error code over RS232 if EEPROM Bad Write
```

```
    //And update The EEPROM 0 location
    if(EEPROM.update(0,i)!=0) {
        Serial.println("EEPROM LOCATION BAD");
        Serial.println("LOCATION POINTER ITERATED");
        Serial.write(0x4);                  //Labview Termination
    }
    //Return EEPROM Value and Location Written
    Serial.print("EEPROM Location: ");
    Serial.println((signed long int) EEPROM.readLong((i+1)*sizeof(long)));
    Serial.print("Memory Chunk: ");
    Serial.println(i);
    Serial.write(0x4);                  //Labview Termination
}
//move steps
if(serialRecieved[0]=='+')
    serialRecieved[0]='0';
    destination += serialRecieved.toInt();
}
if(serialRecieved[0]=='-') {
    serialRecieved[0]='0';
    destination -= ((serialRecieved.toInt()+slk_steps));
    slk=1;
}
//Set destination
if(serialRecieved[0]=='S' && serialRecieved[1]==',') {
    serialRecieved[0]='0';              //replace first two non-numerical values
    serialRecieved[1]='0';              //so toInt parse will function
    if(serialRecieved[2]=='-') {
        serialRecieved[2]='0';
        destination = -1;
    }
    else destination = 1;
    destination = destination*serialRecieved.toInt();
    if((destination - location) < 0) {
        destination -= slk_steps;
        slk=1;
    }
```

```
}

//Set Scan Speed
if(serialRecieved[0]=='D' && serialRecieved[1]=='L' && serialRecieved[2]=='Y' &&
serialRecieved[3]==',') {
    for(char i=0;i<4;++i) serialRecieved[i]='0';          //replace first 3 non-numerical values
    int spd = serialRecieved.toInt();          //scan out speed multiplyer
    if((spd < 100) && (spd > 0)) {
        step_delay_ms = spd;
        Serial.print("Delay = ");
        Serial.print(spd);
        Serial.println(" ms");
        Serial.write(0x4);                     //Labview Termination
    }
    else Serial.println("Range 3 to 99");
    Serial.write(0x4);                         //Labview Termination
}
//Respond to motion query
if(serialRecieved=="STP?\n") {
    if((location==destination) && (slk==0)) {
        Serial.println('Y');
        Serial.write(0x4);                     //Labview Termination
    }
    else {
        Serial.println('N');
        Serial.write(0x4);                     //Labview Termination
    }
}
if(serialRecieved=="STP\n") {
    destination = location;
    slk=0;
}
//Respond to Set Echo
if(serialRecieved=="E\n") {
    if(echo) {
        echo=false;
        Serial.println("ECHO OFF");
```

```
            Serial.write(0x4);                  //Labview Termination
        }
        else {
            echo=true;
            Serial.println("ECHO ON");
            Serial.write(0x4);                  //Labview Termination
        }
    }
    //Last thing to do is reset the incoming string to NULL and reset the flag
    serialRecieved = "";
    stringComplete = false;                     // reset the string complete
}
```